

MATH 205A NOTES

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ABSTRACT. Live-TeX notes from MATH 205A in Fall 2024, taught by Andrea Montanari. Lectures are labeled by week number, then Tu/Th.

These notes may contain typos and definitely contain omissions, but hopefully they are a reasonably faithful rendition of the contents of the course.

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1. 1TU – σ -ALGEBRAS, MEASURE SPACES, CARATHÉODORY EXTENSION THEOREM (STATEMENT)

1.1. Course Logistics.

- (1) HW: one problem set / week, due Thursdays on Gradescope
- (2) Exams: one midterm (80min), one final (3h) both open book (any notes + course texts, no computer) and in person.
- (3) Textbooks: Cohn, *Measure Theory*; Billingsley, *Probability and Measure*
- (4) Note on other courses: 172 and 230A cover similar content. 172 is a slower treatment of similar material; 230A overlaps significantly (particularly in the first half).

1.2. Motivation: Banach–Tarski.

Let Ω be a set, and let 2^Ω denote its power set (i.e., the set of all subsets of Ω). We want a function $\mu: 2^\Omega \rightarrow [0, \infty]$, called a *measure*, assigning each subset A to its measure $\mu(A)$. For example, consider $\Omega = \mathbb{R}^3$ and $\mu = \text{vol}$.

But, in many interesting cases, this is impossible, by the Banach–Tarski paradox.

Let $B^d(r) = \{x \in \mathbb{R}^d: \|x\| < r\}$ be the ball of radius r in $\mathbb{R}^d$. Let $A \sqcup B$ denote the disjoint union of two sets, and $\bigsqcup_{\alpha \in I} A_\alpha$ the disjoint union of a collection of sets. (In this class, I take this to mean that this is a union of sets assumed to be disjoint, not the abstract disjoint union of some collection of sets (that might not be disjoint).)

Definition 1.1 (Equidecomposable sets). The subsets $A, B \subseteq \mathbb{R}^d$ are *equidecomposable* if

$$A = \bigsqcup_{i=1}^k A_i, \quad B = \bigsqcup_{i=1}^k B_i,$$

such that A_i is a rotation and translation away from B_i .

Theorem 1.2 (Banach–Tarski). $B^3(1)$ and $B^3(2)$ are equidecomposable.

This gives us three options to salvage any reasonable notion of measure:

- (1) Give up additivity,
- (2) Give up invariance under the Euclidean group (translations, rotations, etc.),
- (3) **Give up the notion that every set has a volume.**

1.3. σ -algebras and measures.

Definition 1.3 (σ -algebra). A set $\mathcal{F} \subseteq 2^\Omega$ is a σ -algebra if:

- (1) $\emptyset \in \mathcal{F}$,
- (2) (complements) If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$,
- (3) (countable unions) If $A_i \in \mathcal{F}$ for $i \in \mathbb{N}$, then $\bigcup_{i=1}^\infty A_i \in \mathcal{F}$.

As a consequence, we get the following properties:

- (3a) If $A, B \in \mathcal{F}$, then $A \cap B \in \mathcal{F}$.

(3b) If $A_i \in \mathcal{F}$ disjoint, $i \in \mathbb{N}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.

(3b') If $B_i \uparrow B$ (i.e., $B_1 \subseteq B_2 \subseteq \dots$ and $B = \bigcup_{i=1}^{\infty} B_i$), and $B_i \in \mathcal{F}$ for all $i \in \mathbb{N}$, then $B \in \mathcal{F}$.

(4) $\Omega \in \mathcal{F}$,

(5) If $A_i \in \mathcal{F}$ for $i \in \mathbb{N}$, then $\bigcap_{i=1}^{\infty} A_i \in \mathcal{F}$.

Note that by (1) and (2), (3a) and (3b) imply (3): define $A'_i = A_i \setminus \bigcup_{j=1}^{i-1} A_j$; these are disjoint.

Definition 1.4 (measurable space). A pair $(\Sigma, \mathcal{F})$ of a set and a σ -algebra on it is called a *measurable space*.

Definition 1.5 (measure, measure space). Let $(\Sigma, \mathcal{F})$ be a measurable space. A *measure* is a function $\mu: \mathcal{F} \rightarrow [0, \infty]$ such that

(1) $\mu(\emptyset) = 0$,

(2) (countably additive) If $(A_i)_{i \in \mathbb{N}}$ disjoint, then $\mu(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mu(A_i)$.

A triple $(\Omega, \mathcal{F}, \mu)$ is called a *measure space*.

Definition 1.6 (finite, σ -finite measure). Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. Then, μ is *finite* if $\mu(\Omega) < \infty$, and *σ -finite* if there exist $A_i \in \mathcal{F}$ with $A_i \uparrow \Omega$ such that $\mu(A_i) < \infty$ for all $i \in \mathbb{N}$.

Example 1.7. Consider $(\mathbb{R}, 2^{\mathbb{R}}, \mu)$ with $\mu(A) = |A|$ the cardinality. Then this space is not σ -finite, since $\mathbb{R}$ is uncountably infinite.

Proposition 1.8 (Properties of measures). Let $(\Omega, \mathcal{F}, \mu)$ be a measure space.

(1) (monotonicity) If $A \subseteq B$, then $\mu(A) \leq \mu(B)$.

(2) (countable subadditivity) If $A_i \in \mathcal{F}$, then $\mu(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \mu(A_i)$.

(3) If $A_i \uparrow A$, then $\mu(A_i) \nearrow \mu(A)$ (where $a_i \nearrow a$ means that $a_i \rightarrow a$ (i.e., $\lim_{i \rightarrow \infty} a_i = a$) and a_i is nondecreasing).

(4) If $A_i \downarrow A$, then $\mu(A_i) \searrow \mu(A)$ if there exists $n \geq 0$ such that $\mu(A_n) < \infty$.

Example 1.9. Counterexample to 1.8(4): $(\mathbb{N}, 2^{\mathbb{N}}, |\cdot|)$, take $A_i = \{2^i k : k \in \mathbb{N}\}$. Each A_i is infinite, but $A_i \downarrow \emptyset$.

Proof of 1.8(4). For $i \geq n$, $\mu(A_i) = \mu(A_n) - \mu(A_n \setminus A_i)$; note all terms are finite. Note that $A_n \setminus A_i \uparrow A_n \setminus A$, so $\mu(A_n \setminus A_i) \nearrow \mu(A_n \setminus A)$. So, in the limit,

$$\mu(A) = \mu(A_n) - \mu(A_n \setminus A) < \infty.$$

□

Example 1.10. Let $(\Omega, \mathcal{F}) = (\mathbb{R}, 2^{\mathbb{R}})$. Some examples of measures:

(1) For some $x \in \mathbb{R}$, let $\mu = \delta_x$ be the *Dirac delta*, where $\delta_x(A) = 1$ if $x \in A$ and $\delta_x(A) = 0$ if $x \notin A$.

(2) Fix an enumeration of the rationals: $\mathbb{Q} = \{q_1, q_2, \dots\}$. Let $\mu = \sum_{i=1}^{\infty} 2^{-i} \delta_{q_i}$, so that

$$\mu(A) = \sum_{q_i \in A} 2^{-i}.$$

Definition 1.11 (σ -algebra generated by a set). Let $\mathcal{G} \subseteq 2^\Omega$. The σ -algebra generated by $\mathcal{G}$, denoted $\sigma(\mathcal{G})$, is the smallest σ -algebra containing $\mathcal{G}$; that is:

$$\sigma(\mathcal{G}) = \bigcap \{ \mathcal{F} : \mathcal{F} \supseteq \mathcal{G}, \mathcal{F} \text{ is a } \sigma\text{-algebra} \}.$$

To ensure that this definition makes sense, we should check:

Exercise 1.12. Show that the arbitrary intersection of σ -algebras is a σ -algebra.

Definition 1.13 (Borel σ -algebra). Let Ω be a topological space. The *Borel σ -algebra*, denoted by $\mathcal{B}_\Omega$, is the σ -algebra generated by the open (equiv. closed) subsets of Ω .

Remark 1.14. The Borel σ -algebra on $\mathbb{R}$, $\mathcal{B}_\mathbb{R}$, is generated by open intervals (a, b) with rational endpoints $a, b \in \mathbb{Q}$.

Proof of 1.14. Let $\mathcal{F} = \sigma(\{(a, b) \subseteq \mathbb{R} : a, b \in \mathbb{Q}, a < b\})$. Obviously $\mathcal{F} \subseteq \mathcal{B}_\mathbb{R}$.

To show that $\mathcal{B}_\mathbb{R} \subseteq \mathcal{F}$, we show that if $A \subseteq \mathbb{R}$ is open, then we prove that $A \in \mathcal{F}$. To do so, it is sufficient to show that

$$A = \bigcup \{(a, b) \subseteq A : a, b \in \mathbb{Q}, a < b\},$$

which follows from the definition of the (usual) topology on $\mathbb{R}$. □

1.4. Carathéodory extension theorem (statement).

Now we want to come up with the “uniform measure” λ on $(\mathbb{R}, \mathcal{B}_\mathbb{R})$, satisfying the following properties:

- (1) (normalization on intervals) $\lambda((a, b)) = b - a$,
- (2) (translation invariance) $\lambda(A + x) = \lambda(A)$.

We ask: does such a measure exist? If it exists, is it unique?

Definition 1.15 (algebra of sets). A subset $\mathcal{A} \subseteq 2^\Omega$ is an *algebra (of sets)* if

- (1) $\emptyset \in \mathcal{A}$,
- (2) If $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$.
- (3) If $A, B \in \mathcal{A}$, then $A \cup B \in \mathcal{A}$.

Example 1.16. An example of an algebra is the collection of finite unions of half-open intervals:

$$\mathcal{A} = \left\{ \bigcup_{i=1}^k (a_i, b_i] : a_1 < b_1 < a_2 < b_2 < \dots, k \in \mathbb{N} \right\}.$$

Theorem 1.17 (Carathéodory extension). *Let $\mathcal{A} \subseteq 2^\Omega$ be an algebra, and a function $\mu_0: \mathcal{A} \rightarrow [0, \infty]$ such that $\mu_0(\emptyset) = 0$ and μ_0 is countably additive (i.e., for $A = \bigsqcup_{i=1}^\infty A_i$, $A_i, A \in \mathcal{A}$,¹ then $\mu_0(A) = \sum_{i=1}^\infty \mu_0(A_i)$.) Then,*

- (1) *there exists a measure $\mu: \sigma(\mathcal{A}) \rightarrow [0, \infty]$ such that $\mu|_{\mathcal{A}} = \mu_0$,*
- (2) *if Ω is σ -finite with respect to $\mathcal{A}$ (i.e., there exist $A_i \in \mathcal{A}$ with $A_i \uparrow \Omega$ with $\mu_0(A_i) < \infty$), then μ is unique.*

Proof. Next time! □

As a consequence, we get the Lebesgue measure on $\mathbb{R}$: define

$$\lambda_0 \left(\bigcup_{i=1}^k (a_i, b_i] \right) = \sum_{i=1}^k (b_i - a_i)$$

on $\mathcal{A} \subseteq \mathcal{B}_{\mathbb{R}}$ as in Example 1.16. Since $\lambda_0(\emptyset) = 0$, we just need to check countable additivity. But, note that if we have $A_i, A \in \mathcal{A}$ with $A = \bigcup_{i=1}^\infty A_i$, then we must have that each interval in A is the union of some countable number of intervals, so it suffices to check in the case of $A = (a, b]$, with $A_i = (a_i, b_i]$ with $a = a_0, a_{i+1} = b_i$, and $b_i \nearrow b$. Then, we get straightforwardly that

$$\lambda_0(A) = b - a = \sum_{i=1}^\infty b_i - a_i = \sum_{i=1}^\infty \lambda_0(A_i),$$

as desired.

To say a bit more about uniqueness:

Definition 1.18 (π -system). A subset $\mathcal{P} \subseteq 2^\Omega$ is a π -system if $A, B \in \mathcal{P}$ implies that $A \cap B \in \mathcal{P}$.

Definition 1.19 (λ -system). A subset $\mathcal{L} \subseteq 2^\Omega$ is a λ -system if

- (1) $\Omega \in \mathcal{L}$,
- (2) If $A, B \in \mathcal{L}$ with $A \subseteq B$, then $B \setminus A \in \mathcal{L}$,
- (3) If $A_i \uparrow A$ with $A_i \in \mathcal{L}$, then $A \in \mathcal{L}$.

One can think of the properties of a λ -system as being the properties that would make a π -system into a σ -algebra. In fact, this is essentially correct, as we will see in Theorem 1.21.

Proposition 1.20 (Uniqueness of measures). *Let $\mathcal{P}$ be a π -system. If μ_1, μ_2 are measures on $\sigma(\mathcal{P})$, then if*

- (1) $\mu_1|_{\mathcal{P}} = \mu_2|_{\mathcal{P}}$,
- (2) (σ -finite with respect to $\mathcal{P}$) *There exists $A_i \in \mathcal{P}$ with $A_i \uparrow \Omega$, with $\mu_1(A_i), \mu_2(A_i) < \infty$,*

then $\mu_1 = \mu_2$.

¹Note here that we must have $A \in \mathcal{A}$; this condition says nothing about collections $A_i \in \mathcal{A}$ such that $\bigsqcup_{i=1}^\infty A_i \notin \mathcal{A}$.

Note that algebras are automatically π -systems. The benefit here for constructing the Lebesgue measure is that now we can simply check the requisite properties on the π -system

$$\mathcal{P} = \{(a, b] : a < b, a, b \in \mathbb{Q}\}.$$

The upshot of these definitions is that one can prove the Carathéodory extension theorem from a much more general result, the π - λ theorem.

Theorem 1.21 ($\sigma = \pi + \lambda$). *Let $\mathcal{F} \subseteq 2^\Omega$. Then,*

$$\mathcal{F} \text{ is a } \sigma\text{-algebra} \iff \mathcal{F} \text{ is a } \pi\text{-system and a } \lambda\text{-system}.$$

Proof. ($\implies$): Follows from the properties of a σ -algebra (Definition 1.3).

($\impliedby$): Note that if $A, B \in \mathcal{F}$, then $A \cup B = (A^c \cap B^c)^c \in \mathcal{F}$, so $\mathcal{F}$ is closed under finite unions. By taking the third property of λ -systems (closure under monotone unions), we can extend this to countable unions. $\square$

2. 1TH – DYNKIN’S π - λ THEOREM AND CARATHÉODORY EXTENSION THEOREM (PROOF)

2.1. Dynkin’s π - λ theorem.

Remark 2.1. The intersection of any number of λ -systems is a λ -system, and hence it makes sense to talk about the smallest λ system containing a set $\mathcal{G}$, denoted $\lambda(\mathcal{G})$.

Theorem 2.2 (Dynkin’s π - λ theorem). *If $\mathcal{P}$ is a π -system, then*

$$\sigma(\mathcal{P}) = \lambda(\mathcal{P}).$$

Remark 2.3. The direction $\sigma(\mathcal{P}) \subseteq \lambda(\mathcal{P})$ is obvious, since every σ -algebra is a λ -system. So, the substantive part of the theorem is to prove the other inclusion.

Proof. From the above remark, it remains to show that $\sigma(\mathcal{P}) \subseteq \lambda(\mathcal{P})$. In particular, it suffices to show that $\lambda(\mathcal{P})$ is a σ -algebra, and hence it suffices to show that $\lambda(\mathcal{P})$ is closed under (finite) intersections.

Let us denote $\mathcal{L} = \lambda(\mathcal{P})$, and let

$$\mathcal{L}_2 := \{A \in \mathcal{L} : A \cap B \in \mathcal{L} \text{ for all } B \in \mathcal{L}\}.$$

Obviously $\mathcal{L}_2 \subseteq \mathcal{L}$; our goal is to show equality. Define

$$\mathcal{L}_1 = \{A \in \mathcal{L} : A \cap B \in \mathcal{L} \text{ for all } B \in \mathcal{P}\}.$$

Of course, $\mathcal{L}_2 \subseteq \mathcal{L}_1 \subseteq \mathcal{L}$, so our task is to show both inclusions are equalities.

($\mathcal{L}_1 = \mathcal{L}$): Since $\mathcal{P}$ is a π -system, $\mathcal{P} \subseteq \mathcal{L}_1$. We want to show that $\mathcal{L}_1$ is a λ -system, which would entail that $\lambda(\mathcal{P}) = \mathcal{L} \subseteq \mathcal{L}_1$, as desired.

(1) $\Omega \in \mathcal{L}_1$ because $\Omega \in \mathcal{L}$, and $\Omega \cap B = B$ for any $B \in \mathcal{P}$, and hence $\Omega \cap B \in \mathcal{L}$.

(2) Let $A_1, A_2 \in \mathcal{L}_1$. Then, since

$$(A_2 \setminus A_1) \cap B = (A_2 \cap B) \setminus (A_1 \cap B),$$

which is in $\mathcal{L}$ since $\mathcal{L}$ is a λ -system.

(3) Let $A_i \in \mathcal{L}_1$, with $A = \bigcup_{i=1}^\infty A_i$ such that $A_i \uparrow A$. Then, since $A_i \cap B \uparrow A \cap B$, and $A_i \cap B \in \mathcal{L}$, it follows that $A \cap B \in \mathcal{L}$, so $A \in \mathcal{L}$, as desired.

But, note that we have just shown that if $A \in \mathcal{L}$ and $B \in \mathcal{P}$, then $A \cap B \in \mathcal{L}$. Therefore, $\mathcal{P} \subseteq \mathcal{L}_2$. But, hence now it suffices to prove that $\mathcal{L}_2$ is a λ -system, since this would show that $\mathcal{L} \subseteq \mathcal{L}_2$, as desired. But, repeating the steps above but with $B \in \mathcal{L}$, we get that $\mathcal{L}_2$ is a λ -system, and we are done. $\square$

Recall Proposition 1.20, which says that if two measures on a measure space $(\Omega, \sigma(\mathcal{P}))$ coincide on a π -system $\mathcal{P}$, and Ω is σ -finite with respect to $\mathcal{P}$, then the set of subsets where μ_1 and μ_2 coincide is a λ -system, and hence they coincide on all of $\sigma(\mathcal{P})$.

Proof of Proposition 1.20. Let

$$\mathcal{L} = \{A \in \sigma(\mathcal{P}) : \mu_1(A) = \mu_2(A)\},$$

and note that $\mathcal{P} \subseteq \mathcal{L}$. Since $\mathcal{P}$ is a π -system, we show that $\mathcal{L}$ is a λ -system, which by Dynkin's π - λ theorem implies that $\sigma(\mathcal{P}) \subseteq \mathcal{L}$ and hence $\sigma(\mathcal{P}) = \mathcal{L}$.

(Ω is finite): First consider the case where $\mu_1(\Omega) = \mu_2(\Omega) < \infty$.

(1) We immediately get that $\Omega \in \mathcal{L}$, as desired.

(2) If $A, B \in \mathcal{L}$, then we have

$$\mu_1(B \setminus A) = \mu_1(B) - \mu_1(A) = \mu_2(B) - \mu_2(A) = \mu_2(B \setminus A),$$

so $B \setminus A \in \mathcal{L}$.

(3) Similarly if $A_i \uparrow A$, where $A_i \in \mathcal{L}$, then

$$\mu_1(A) = \lim_{i \rightarrow \infty} \mu_1(A_i) = \lim_{i \rightarrow \infty} \mu_2(A_i) = \mu_2(A).$$

(Ω is σ -finite with respect to $\mathcal{P}$): Fix an exhaustion $\Omega_k \uparrow \Omega$, $\Omega_k \in \mathcal{P}$ and $\mu_1(\Omega_k) = \mu_2(\Omega_k) < \infty$. Define

$$\mu_1^{(k)}(A) = \mu_1(A \cap \Omega_k) \quad \text{and} \quad \mu_2^{(k)}(A) = \mu_2(A \cap \Omega_k).$$

By the finite case, we get that $\mu_1^{(k)} = \mu_2^{(k)}$ everywhere. So, now for any $A \in \sigma(\mathcal{P})$, we have that

$$\mu_1(A) = \lim_{k \rightarrow \infty} \mu_1(A \cap \Omega_k) = \lim_{k \rightarrow \infty} \mu_2(A \cap \Omega_k) = \mu_2(A).$$

$\square$

This concludes the proof of the uniqueness of measures agreeing on a π -system.

2.2. Carathéodory extension theorem.

We now move to existence, which is the Carathéodory extension theorem (Theorem 1.17). In fact, we have a slightly more general result:

Theorem 2.4. *Let $\mathcal{A} \subseteq 2^\Omega$ be an algebra, and $\mu_0 : \mathcal{A} \rightarrow [0, \infty]$ a function with $\mu_0(\emptyset) = 0$ and μ_0 is countably additive. Then, there exists a σ -algebra $\mathcal{G} \supseteq \mathcal{A}$ and a measure $\mu : \mathcal{G} \rightarrow [0, \infty]$ extending μ_0 .*

Remark 2.5. Our primary application will be to the algebra of finite unions of half-open intervals $(a, b]$. The σ -algebra generated by this algebra is the Borel σ -algebra, but the existence theorem extends to the entire σ -algebra of Lebesgue-measurable sets.

Definition 2.6 (outer measure). An *outer measure* on a set Ω is a function $\bar{\mu}: 2^\Omega \rightarrow [0, \infty]$ such that

- (1) $\bar{\mu}(\emptyset) = 0$,
- (2) (monotonicity) If $A \subseteq B$, then $\bar{\mu}(A) \leq \bar{\mu}(B)$,
- (3) (countably subadditive) If $A_i \subseteq \Omega$, then $\bar{\mu}(\bigcup_{i=1}^\infty A_i) \leq \sum_{i=1}^\infty \bar{\mu}(A_i)$.

Proof of Theorem 2.4. The proof of Theorem 2.4 follows the following outline:

- (1) Define an outer measure μ^* on 2^Ω ,
- (2) Prove that $\mu^*|_{\mathcal{A}} = \mu_0$,
- (3) Define the σ -algebra of μ^* -measurable sets

$$\mathcal{G} = \{A \subseteq \Omega: \mu^*(F \cap A) + \mu^*(F \cap A^c) = \mu^*(F) \text{ for all } F \subseteq \Omega\},$$

- (4) Prove that $\mathcal{G}$ is a σ -algebra,
- (4.5) Prove that $\mathcal{A} \subseteq \mathcal{G}$,
- (5) Prove that $\mu^*|_{\mathcal{G}}$ is a measure.

(Step 1: outer measure): Define

$$\mu^*(A) = \inf \left\{ \sum_{i=1}^\infty \mu_0(A_i): A_i \in \mathcal{A}, A \subseteq \bigcup_{i=1}^\infty A_i \right\}.$$

We show that μ^* is an outer measure. Properties (1) and (2) are obvious by definition. For property (3), suppose for the sake of contradiction that we have $A = \bigcup_{i=1}^\infty A_i$ with $\sum_{i=1}^\infty \mu^*(A_i) \leq \mu^*(A) - \varepsilon$ for some $\varepsilon > 0$. For each A_i , we can find an approximating cover $A_{i,k} \in \mathcal{A}$ such that for all i , we have

$$\sum_{k=1}^\infty \mu_0(A_{i,k}) \leq \mu^*(A_i) + 2^{-(i+1)}\varepsilon.$$

Since $\bigcup_{i,k=1}^\infty A_{i,k} \supseteq A$, we have that $\mu^*(A) \leq \sum_{i,k=1}^\infty \mu_0(A_{i,k})$, so

$$\mu^*(A) \leq \sum_{i,k=1}^\infty \mu_0(A_{i,k}) \leq \sum_{i=1}^\infty \mu^*(A_i) + \frac{\varepsilon}{2} \leq \mu^*(A) - \frac{\varepsilon}{2},$$

a contradiction.

(Step 2: μ^* extends μ_0): If $A \in \mathcal{A}$, then $A \subseteq A$, and hence $\mu^*(A) \leq \mu_0(A)$. Thus it remains to show the reverse inequality.

For the sake of contradiction, suppose that there exists a cover $A_i \in \mathcal{A}$ with $A \subseteq \bigcup_{i=1}^\infty A_i$ with $\sum_{i=1}^\infty \mu_0(A_i) \leq \mu_0(A) - \varepsilon$. Without loss of generality, we may assume that the A_i are disjoint. Note that since $\mathcal{A}$ is an algebra, $A_i \cap A \in \mathcal{A}$, so by monotonicity of μ_0 on $\mathcal{A}$, we have that $\mu_0(A_i \cap A) \leq \mu_0(A_i)$. By (disjoint) countable additivity, we get that

$$\mu_0(A) = \mu_0\left(\bigcup_{i=1}^\infty A_i \cap A\right) = \sum_{i=1}^\infty \mu_0(A_i \cap A) \leq \sum_{i=1}^\infty \mu_0(A_i) \leq \mu_0(A) - \varepsilon,$$

a contradiction.

(Step 3: Define the σ -algebra of μ^* -measurable sets): Let us first consider the finite case: $\mu^*(\Omega) < \infty$. Then, we can define the inner measure

$$\mu_*(A) = \mu^*(\Omega) - \mu^*(\Omega \setminus A).$$

Note that $A \in \mathcal{G}$ if and only if $\mu_*(A) = \mu^*(A)$, so $\mathcal{G}$ is the σ -algebra on which inner approximations and outer approximations by sets in $\mathcal{A}$ agree. Alternatively, we may think of $\mathcal{G}$ as the sets for which μ^* “cuts them additively” with respect to any other set $F \subseteq \Omega$.

(Step 4: $\mathcal{G}$ is a σ -algebra):

(1) $\Omega \in \mathcal{G}$ is obvious, since

$$\mu^*(F) + \mu^*(\emptyset) = \mu^*(F).$$

(2) $\mathcal{G}$ is clearly closed under complements, since the expression

$$\mu^*(F) = \mu^*(F \cap A) + \mu^*(F \cap A^c)$$

is symmetric about A and A^c .

(3a) Let $B, C \in \mathcal{G}$, and denote $A = B \cap C$. Then,

$$\begin{aligned} \mu^*(A \cap F) + \mu^*(A^c \cap F) &= \mu^*(B \cap C \cap F) + \mu^*((B^c \cup C^c) \cap F) \\ &\leq \mu^*(B \cap C \cap F) + \mu^*(B \cap C^c \cap F) + \mu^*(B^c \cap C \cap F) + \mu^*(B^c \cap C^c \cap F) \\ &= \mu^*(B \cap F) + \mu^*(B^c \cap F) \\ &= \mu^*(F). \end{aligned}$$

The other direction follows immediately from subadditivity.

(3b) Let $A_i \in \mathcal{G}$ be disjoint, and denote $A = \bigsqcup_{i=1}^{\infty} A_i$. By subadditivity, we immediately get that

$$\mu^*(F \cap A) \leq \sum_{i=1}^{\infty} \mu^*(F \cap A_i).$$

For the other direction, we begin by proving the desired inequality for finite unions.

$$\begin{aligned} \mu^*(F \cap (A_1 \cap A_2)) &= \mu^*(F \cap A_1) + \mu^*(F \cap A_2) \\ &= \mu^*(B \cap A_1) + \mu^*(B \cap A_2), \end{aligned}$$

where we write $B = F \cap (A_1 \cap A_2)$ and the first step is splitting by A_1 . Now for countable unions, note that at each finite step we get that

$$\mu^*(F \cap A) \geq \mu^*\left(F \cap \left(\bigcup_{i=1}^n A_i\right)\right) = \sum_{i=1}^n \mu^*(F \cap A_i)$$

for each n , so taking $n \rightarrow \infty$ we get the desired inequality, and hence

$$(*) \quad \mu^*(F \cap A) = \sum_{i=1}^{\infty} \mu^*(F \cap A_i).$$

Now, to show that $A \in \mathcal{G}$, we want to show that

$$\mu^*(F \cap A) + \mu^*(F \cap A^c) \leq \mu^*(F).$$

Let $B_n = (\bigcup_{i=1}^n A_i)^c \in \mathcal{G}$. We can split F by each of these B_n :

$$\begin{aligned}\mu^*(F) &= \mu^*\left(F \cap \left(\bigcup_{i=1}^n A_i\right)\right) + \mu^*(F \cap B_n) \\ &\geq \sum_{i=1}^n \mu^*(F \cap A_i) + \mu^*(F \cap A^c),\end{aligned}$$

since $B_n \supseteq A^c$. Taking the limit as $n \rightarrow \infty$, we get the desired inequality.

(Step 4.5: $\mathcal{A} \subseteq \mathcal{G}$): Fix $A \in \mathcal{A}$ and $F \in 2^\Omega$. Then, we want to show that

$$\mu^*(F) \geq \mu^*(F \cap A) + \mu^*(F \cap A^c),$$

as the other inequality follows from subadditivity.

Pick $A_i \in \mathcal{A}$ approximating F : $\sum_{i=1}^\infty \mu_0(A_i) \leq \mu^*(F) + \varepsilon$, and $F \subseteq \bigcup_{i=1}^\infty A_i$. Then, note that

$$F \cap A \subseteq \bigcup_{i=1}^\infty A_i \cap A \quad \text{and} \quad F \cap A^c \subseteq \bigcup_{i=1}^\infty A_i \cap A^c,$$

which are both covers by elements in $\mathcal{A}$. Then,

$$\begin{aligned}\mu^*(F \cap A) + \mu^*(F \cap A^c) &\leq \sum_{i=1}^\infty \mu_0(A_i \cap A) + \mu_0(A_i \cap A^c) \\ &\leq \sum_{i=1}^\infty \mu_0(A_i) \\ &\leq \mu^*(F) + \varepsilon.\end{aligned}$$

Note that in the second inequality we use μ_0 -measurability of A_i . Taking $\varepsilon \rightarrow 0$, we get the desired result.

(Step 5: $\mu^*|_{\mathcal{G}}$ is a measure): The only property we are left to check is countable additivity. Taking $F = \Omega$ in (*), we get this immediately. $\square$

3. 2TU – COMPLETIONS

We begin by completing the proof of the extension theorem from last time: see Step 4.5 in the proof of Theorem 2.4.

3.1. Complete measure spaces.

Remark 3.1. If $\mu^*(A) = 0$, then $A \in \mathcal{G}_{\mu^*}$ is μ^* -measurable.

To see this, note that by monotonicity, since $F \cap A \subseteq A$,

$$\mu^*(F \cap A) + \mu^*(F \cap A^c) = \mu^*(F \cap A^c) \leq \mu^*(F).$$

Definition 3.2. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. A set $A \in 2^\Omega$ is μ -null or μ -negligible if $\mu(A) = 0$.

Definition 3.3. A measure space $(\Omega, \mathcal{F}, \mu)$ is *complete* if for every $A \subseteq B$, $B \in \mathcal{F}$, $\mu(B) = 0$, then $A \in \mathcal{F}$ and $\mu(A) = 0$.

That is, null subsets of measurable null sets are measurable.

By Remark 3.1, $(\Omega, \mathcal{G}_{\mu^*}, \mu^*|_{\mathcal{G}_{\mu^*}})$ is complete, so in particular $(\mathbb{R}^d, \mathcal{B}_{\mathbb{R}^d}, \lambda)$ is complete.

Given a measure space, we can define its completion:

Definition 3.4. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space. Its *completion*, denoted $(\Omega, \mathcal{F}_\mu, \bar{\mu})$, is given as follows.

- (1) $\mathcal{F}_\mu = \{A \in 2^\Omega : \exists A_-, A_+ \in \mathcal{F} \text{ s.t. (1) } A_- \subseteq A \subseteq A_+ \text{ and (2) } \mu(A_+ \setminus A_-) = 0\}$,
- (2) $\bar{\mu}(A) = \bar{\mu}(A_-) = \bar{\mu}(A_+)$.

To see that the measure is well-defined, we show that $\bar{\mu} = \hat{\mu}$, where

$$\hat{\mu}(A) = \sup \{\mu(B) : B \in \mathcal{F}, B \subseteq A\}$$

is the inner approximation by sets in $\mathcal{F}$. But, since

$$\mu(A_-) \leq \hat{\mu}(A) \leq \mu(A_+),$$

we immediately get $\bar{\mu} = \hat{\mu}$.

We now check that $\mathcal{F}_\mu$ is a σ -algebra.

- (1) Obviously $\mathcal{F} \subseteq \mathcal{F}_\mu$, so we automatically have $\emptyset \in \mathcal{F}_\mu$.
- (2) Let $A \in \mathcal{F}_\mu$. Then, we have $A_-, A_+ \in \mathcal{F}$ as defined, but note that then $A_+^c \subseteq A^c \subseteq A_-^c$ with $\mu(A_-^c \setminus A_+^c) = 0$, so $A^c \in \mathcal{F}_\mu$.
- (3) Let $A_i \in \mathcal{F}_\mu$, take A_{i+}, A_{i-} as defined, and denote $A = \bigcup_{i=1}^\infty A_i$. Then, defining

$$A_+ = \bigcup_{i=1}^\infty A_{i+} \quad \text{and} \quad A_- = \bigcup_{i=1}^\infty A_{i-},$$

we get that

$$A_+ \setminus A_- \subseteq \bigcup_{i=1}^\infty (A_{i+} \setminus A_{i-}),$$

and therefore by monotonicity and countable subadditivity, $\mu(A_+ \setminus A_-) = 0$.

3.2. Completion of the Borel σ -algebra.

Let us denote $(\mathbb{R}^d, \mathcal{L}, \lambda_{\mathcal{L}})$ for the Lebesgue measure space with the (entire) Lebesgue σ -algebra constructed by the Carathéodory extension theorem, $(\mathbb{R}^d, \mathcal{B}_{\mathbb{R}^d}, \lambda)$ the Borel measure space with $\lambda = \lambda|_{\mathcal{L}}$, and $(\mathbb{R}^d, \mathcal{B}_\lambda, \bar{\lambda})$ the completion of the Borel measure space. We show that $\mathcal{B}_{\mathbb{R}^d}$ is not complete, but that its completion and the Lebesgue σ -algebra coincide.

Proposition 3.5.

- (1) $(\mathcal{L} \subseteq \mathcal{B}_\lambda)$ For all $A \in \mathcal{L}$, there exist $A_-, A_+ \in \mathcal{B}$ such that $A_- \subseteq A \subseteq A_+$ with $\lambda(A_+ \setminus A_-) = 0$,
- (2) $(\mathbb{R}^d, \mathcal{L}, \lambda_{\mathcal{L}}) = (\mathbb{R}^d, \mathcal{B}_\lambda, \bar{\lambda})$,
- (3) $\mathcal{L} \setminus \mathcal{B} \neq \emptyset$.

The proof of this proposition will follow from the following regularity result:

Proposition 3.6 (Inner and outer regularity). *Let $(\mathbb{R}^d, \mathcal{L}, \lambda_{\mathcal{L}})$ be the Lebesgue measure space, and let $A \in \mathcal{L}$.*

- (1) *(outer regularity by open sets)* $\lambda(A) = \inf \{ \lambda(U) : U \text{ open}, A \subseteq U \},$
- (2) *(inner regularity by compact sets)* $\lambda(A) = \sup \{ \lambda(K) : K \text{ compact}, K \subseteq A \}.$

Proof. We begin with the proof of (1) for $d = 1$. Note that

$$\begin{aligned} \lambda(A) &= \inf \left\{ \sum_{i=1}^{\infty} \lambda_0(A_i), A_i = \bigcup_{j=1}^{n_i} J_{ij}, A \subseteq \bigcup_{i,j} J_{ij} \right\} \\ &= \inf \left\{ \sum_{i,j} |J_{ij}|, A \subseteq \bigcup_{i,j} J_{ij} \right\}. \end{aligned}$$

So, we can reindex to get

$$\lambda(A) = \inf \left\{ \sum_{i=1}^{\infty} (b_i - a_i) : a_i < b_i, A \subseteq (a_i, b_i) \right\}.$$

Note that the open intervals do not form an algebra, but since we can always approximate any cover by half-open intervals $(a_i, b_i]$ by an arbitrary thickening $(a_i, b_i + 2^{-i}\varepsilon)$ to form a cover by open intervals, we lose no generality. Then,

$$\begin{aligned} \lambda(A) &= \inf \left\{ \sum_{i=1}^{\infty} (b_i - a_i) : a_i < b_i, A \subseteq (a_i, b_i) \right\} \\ &\geq \inf \left\{ \lambda(U) : U = \bigcup_{i=1}^{\infty} (a_i, b_i), A \subseteq U \right\} \\ &\geq \inf \{ \lambda(U) : U \text{ open}, A \subseteq U \}. \end{aligned}$$

The reverse inequality follows by monotonicity, so we are done.

We now do the proof of (2) for $d = 1$. First assume $A \in \mathcal{L}$ is bounded, so $A \subseteq [-M, M] = C$. Take $C \setminus A$, which is in $\mathcal{L}$ and therefore by (1) we can approximate by open sets:

$$\lambda(C \setminus A) = \inf \{ \lambda(U) : U \supseteq C \setminus A \}.$$

Note that if $C \setminus A \subseteq U$, then $C \setminus U \subseteq A$ is compact. By subadditivity, $\lambda(U) \geq \lambda(C) - \lambda(C \setminus U)$. So,

$$\begin{aligned} \lambda(C \setminus A) &= \inf \{ \lambda(U) : U \supseteq C \setminus A \} \\ &\geq \inf \{ \lambda(C) - \lambda(C \setminus U) : C \setminus U \subseteq A \text{ compact} \} \\ &= \lambda(C) - \sup \{ \lambda(K) : K \subseteq A \text{ compact} \}, \end{aligned}$$

so we get that

$$\lambda(A) \leq \sup \{ \lambda(K) : K \subseteq A \text{ compact} \}.$$

As before, the reverse inequality follows by monotonicity.

Now we do the case of (2) where A is unbounded but $\lambda(A) < \infty$. Then, picking a sequence $M_n \rightarrow \infty$, let $A_n = A \cap [-M_n, M_n]$, so that $A_n \uparrow A$ and $\lambda(A_n) \geq \lambda(A) - \frac{1}{n}$. By the finite

case, for each n we can produce a K_n compact such that $\lambda(K_n) \geq \lambda(A_n) - \frac{1}{n}$. Then,

$$\lambda(A) \leq \lambda(A_n) + \frac{1}{n} \leq \lambda(K_n) + \frac{2}{n},$$

so

$$\lambda(A) \leq \sup \{ \lambda(K) : K \subseteq A \text{ compact} \}.$$

Finally, we check the case of (2) where $\lambda(A) = \infty$. Then, we can find a sequence K_n such that $\lambda(K_n) \geq n$ and $K_n \uparrow A$, so we are done. $\square$

Proof of Proposition 3.5. (1): First suppose $\lambda(A) < \infty$. Using outer regularity, pick open $U_n \supseteq A$ such that $\lambda(U_n) \leq \lambda(A) + \frac{1}{n}$. Then, pick $A_+ = \bigcap_{n=1}^{\infty} U_n$; this is Borel and hence

$$\lambda(A_+) \leq \lambda(U_n) \leq \lambda(A) + \frac{1}{n},$$

so since this is true of all n , $\lambda(A_+) = \lambda(A)$. On the other end, using inner regularity we can pick compact $K_n \subseteq A$ with $\lambda(K_n) \geq \lambda(A) - \frac{1}{n}$ and take $A_- = \bigcup_{n=1}^{\infty} K_n$. Then, similarly we get $\lambda(A_-) = \lambda(A)$, so $\lambda(A_+ \setminus A_-) = \lambda(A) - \lambda(A) = 0$.

If $\lambda(A) = \infty$, then we can split $A = \bigsqcup_{i=1}^{\infty} A_i$, where

$$A_i = A \cap (B_i(0) \setminus B_{i-1}(0))$$

and hence $\lambda(A_i) \leq \lambda(B_i(0)) < \infty$. Then, approximating at each finite stage and taking the union, we get inner and outer Borel approximations to A . $\square$

Note that Proposition 3.5(1) proves that $\mathcal{L} \subseteq \mathcal{B}_\lambda$ and $\lambda_{\mathcal{L}}|_{\mathcal{B}} = \lambda$. Now for Proposition 3.5(2), we want to show the other inclusion $\mathcal{B}_\lambda \subseteq \mathcal{L}$. This is a consequence of the following lemma, which expresses a sort of universal property for measurable maps into a complete measure space:

Lemma 3.7. *Let $(\Omega, \mathcal{F}, \mu) \subseteq (\Omega, \mathcal{G}, \nu)^2$ are measure spaces, where $(\Omega, \mathcal{G}, \nu)$ is complete. Then, $(\Omega, \mathcal{F}_\mu, \bar{\mu}) \subseteq (\Omega, \mathcal{G}, \nu)$.*

Proof. Let $A \in \mathcal{F}_\mu$. Then, by definition we have $A_-, A_+ \in \mathcal{F}$ with $A_- \subseteq A \subseteq A_+$ and $\mu(A_+ \setminus A_-) = 0$. Now note that $A_+ \setminus A \subseteq A_+ \setminus A_-$, and since $A_+ \setminus A_- \in \mathcal{F}$ is μ -null, we have $A_+ \setminus A \in \mathcal{G}$ is ν -null, and therefore $A_+ \setminus A \subseteq A_+ \setminus A_-$ is ν -null by completeness.

Therefore, $A = A_+ \cap (A_+ \setminus A)^c \in \mathcal{G}$, and moreover

$$\nu(A) = \nu(A_+) = \mu(A_+) = \bar{\mu}(A_+),$$

so $\nu|_{\mathcal{F}_\mu} = \bar{\mu}$, as desired. $\square$

3.3. Lebesgue measure.

The Lebesgue measure is uniquely characterized by the properties:

- (1) (translation invariance) $\lambda(A) = \lambda(A + x)$ for all $x \in \mathbb{R}^d$,
- (2) (scaling) $\lambda(cA) = c^d \lambda(A)$ for all $c > 0$,
- (3) (normalization) $\lambda([0, 1]^d) = 1$.

²We say $(\Omega, \mathcal{F}, \mu) \subseteq (\Omega, \mathcal{G}, \nu)$ when $\mathcal{F} \subseteq \mathcal{G}$ and $\nu|_{\mathcal{F}} = \mu$.

In particular, for any measure μ on $(\mathbb{R}^d, \mathcal{B}_{\mathbb{R}^d})$ satisfying (1) and (2), and if $\mu([0, 1]^d) = a$ for some $a > 0$, then $\mu = a\lambda$.

Note: no lecture on Thursday!

4. 3TU – MEASURABLE FUNCTIONS

4.1. Measurable functions.

Definition 4.1. Let $(X, \mathcal{F})$ and $(Y, \mathcal{G})$ be measurable spaces. A function $f: X \rightarrow Y$ is called *measurable* if $f^{-1}(A) \in \mathcal{F}$ for all $A \in \mathcal{G}$. If $f: (X, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ is measurable, we say that it is *Borel*.

As a matter of convention, we consider any $f: (X, \mathcal{F}) \rightarrow (Y, \mathcal{G})$ to denote a measurable function. We may also say that $f: X \rightarrow Y$ is measurable if the σ -algebras are understood by context (e.g., if both X, Y are topological spaces with the Borel σ -algebra).

Proposition 4.2. If $\mathcal{A} \subseteq 2^Y$, and $f: X \rightarrow Y$ is a set-function such that $f^{-1}(A) \in \mathcal{F}$ for all $A \in \mathcal{A}$, then $f: (X, \mathcal{F}) \rightarrow (Y, \sigma(\mathcal{A}))$ is measurable.

That is, to check that a function is measurable it suffices to check the property on a generating set.

Proof. Let $\mathcal{G} = \{A \in \sigma(\mathcal{A}): f^{-1}(A) \in \mathcal{F}\}$; we want to show that $\mathcal{G} = \sigma(\mathcal{A})$. By assumption, $\mathcal{A} \subseteq \mathcal{G} \subseteq \sigma(\mathcal{A})$, so it remains to check that $\mathcal{G}$ is a σ -algebra, which would show that $\sigma(\mathcal{A}) \subseteq \mathcal{G}$ and hence $\mathcal{G} = \sigma(\mathcal{A})$.

- (1) $\emptyset \in \mathcal{G}$ since $f^{-1}(\emptyset) = \emptyset \in \mathcal{F}$.
- (2) If $A \in \mathcal{G}$, then $f^{-1}(A^c) = (f^{-1}(A))^c$, so $A^c \in \mathcal{G}$.
- (3) If $A_i \in \mathcal{G}$ and $A = \bigcup_{i=1}^{\infty} A_i$, then $f^{-1}(A) = \bigcup_{i=1}^{\infty} f^{-1}(A_i)$, but since each set $f^{-1}(A_i) \in \mathcal{F}$ by design and $\mathcal{F}$ is a σ -algebra, $f^{-1}(A) \in \mathcal{F}$ and hence $A \in \mathcal{G}$.

□

Example 4.3. For a function $f: (X, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$, to show that it is Borel it suffices to check that sublevel sets $f^{-1}((-\infty, a]) \in \mathcal{F}$ for all $a \in \mathbb{R}$, since the half-spaces $(-\infty, a]$ generate the Borel σ -algebra $\mathcal{B}_{\mathbb{R}}$.

Definition 4.4. Let $f_{\alpha}: X \rightarrow \mathbb{R}$ be a collection of functions indexed by $\alpha \in \Gamma$. The σ -algebra generated by $\{f_{\alpha}\}$ is the smallest σ -algebra on X such that all f_{α} are measurable:

$$\sigma(\{f_{\alpha}\}_{\alpha \in \Gamma}) = \sigma\left(\bigcup_{\alpha \in \Gamma} \sigma(f_{\alpha})\right).$$

Example 4.5. If $X = \mathbb{R}^2$ and $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ is $f(x_1, x_2) = x_1$, then

$$\sigma(f) = \mathcal{B}_{\mathbb{R}} \times \{\mathbb{R}\} = \{A \times \mathbb{R} \subseteq \mathbb{R}^2: A \in \mathcal{B}_{\mathbb{R}}\}.$$

Definition 4.6. A function $f: (X, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ is *simple* if it is a linear combination of characteristic functions of measurable subsets:

$$f(x) = \sum_{i=1}^N c_i \mathbb{1}_{A_i}(x),$$

where

$$\mathbb{1}_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A. \end{cases}$$

We denote by $\text{SF}(X, \mathbb{R})$ the set of simple functions $f: (X, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B})$, where the σ -algebra $\mathcal{F}$ is usually fixed by context.

Example 4.7. If $f \in \text{SF}(X, \mathbb{R})$, with $f = \sum_{i=1}^N c_i \mathbb{1}_{A_i}$, then

$$\sigma(f) = \sigma(\{A_i\}_{i=1}^N).$$

Proposition 4.8 (Pointwise approximation of measurable functions by simple functions).

Let $f: X \rightarrow \mathbb{R}$ be Borel. Then, there exists a sequence $f_n \in \text{SF}(X, \mathbb{R})$ such that $f_n(x) \rightarrow f(x)$ for all $x \in X$. Furthermore, if $f \geq 0$, then we can take $f_n \nearrow f$ and $f_n \geq 0$.

Proof. We start with the nonnegative case: $f \geq 0$. The idea is to slice up the codomain into $L_n = 4^n$ small intervals of length $\varepsilon_n = 2^{-n}$, so that in the limit $L_n \varepsilon_n \rightarrow \infty$ and the entire codomain $[0, \infty)$ is covered by intervals of gradually smaller length. Then, define the simple approximation

$$f_n(x) = \sum_{k=0}^{L_n-1} k\varepsilon_n \mathbb{1}_{f^{-1}([k\varepsilon_n, (k+1)\varepsilon_n))}(x) + L_n \varepsilon_n \mathbb{1}_{f^{-1}([L_n \varepsilon_n, \infty))}(x).$$

Clearly $f_n \leq f$ and $f_n \leq f_{n+1}$, and pointwise for all $x \in X$ we have $n_0(x)$ such that for all $n > n_0$, $|f(x) - f_n(x)| \leq \varepsilon_n$, and hence $f_n \nearrow f$ pointwise.

In general, we write $f = f_+ - f_-$, where

$$f_+(x) = \max\{f(x), 0\} \quad \text{and} \quad f_-(x) = \max\{-f(x), 0\}.$$

Then, from the nonnegative case we have two sequences of simple functions f_{n+} and f_{n-} converging pointwise to f_+ and f_- , so we have $f_n \rightarrow f$ pointwise with $f_n = f_{n+} - f_{n-}$. $\square$

4.2. Closure properties.

Theorem 4.9. Let Ω be a topological space.

- (1) If $f: (\Omega, \mathcal{B}_\Omega) \rightarrow (\mathbb{R}, \mathcal{B}_\mathbb{R})$ is continuous, it is measurable.
- (2) If $f: (X, \mathcal{F}) \rightarrow (Y, \mathcal{G})$ and $g: (Y, \mathcal{G}) \rightarrow (W, \mathcal{H})$ are measurable, then $g \circ f: (X, \mathcal{F}) \rightarrow (W, \mathcal{H})$ is measurable.
- (3) If $f: \mathbb{R} \rightarrow \mathbb{R}$ is monotone (e.g., $x \leq y \implies f(x) \leq f(y)$), then it is measurable.

Corollary 4.10. If $f_1, f_2: X \rightarrow \mathbb{R}$ are measurable, then $\alpha_1 f_1 + \alpha_2 f_2$ is measurable for all $\alpha_1, \alpha_2 \in \mathbb{R}$.

Proof. Take $F: X \rightarrow \mathbb{R}^2$ and $G: \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$F(x) = (f_1(x), f_2(x)) \quad \text{and} \quad G(z_1, z_2) = \alpha_1 z_1 + \alpha_2 z_2.$$

Then, $\alpha_1 f_1 + \alpha_2 f_2 = G \circ F$. Note that G is continuous and hence measurable. To see that F is measurable, note that it suffices to check on Borel rectangles $A_1 \times A_2 \in \mathcal{B}_\mathbb{R} \times \mathcal{B}_\mathbb{R}$ (recall HW 1):

$$F^{-1}(A_1 \times A_2) = \{x \in X: f_1(x) \in A_1 \text{ and } f_2(x) \in A_2\} = f_1^{-1}(A_1) \cap f_2^{-1}(A_2).$$

□

Definition 4.11. Let $\overline{\mathbb{R}} = \mathbb{R} \cup \{\pm\infty\} = [-\infty, \infty]$ denote the *extended real line*, with $\mathcal{B}_{\overline{\mathbb{R}}} = \sigma(\{[-\infty, a) : a \in \mathbb{R}\})$.

Proposition 4.12. Let $f_n : (X, \mathcal{F}) \rightarrow (\overline{\mathbb{R}}, \mathcal{B}_{\overline{\mathbb{R}}})$ be a sequence of measurable functions. Then,

$$\inf_n f_n, \quad \sup_n f_n, \quad \liminf_n f_n, \quad \limsup_n f_n$$

are all measurable.

Proof. Since for any $f : X \rightarrow \mathbb{R}$, $\inf f = -\sup(-f)$, and

$$\liminf_n f_n = \sup_n \inf_{k \geq n} f_k,$$

it suffices to prove for $\inf$.

Let $f = \inf_n f_n$. Then, for $a \in \overline{\mathbb{R}}$ arbitrary,

$$\begin{aligned} f^{-1}([-\infty, a)) &= \{x \in X : \exists n \text{ such that } f_n(x) < a\} \\ &= \bigcup_n \{x : f_n(x) < a\} \\ &= \bigcup_n f_n^{-1}([-\infty, a)), \end{aligned}$$

and since $f_n^{-1}([-\infty, a)) \in \mathcal{F}$ for all n , $f^{-1}([-\infty, a)) \in \mathcal{F}$. □

Definition 4.13. Let $(X, \mathcal{F}, \mu)$ be a measure space, and let $f_n, f : X \rightarrow \mathbb{R}$ for $n \geq 1$ be measurable. Then, f_n *converges to f almost everywhere (a.e.)*, denoted $f_n \rightarrow f$ a.e., if

$$\mu(\{x \in X : f_n(x) \not\rightarrow f(x)\}) = 0.$$

Note that this set is measurable, since

$$\{x \in X : f_n(x) \not\rightarrow f(x)\} = \left\{x \in X : \liminf_n f_n(x) \neq f(x)\right\} \cup \left\{x \in X : \limsup_n f_n(x) \neq f(x)\right\}.$$

In general, we say that a property p holds a.e. if its negation is true only on a set of measure zero:

$$\mu(\{x \in X : \neg p(x)\}) = 0.$$

This definition of a.e. convergence is intuitively straightforward but in practice hard to use. But, we have the following theorem, which tells us that [TODO: what]

Theorem 4.14 (Monotone class theorem³). Let $\mathcal{P} \subseteq 2^X$ be a π -system and let $\mathcal{H}$ be a set of functions $f : (X, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ such that

- (1) $1 = \mathbb{1}_X \in \mathcal{H}$,
- (2) $\mathbb{1}_A \in \mathcal{H}$ for all $A \in \mathcal{P}$,
- (3) $\mathcal{H}$ is a vector space,
- (4) (closed under monotone limits) If $h_n \in \mathcal{H}$, $h_n \geq 0$, $h_n \nearrow h$, then $h \in \mathcal{H}$.

³Cohn, Exercise 1.6.–

Then, $\mathcal{H} \supseteq m\sigma(\mathcal{P})^4$.

Proof. Let $\mathcal{L} = \{A \in \sigma(\mathcal{P}) : \mathbb{1}_A \in \mathcal{H}\}$. By (2) and by construction, $\mathcal{P} \subseteq \mathcal{L} \subseteq \sigma(\mathcal{P})$. By Dynkin's π - λ theorem,

We want to show that $\mathcal{L}$ is a λ -system. [TODO: Why?]

- (1) Since $1 = \mathbb{1}_X \in \mathcal{H}$, $X \in \mathcal{L}$.
- (2) If $A, B \in \mathcal{L}$ with $A \subseteq B$, then $\mathbb{1}_{A \setminus B} = \mathbb{1}_A - \mathbb{1}_B \in \mathcal{H}$ by the vector space property, so $A \setminus B \in \mathcal{L}$.
- (3) If $A_i \uparrow A$ and $\mathbb{1}_{A_i} \in \mathcal{H}$, then $\mathbb{1}_{A_i} \nearrow \mathbb{1}_A$, and since $\mathcal{H}$ is closed under monotone limits, $\mathbb{1}_A \in \mathcal{H}$ and hence $A \in \mathcal{L}$.

Show that $\text{SF}(X) \subseteq \mathcal{H}$. Then, use simple approximation to show the result. [TODO: finish] $\square$

4.3. The Cantor set.

Definition 4.15. Let $K_n \subseteq [0, 1]$ be defined inductively for $n \geq 1$ by

$$K_0 = [0, 1] \quad \text{and} \quad K_n = \frac{1}{3}K_{n-1} \cup \left(\frac{1}{3}K_{n-1} + \frac{2}{3} \right).$$

Then, the *Cantor set* is defined by $K = \bigcap_{n \geq 0} K_n$, which can alternatively be written as

$$K = \left\{ \sum_{n=1}^{\infty} \frac{2b_n}{3^n} : b_n \in \{0, 1\} \right\}.$$

The Cantor set is closed (intersection of closed sets) and bounded and hence compact, and also is uncountable. Moreover, since $\lambda(K_n) = \left(\frac{2}{3}\right)^n$, $\lambda(K) = 0$.

Define the *Cantor step function* $f : [0, 1] \rightarrow [0, 1]$ by

$$f_0(x) = x \quad \text{and} \quad f_{n+1}(x) = \begin{cases} \frac{1}{2}f_n(3x) & \text{if } x \in [0, \frac{1}{3}] \\ \frac{1}{2} & \text{if } x \in (\frac{1}{3}, \frac{2}{3}) \\ \frac{1}{2} + \frac{1}{2}f_n(3(x - \frac{2}{3})) & \text{if } x \in [\frac{2}{3}, 1] \end{cases}.$$

Then, we have a uniform limit $f_n \rightarrow f$; we can think of this function as taking a point $x \in K$ to its ternary expansion $\sum_{n=1}^{\infty} \frac{2b_n}{3^n}$ and converting the 2's to 1's: $f(x) = \sum_{n=1}^{\infty} \frac{b_n}{2^n}$. (Note that there is a small ambiguity since any real number can have up to two d -ary representations, but we take the convention that the preferred representation is that with trailing 0's.)

This function $f : [0, 1] \rightarrow [0, 1]$ is continuous, monotone, constant a.e., and measurable. Consider the function $g : [0, 1] \rightarrow [0, 1]$ defined by

$$g(y) = \inf \{x \in [0, 1] : f(x) \geq y\}.$$

Then, we can see that the image $g([0, 1]) \subseteq K$ [TODO: why?].

Theorem 4.16. *There exists a set $S \subseteq [0, 1]$, that is Lebesgue but not Borel measurable: $S \in \mathcal{L} \setminus \mathcal{B}$.*

⁴ $m\mathcal{F} = \text{Hom}((X, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}_{\mathbb{R}}))$

Proof. Take $A \subseteq [0, 1]$ with $A \notin \mathcal{B}$, which exists for instance since $|2^{[0,1]}| = 2^{2^{\aleph_0}}$ and $|\mathcal{B}_{[0,1]}| = 2^{\aleph_0}$. Alternatively, we can take the Vitali set, which is not even Lebesgue measurable.

Define $S = g(A)$, and since $S \subseteq K$ and $\lambda(K) = 0$, for any $F \subseteq [0, 1]$ we have

$$\lambda(F) \geq \lambda(F \cap S^c) = \lambda(F \cap S^c) + \lambda(K) \geq \lambda(F \cap S^c) + \lambda(F \cap S)$$

and hence S is Lebesgue measurable. [TODO: check]

But, $g^{-1}(S) = A$, which is not Borel. Since g is measurable, S cannot be Borel. $\square$

5. 3TH – INTEGRATION, MONOTONE AND DOMINATED CONVERGENCE THEOREMS

5.1. Lebesgue integral.

Our setup is as follows: $(X, \mathcal{F}, \mu)$ is a measure space, and $f: X \rightarrow \mathbb{R}$ is measurable. We want to define the *Lebesgue integral*, denoted in any of the following:

$$\int f \, d\mu, \quad \int f(x) \mu(dx), \quad \mu(f), \quad \int f(x) \, d\mu(x), \quad \int f(x) \, d\mu_x.$$

We will start with $f \geq 0$, then generalize to $f = f_+ - f_-$ with $\mu(f_+) < \infty$ or $\mu(f_-) < \infty$ by $\mu(f) = \mu(f_+) - \mu(f_-)$. Note that when

$$\mu(|f|) = \mu(f_+) + \mu(f_-) < \infty,$$

we say that f is *integrable*, and $f \in \mathcal{L}^1(X, \mathcal{F}, \mu)$. (Note: As we will see later, this is distinct from $L^1(X, \mathcal{F}, \mu)$, which is the quotient of $\mathcal{L}^1(X, \mathcal{F}, \mu)$ by μ -a.e. equivalence.)

Denote by $\text{SF}_+(X) = \{f = \sum_{i=1}^n a_i \mathbb{1}_{A_i} \in \text{SF}(X) : a_i \geq 0\}$.

Definition 5.1 (Integral of simple function). Let $\varphi \in \text{SF}_+(X)$, so that $\varphi = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$, with $a_i \geq 0$ and $A_i \in \mathcal{F}$. Define

$$\mu_0(\varphi) = \sum_{i=1}^n a_i \mu(A_i).$$

Remark 5.2. Note that all of these definitions work fine with the extended nonnegative reals $[0, \infty]$, with the convention that $0 \cdot \infty = \infty \cdot 0 = 0$.

Definition 5.3 (Integral of measurable functions). Let $f \in \text{Hom}_{\text{Meas}}(X, [0, \infty])$.⁵ The *Lebesgue integral* of f is defined by

$$\mu(f) = \sup \{\mu_0(\varphi) : \varphi \in \text{SF}_+(X), \varphi \leq f\}.$$

We now check that μ_0 is well-defined: that if

$$\varphi = \sum_{i=1}^n a_i \mathbb{1}_{A_i} = \sum_{j=1}^m b_j \mathbb{1}_{B_j},$$

then

$$\sum_{i=1}^n a_i \mu(A_i) = \sum_{j=1}^m b_j \mu(B_j).$$

⁵By this, I mean the homset of measurable functions from $(X, \mathcal{F})$ to $([0, \infty], \mathcal{B}_{[0, \infty]})$ in the category **Meas** of measurable spaces.

We first reduce to the disjoint case. The idea is to break up the A_i 's and B_j 's into disjoint sets, then take their pairwise intersections $A_i \cap B_j$ (a refinement of the two disjoint covers), then if the constants are handled correctly one can hope they will be equal. [TODO: finish typing up the proof]

Lemma 5.4. *Let $\varphi, \psi \in \text{SF}_+(X)$, $c \geq 0$.*

- (1) *If $\varphi = \psi$ μ -a.e., then $\mu_0(\varphi) = \mu_0(\psi)$.*
- (2) *$\mu_0(\varphi + \psi) = \mu_0(\varphi) + \mu_0(\psi)$*
- (3) *$\mu_0(c\varphi) = c\mu_0(\varphi)$*
- (4) *$\varphi \leq \psi \implies \mu_0(\varphi) = \mu_0(\psi)$.*

For the proofs, take $\varphi = \sum_{i=1}^n a_i \mathbb{1}_{A_i}$ and $\psi = \sum_{j=1}^m b_j \mathbb{1}_{B_j}$. Without loss of generality, we can assume A_i and B_j are both pairwise disjoint, and passing to a refinement we can assume $\{A_i\} = \{B_j\}$ are the same collection. Then, all of these results are a simple consequence of the corresponding results on characteristic functions.

Passing to the Lebesgue integral, we get the following analogous set of properties:

Lemma 5.5. *Let $f, g \in \mathcal{L}^1(X, \mathcal{F}, \mu)$ or $f, g \in \text{Hom}_{\text{Meas}}(X, [0, \infty])$, and $c \in \mathbb{R}$. Then,*

- (1) *If $f = g$ μ -a.e., then $\mu(f) = \mu(g)$.*
- (2) *$\mu(f + g) = \mu(f) + \mu(g)$*
- (3) *$\mu(cf) = c\mu(f)$*
- (4) *$f \leq g \implies \mu(f) = \mu(g)$*
- (5) *$f \in \text{SF}_+(X) \implies \mu(f) = \mu_0(f)$.*

5.2. Monotone convergence theorem.

Theorem 5.6 (Monotone convergence theorem). *Let $f_n \in \text{Hom}_{\text{Meas}}(X, [0, \infty])$, with $f_n \nearrow f$ pointwise. Then,*

$$\int f_n d\mu \nearrow \int f d\mu.$$

Proof. By monotonicity, $\mu(f_n) \leq \mu(f)$, so $\sup_n \mu(f_n) \leq \mu(f)$. We now want to show the converse.

We thus want to show that for all $\lambda \in (0, 1)$,

$$\begin{aligned} \sup_n \sup_{\varphi \in \text{SF}_+(X)} \{\mu_0(\varphi) : \varphi \leq f_n\} &= \sup_n \sup_{\varphi \in \text{SF}_+(X)} \{\mu_0(\varphi) : \varphi \leq f_n\} \\ &\geq \lambda \sup_{\varphi \in \text{SF}_+} \{\mu_0(\varphi) : \varphi \leq f\}. \end{aligned}$$

[todo: i don't get this]

Fix $\varphi \leq f$, $\lambda \in (0, 1)$. It suffices to show that there exists a sequence $\varphi_n \in \text{SF}_+(X)$ with $\varphi_n \leq f_n$ such that

$$\sup_n \mu_0(\varphi_n) \geq \lambda \mu_0(\varphi).$$

Without loss of generality, assume that

$$\varphi = \sum_{i=1}^N c_i \mathbb{1}_{A_i}$$

with A_i disjoint, and define

$$f_*(A_i) = \inf \{f(x) : x \in A_i\}.$$

Assume for the moment that $f_*(A_i) < \infty$ for all i . Of course in general we should have $f_*(A_i) \leq c_i$, but without loss of generality we can assume $c_i = f_*(A_i)$.

Now define φ_n by

$$\varphi_n = \sum_{i=1}^N \lambda f_*(A_i) \mathbb{1}_{A_{i,n}},$$

where we define

$$A_{i,n} = \{x \in A_i : f_n(x) \leq \lambda f_*(A_i)\}.$$

That is, at each step n , we thin out each of the sets A_i so that $\lambda f_*(A_i) \mathbb{1}_{A_{i,n}} \leq f_n$ on A_i , and doing this on each i we get a simple lower approximation φ_n to f_n .

Now note that as $n \nearrow \infty$, we get $A_{i,n} \uparrow A_i$, so $\mu(A_{i,n}) \nearrow \mu(A_i)$, meaning that

$$\mu_0(\varphi_n) = \lambda \sum_{i=1}^N c_i \mu(A_{i,n}) \rightarrow \lambda \sum_{i=1}^N c_i \mu(A_i) = \lambda \mu_0(\varphi).$$

Since φ_n is monotone, we have

$$\lim_{n \nearrow \infty} \mu_0(\varphi_n) = \sup_n \mu_0(\varphi_n) = \lambda \mu_0(\varphi).$$

Recall that we assumed earlier that $f_*(A_i) < \infty$. Now let us assume that $f_*(A_i) = \infty$; if $\mu(A_i)$ we can safely ignore A_i by setting $\varphi_n = 0$ on A_i .

But, if $\mu(A_i) > 0$, then we must have $\mu(f) \geq \mu_0(A_i)c \geq \mu(c \mathbb{1}_{A_{i,n}})$ for all $c > 0$ with $A_{i,n}$ defined as above and hence $\mu(f) = \infty$. [TODO: clean this up] $\square$

[TODO: musings about applicability of the MCT]

Lemma 5.7 (Linearity of the integral). *Let $f, g \in \mathcal{L}^1(X, \mathcal{F}, \mu)$. Then,*

$$\mu(f) + \mu(g) = \mu(f + g).$$

Proof. Without loss of generality, take $f, g \geq 0$ with simple approximations $f_n \nearrow f$ and $g_n \nearrow g$. Note that then $f_n + g_n \nearrow f + g$. Then,

$$\mu(f + g) = \lim_{n \rightarrow \infty} \mu(f_n + g_n) = \lim_{n \rightarrow \infty} \mu(f_n) + \mu(g_n) = \mu(f) + \mu(g),$$

where the first and third equalities are a consequence of the monotone convergence theorem. $\square$

Proposition 5.8 (Markov's inequality). *Let $f \in \text{Hom}_{\text{Meas}}(X, [0, \infty])$, $t > 0$. Then,*

$$\mu(f^{-1}([t, \infty])) \leq \frac{1}{t} \mu(f).$$

Proof. Note that $t \mathbb{1}_{f^{-1}([t, \infty])} \leq f$. Then, use the MCT. $\square$

Proposition 5.9. Let $f \in \mathcal{L}^1(X, \mathcal{F}, \mu)$. Then, $A = \{x \in X : f(x) \neq 0\}$ is σ -finite.

Proof. Let $A_n = \{x \in X : |f(x)| \geq \frac{1}{n}\}$; note that $A_n \uparrow A$. Then, by Markov's inequality, $\mu(A_n) \leq n\mu(|f|) < \infty$. [TODO: finish understanding] $\square$

Proposition 5.10. Let $f \in \mathcal{L}^1(X, \mathcal{F}, \mu)$, with $\mu(f) = 0$. Then, $f = 0$ μ -a.e.

Proposition 5.11. Let $f \in \mathcal{L}^1(X, \mathcal{F}, \mu)$ with

$$\int_A f d\mu \geq 0$$

for all $A \in \mathcal{F}$. [TODO: finish this i missed the statement]

5.3. Fatou's lemma and the dominated convergence theorem.

Theorem 5.12 (Fatou's lemma). Let $f_n \in \text{Hom}_{\text{Meas}}(X, [0, \infty])$. Then,

$$\int \liminf_{n \rightarrow \infty} f_n d\mu \leq \liminf_{n \leq \infty} \int f_n d\mu.$$

Proof. Let $f = \liminf_{n \rightarrow \infty} f_n$, and take $g_n = \inf_{k \geq n} f_k$; note that this is measurable and monotone, with $g_n \nearrow f$. By the monotone convergence theorem,

$$\int f d\mu = \lim_{n \rightarrow \infty} \int g_n d\mu = \liminf_{n \rightarrow \infty} \int g_n d\mu \leq \liminf_{n \rightarrow \infty} \int f_n d\mu,$$

since $g_n \leq f_n$. $\square$

Remark 5.13. Note that we can generalize to the case where f_n is dominated below by a measurable function with finite negative mass: if $f_n \geq g$ for $g \in \mathcal{L}^1(X)$ with $\int g_- d\mu < \infty$, then we can just apply Fatou's lemma to $f_n - g_-$.

Corollary 5.14 (Reverse Fatou's lemma). Let $f_n, g \in \text{Hom}_{\text{Meas}}(X, \mathbb{R})$ with $f_n \leq g$ and $\int g_+ d\mu < \infty$. Then,

$$\int \limsup_{n \rightarrow \infty} f_n d\mu \geq \limsup_{n \rightarrow \infty} \int f_n d\mu.$$

Theorem 5.15 (Dominated convergence theorem). Let $f_n, g \in \mathcal{L}^1(X, \mathcal{F}, \mu)$ with $|f_n| \leq g$ for all n . Then, if $f_n \rightarrow f$ pointwise, then $f_n \rightarrow f$ in L^1 and in integral:

$$\|f_n - f\|_{L^1} = \int |f_n - f| d\mu \rightarrow 0 \quad \text{and} \quad \int f_n d\mu \rightarrow \int f d\mu.$$

Proof. Note that if we have L^1 convergence, then we get convergence in integral, since

$$\left| \int f_n d\mu - \int f d\mu \right| \leq \int |f_n - f| d\mu.$$

Note that $|f_n - f| \leq 2g$, with $2g = 2g_+ \in \mathcal{L}^1(X)$, so by reverse Fatou,

$$\limsup_{n \rightarrow \infty} \int |f_n - f| d\mu \leq \int \limsup_{n \rightarrow \infty} |f_n - f| d\mu = 0,$$

since $f_n \rightarrow f$ pointwise. $\square$

6. 4TU – CONVERGENCE OF MEASURABLE FUNCTIONS

6.1. Constructing measures.

We first describe two standard constructions of measures.

Example 6.1. Consider the measure space $(\mathbb{R}, \mathcal{B}_{\mathbb{R}})$. For any nondecreasing right-continuous function $F: \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ with $F(x) \searrow 0$ as $x \searrow -\infty$, consider the algebra

$$\mathcal{A} = \left\{ \bigcup_{i=1}^n (a_i, b_i] : a_1 < b_1 < \cdots < a_n < b_n, n \in \mathbb{N} \right\},$$

and define

$$\mu \left(\bigcup_{i=1}^n (a_i, b_i] \right) = \sum_{i=1}^n F(b_i) - F(a_i).$$

Then, we see that

$$F(x) = \mu((-\infty, x]).$$

We can check that μ is countably additive on $\mathcal{A}$, and hence is uniquely defined on $\mathcal{B}_{\mathbb{R}}$.

Example 6.2. Let $(X, \mathcal{F}, \mu)$ be a measure space and consider $f: (X, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$. Then, define

$$f\mu(A) = \mu(f\mathbb{1}_A) = \int_A f d\mu.$$

We have the following properties:

- (1) $f\mu$ is a measure on $(X, \mathcal{F})$,
- (2) $h(f\mu) = (hf)\mu$,
- (3) $h \in \mathcal{L}^1(X, \mathcal{F}, f\mu)$ if and only if $hf \in \mathcal{L}^1(X, \mathcal{F}, \mu)$ and $(f\mu)(h) = \mu(hf)$:

$$\int h df\mu = \int hf d\mu.$$

Proof of (2). We want to show that for all $A \in \mathcal{F}$,

$$\int_A h df\mu = \int_A fh d\mu.$$

First consider the case where $h = \mathbb{1}_B$. Then,

$$\int_A h df\mu = \int_{A \cap B} df\mu = \int_{A \cap B} f d\mu = \int_A fh d\mu.$$

By using linearity of the integral we can easily prove the case where h is simple, then using the monotone convergence theorem we can generalize to h measurable. $\square$

Definition 6.3. Let $(X, \mathcal{F}, \mu)$ be a measure space and $f: X \rightarrow \mathbb{R}$ Borel. If $\nu = f\mu$, then we say that f is the *Radon-Nikodym derivative* of ν with respect to μ .

6.2. Convergence of measurable functions.

Definition 6.4. Let $f_n, f: (X, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ be measurable, for $n \in \mathbb{N}$. We say that $f_n \rightarrow f$ *in measure* if for all $\varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \mu(\{x \in X : |f_n(x) - f(x)| \geq \varepsilon\}) = 0.$$

Remark 6.5. Note that convergence a.e. does not imply convergence in measure: $\mathbb{1}_{[n,\infty)} \rightarrow 0$ a.e., but not in measure.

Remark 6.6. But, this is the only way that this fails: if μ is finite, then $f_n \rightarrow f$ a.e. implies that $f_n \rightarrow f$ in measure.

To see this, note by HW 3 Problem 4(a) that $f_n \rightarrow f$ a.e. if and only if for all $\varepsilon > 0$,

$$\mu \left(\bigcap_{n=1}^{\infty} \bigcup_{k \geq n} \{x \in X : |f_k(x) - f(x)| > \varepsilon\} \right) = 0.$$

Of course,

$$\begin{aligned} \mu \left(\bigcap_{n=1}^{\infty} \bigcup_{k \geq n} \{x \in X : |f_k(x) - f(x)| \geq \varepsilon\} \right) &= \lim_{n \rightarrow \infty} \mu \left(\bigcup_{k \geq n} \{x \in X : |f_k(x) - f(x)| \geq \varepsilon\} \right) \\ &\geq \lim_{n \rightarrow \infty} \mu(\{x \in X : |f_n(x) - f(x)| \geq \varepsilon\}), \end{aligned}$$

which goes to 0 because $f_n \rightarrow f$ in measure. Note that we use the μ finite hypothesis to commute μ with the intersection (which is a downward limit), and in the second line we simply use the set-theoretic statement that $\bigcup_{k \geq n} U_k \supseteq U_n$.

Remark 6.7. Note that $f_n \rightarrow f$ in measure does not imply that $f_n \rightarrow f$ a.e., even in finite spaces. Take the measure space $((0, 1], \mathcal{B}, \lambda)$, with functions $f_1 = \mathbb{1}_{(0, 1/2]}$, $f_2 = \mathbb{1}_{(1/2, 1]}$, $f_3 = \mathbb{1}_{(0, 1/4]}$, $f_4 = \mathbb{1}_{(1/4, 1/2]}$, and so on. Then, $f_n \rightarrow 0$ in measure but $\limsup_{n \rightarrow \infty} f_n = 1$, so f_n does not converge to 0 a.e..

So, generally we have that on finite spaces, a.e. convergence strictly is stronger than convergence in measure. But, we have a compactness result:

Proposition 6.8. *Let $f_n, f: X \rightarrow \mathbb{R}$ measurable, with $f_n \rightarrow f$ in measure. Then, there is a subsequence that converges a.e..*

Proof. Define n_k recursively so that $n_k > n_{k-1}$,

$$\mu \left(\left\{ x \in X : |f_{n_k}(x) - f(x)| \geq \frac{1}{k} \right\} \right) \leq \frac{1}{2^k},$$

which is possible since $f_n \rightarrow f$ in measure by assumption. Now, fix $\varepsilon > 0$; we compute

$$\begin{aligned} \mu \left(\bigcap_{m=1}^{\infty} \bigcup_{k \geq m} \{x \in X : |f_{n_k}(x) - f(x)| \geq \varepsilon\} \right) &\leq \lim_{m \rightarrow \infty} \mu \left(\bigcup_{n_k \geq m} \{x \in X : |f_{n_k}(x) - f(x)| \geq \varepsilon\} \right) \\ &\leq \lim_{K \rightarrow \infty} \sum_{k \geq K} \mu(\{x \in X : |f_{n_k}(x) - f(x)| \geq \varepsilon\}) \\ &\leq \lim_{K \rightarrow \infty} \sum_{k \geq K} \frac{1}{2^k} \\ &= \lim_{K \rightarrow \infty} \frac{1}{2^{K-1}} \\ &= 0. \end{aligned}$$

[TODO: make this clearer]

□

Theorem 6.9 (Egoroff's Theorem). *Let $(X, \mathcal{F}, \mu)$ be a finite measure space, and $f_n, f: (X, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B})$ with $f_n \rightarrow f$ a.e.. Then, for all $\varepsilon > 0$, there exists $B \in \mathcal{F}$ such that $\mu(X \setminus B) < \varepsilon$, and $f_n \rightarrow f$ uniformly on B :*

$$\lim_{n \rightarrow \infty} \sup_{x \in X \setminus B} |f_n(x) - f(x)| = 0.$$

Proof. Take $g_n(x) = \sup_{k \geq n} |f_k(x) - f(x)|$, and by assumption $g_n \searrow 0$ a.e., and hence in measure. For a given $\varepsilon > 0$, choose n_k inductively such that

$$\mu \left(\left\{ x \in X : g_{n_k} > \frac{1}{k} \right\} \right) \leq \frac{\varepsilon}{2^k}.$$

Then, choose

$$B^c = \bigcup_k \left\{ x \in X : g_{n_k}(x) > \frac{1}{k} \right\}.$$

By the countable subadditivity, we have that

$$\mu(B^c) \leq \sum_{k=1}^{\infty} \frac{\varepsilon}{2^k} = \varepsilon.$$

Thus for all $x \in B$, for all k we have $g_{n_k} \leq \frac{1}{k}$. Thus, for each k ,

$$\lim_{n \rightarrow \infty} \sup_{x \in B} |f_n(x) - f(x)| \leq \sup_{x \in B} g_{n_k}(x) \leq \frac{1}{k},$$

and hence $\sup_{x \in B} |f_n(x) - f(x)| \rightarrow 0$. □

Definition 6.10. Let $f_n, f: (X, \mathcal{F}) \rightarrow (\mathbb{R}, \mathcal{B}_{\mathbb{R}})$ be measurable, for $n \in \mathbb{N}$. We say that $f_n \rightarrow f$ in mean (or in L^1) if

$$\int_X |f_n - f| \rightarrow 0.$$

Remark 6.11. For all $\varepsilon > 0$, by Markov's inequality we have

$$\mu(\{x \in X : |f_n - f| \geq \varepsilon\}) \leq \frac{1}{\varepsilon} \int_X |f_n - f|,$$

so convergence in mean implies convergence in measure.

Remark 6.12. Of course, the converse is not true: $n^3 \mathbb{1}_{[0, 1/n]} \rightarrow 0$ in measure but not in mean.

Remark 6.13. If $f_n \rightarrow f$ a.e. or in measure and $|f_n| \leq g$ and $|f| \leq g$ for some $g \in \mathcal{L}^1$, then $f_n \rightarrow f$ in L^1 .

This is obvious in the a.e. case by the dominated convergence theorem.

In the measure case, assume by way of contradiction that $f_n \not\rightarrow f$ in L^1 . Passing to a subsequence, we can assume that $\int_X |f_n - f| \geq \varepsilon$ for some $\varepsilon > 0$. Passing to a further subsequence, since $f_n \rightarrow f$ in measure, we can get $f_n \rightarrow f$ a.e.. But, by the dominated convergence theorem, this implies that we have convergence in L^1 , a contradiction.

[TODO: organize these remarks into actual propositions]

7. 4TH – DEFINITION OF L^p SPACES

The setup for today is as follows: let $(X, \mathcal{F}, \mu)$ be a measure space. Then, for $p \in [1, \infty)$, we define $\mathcal{L}^p$ as follows, after a preliminary definition.

Definition 7.1 (seminorms and norms). Let V be a vector space over $\mathbb{R}$. A map $\|-\| : V \rightarrow \mathbb{R}$ is a *seminorm* if it satisfies the following properties:

- (1) (absolute homogeneity) $\|cv\| = |c| \|v\|$ for $c \in \mathbb{R}, v \in V$,
- (2) (triangle inequality / subadditivity) $\|v + w\| \leq \|v\| + \|w\|$ for $v, w \in V$.

In addition, if $\|-\|$ satisfies the property

- (3) (positive definite / point-separating) $\|v\| = 0 \implies v = 0$,

then $\|-\|$ is called a *norm*.

If $\|-\|$ is a norm (resp. seminorm) on V , we may call V a *normed* (resp. *seminormed*) *space*.

Note that the converse of (3) holds automatically for seminorms: if $v = 0$, then $0v = 0 = v$ and hence

$$\|v\| = |0| \|v\| = 0.$$

Moreover, some authors (including Cohn) will add the stipulation that $\|-\|$ is nonnegative: $\|v\| \geq 0$ for all $v \in V$. However, this follows from the triangle inequality: for any $v \in V$,

$$0 = \|0\| = \|v - v\| \leq \|v\| + \|-v\| = 2\|v\|.$$

Definition 7.2 ($\mathcal{L}^p$ spaces, $p \in [1, \infty)$). Let $(X, \mathcal{F}, \mu)$ be a measure space, and $p \in [1, \infty)$. For $p \in [1, \infty)$ define the $\mathcal{L}^p$ *seminorm*

$$\|f\|_p = \left(\int_X |f|^p d\mu \right)^{1/p}.$$

Define the $\mathcal{L}^p$ *spaces* by

$$\mathcal{L}^p(X, \mathcal{F}, \mu) = \left\{ f \in \text{Hom}_{\text{Meas}}(X, \mathbb{R}) : \|f\|_p < \infty \right\}.$$

As always, we may omit any portion of $(X, \mathcal{F}, \mu)$ if they are clear from context.

[TODO: refer to proof of seminorm]

Note that for all $p \in [1, \infty)$, $\mathcal{L}^p(X)$ is a vector space over $\mathbb{R}$. For $p < \infty$ this follows straightforwardly from the fact that

$$|f + g|^p \leq (|f| + |g|)^p \leq (2 \max\{|f|, |g|\})^p \leq 2^p |f|^p + 2^p |g|^p.$$

Note that in the first step, we need $p \geq 1$ to pass the triangle inequality through the exponent.

In the $p = \infty$ case, this is a bit more complicated. As in the case for $C([0, 1]) = \text{Hom}_{\text{Top}}([0, 1], \mathbb{R})$,⁶ we would like to use the *sup norm* (or uniform norm)

$$\|f\|_\infty = \sup_{x \in [0, 1]} \|f(x)\|.$$

⁶Top is the category of topological spaces with continuous maps, so $\text{Hom}_{\text{Top}}(X, Y)$ is the set of continuous maps $f: X \rightarrow Y$.

But, we may have two functions which disagree on a set of μ -measure zero, which can have different suprema—one of them may not even have a finite supremum. But, from the perspective of integration, we wish to treat them as “the same function,” which motivates the following definitions.

Definition 7.3. Let $(X, \mathcal{F}, \mu)$ be a measure space. A subset $A \in \mathcal{F}$ is *locally μ -null* (or *locally null* if μ is clear from context) if for all $S \in \mathcal{F}$ with $\mu(S) < \infty$, we have $\mu(S \cap A) = 0$.

Remark 7.4 (null implies locally null, converse if σ -finite). Note that if A is μ -null (i.e., $\mu(A) = 0$), then A is locally μ -null. Moreover, if μ is σ -finite, then A locally μ -null implies A μ -null: Let $X = \bigcup_{i=1}^{\infty} S_i$, with $\mu(S_i) < \infty$. Then, if A is locally μ -null,

$$\mu(A) = \mu\left(\bigcup_{i=1}^{\infty} S_i \cap A\right) \leq \sum_{i=1}^{\infty} \mu(S_i \cap A) = 0.$$

The consequence of this is that locally μ -null and μ -null sets are only different if the space is not σ -finite:

Example 7.5. Let $(X, \mathcal{F}, \mu) = (\mathbb{R}, 2^{\mathbb{R}}, \mu)$ where $\mu(A) = 0$ only when A is countable and $\mu(A) = \infty$ otherwise. Of course, there exists no countable cover of $\mathbb{R}$ by countable sets, so $\mathbb{R}$ is not σ -finite.

Pick any subset $S \subseteq \mathbb{R}$. Then, if A is countable then $A \cap S$ is countable, so $\mu(S \cap A) = 0$. Hence, all subsets of $\mathbb{R}$ are locally μ -null, even the uncountable ones, so this measure space has locally μ -null sets which are not μ -null.

With this in hand, we can define the essential supremum, a weaker notion of supremum which is in essence a “supremum up to a set of local measure zero,” which fixes the aforementioned problems.

Definition 7.6. Let $(X, \mathcal{F}, \mu)$ be a measure space. Then, the *essential supremum* is

$$\|f\|_{\infty} = \text{ess sup } f = \inf \left\{ M \in \mathbb{R} : f^{-1}([M, \infty)) \subseteq X \text{ locally } \mu\text{-null} \right\}.$$

If μ is σ -finite, then locally μ -null is equivalent to μ -null and hence we can write

$$\|f\|_{\infty} = \text{ess sup } f = \inf \left\{ L \in \mathbb{R} : \mu(\{x \in X : |f(x)| > L\}) = 0 \right\}.$$

Note that if X is a topological space with its Borel σ -algebra and $f : X \rightarrow \mathbb{R}$ is continuous, then the supremum of $|f|$ is equal to the essential supremum, which justifies the abuse of notation $\|f\|_{\infty}$.

Remark 7.7. Denote by

$$A_L = \{x \in X : |f(x)| \geq L\} \quad \text{and} \quad A_* = \{x \in X : |f(x)| > \|f\|_{L^{\infty}}\}.$$

We show that A_* is locally μ -null: for any $S \in \mathcal{F}$ with $\mu(S) < \infty$,

$$\begin{aligned} \mu(A_* \cap S) &= \mu\left(\bigcup_{n=1}^{\infty} A_{\|f\|_{L^{\infty}} + \frac{1}{n}} \cap S\right) \\ &\leq \lim_{n \rightarrow \infty} \mu(A_{\|f\|_{L^{\infty}} + \frac{1}{n}} \cap S) = 0. \end{aligned}$$

Therefore, $\|f\|_{L^{\infty}}$ achieves the infimum above.

Definition 7.8. Let $p, q \in [1, \infty]$. We say that they are *conjugate exponents* if

$$\frac{1}{p} + \frac{1}{q} = 1,$$

or if $(p, q) = (1, \infty)$ or $(p, q) = (\infty, 1)$.

Remark 7.9. Some handy identities: if p, q are conjugate exponents, then

$$p + q = pq, \quad p = q(p - 1), \quad q = p(q - 1), \quad q = \frac{p}{p - 1}, \quad p = \frac{q}{q - 1}.$$

Also, $p = q$ if and only if $p = q = 2$, which turns out to give $\mathcal{L}^2$ the additional structure of an inner product space, so L^2 is a Hilbert space.

Theorem 7.10 (Hölder's inequality). *Let $(X, \mathcal{F}, \mu)$ be a measure space, let $p, q \in [1, \infty]$ be conjugate exponents, and let $f \in \mathcal{L}^p(X, \mathcal{F}, \mu)$ and $g \in \mathcal{L}^q(X, \mathcal{F}, \mu)$. Then, $fg \in \mathcal{L}^1(X, \mathcal{F}, \mu)$ and*

$$\|fg\|_1 \leq \|f\|_p \|g\|_q.$$

The proof is essentially an application of the following lemma:

Lemma 7.11. *Let $x, y \in \mathbb{R}$, $x, y \geq 0$. Then, for p, q conjugate,*

$$xy \leq \frac{x^p}{p} + \frac{y^q}{q}.$$

Proof. If (without loss of generality) $y = 0$, then the inequality reduces to $0 \leq \frac{x^p}{p}$, which is clear.

By substituting $u = x^p$ and $v = y^q$, this is equivalent to proving that for $u, v > 0$,

$$u^{1/p} v^{1/q} \leq \frac{u}{p} + \frac{v}{q}.$$

Rearranging (dividing both sides by v), this is equivalent to

$$\frac{u^{1/p}}{v^{1/p}} = \frac{u^{1/p}}{v^{1-1/q}} \leq \frac{u/v}{p} + \frac{1}{q},$$

which by taking $t = u/v$ is equivalent to

$$t^{1/p} \leq \frac{t}{p} + \frac{1}{q}.$$

But, using single-variable calculus, we can show that this is true for all $t > 0$: the function $f(t) = \frac{t}{p} + \frac{1}{q} - t^{1/p}$ has derivatives

$$f'(t) = \frac{1}{p} - \frac{1}{p} t^{1/p-1} = \frac{1}{p} (1 - t^{-1/q}) \quad \text{and} \quad f''(t) = \frac{1}{pq} t^{-1-1/q},$$

with one critical point at $t = 1$ and $f''(1) > 0$, so since $f(1) = \frac{1}{p} + \frac{1}{q} - 1 = 0$, f attains a minimum at 0 and hence is nonnegative for all $t > 0$.

□

Proof of Theorem 7.10 (Hölder's inequality). Without loss of generality, assume $\|f\|_p, \|g\|_q > 0$. If either were equal to 0, then we would have $|fg| = 0$ a.e., and hence both sides of the inequality would be equal to zero, so we have nothing to prove.

Since both sides are invariant by scaling by constants, we can assume that $\|f\|_p = \|g\|_q = 1$. So, using our lemma, we have that

$$\int_X |fg| d\mu \leq \int_X \frac{|f|^p}{p} d\mu + \int_X \frac{|g|^q}{q} d\mu = \frac{1}{p} \int_X |f|^p d\mu + \frac{1}{q} \int_X |g|^q d\mu = \frac{1}{p} + \frac{1}{q} = 1,$$

as desired.

Now let us consider the case where $(p, q) = (1, \infty)$. The idea here is that we should cut away the set where $|g| > \|g\|_\infty$, which is locally μ -null, by cutting it off with respect to a σ -finite set, as follows: define $A = X \setminus f^{-1}(0)$, and note that we can write

$$A = \bigcup_{n=1}^{\infty} A_n = \bigcup_{n=1}^{\infty} \left\{ x \in X : |f(x)| > \frac{1}{n} \right\},$$

where by Markov's inequality,

$$\mu(A_n) \leq n \int_X |f| d\mu < \infty$$

because $f \in \mathcal{L}^1$. Hence, A is σ -finite.

Then, since $|f| = |f| \mathbb{1}_A$, we can split the integral

$$\int_X |fg| d\mu = \int_X \mathbb{1}_A |fg| d\mu = \int_X |fg| \mathbb{1}_A \mathbb{1}_{\{|g| \leq \|g\|_\infty\}} d\mu + \int_X |fg| \mathbb{1}_A \mathbb{1}_{\{|g| > \|g\|_\infty\}} d\mu.$$

Since $\{|g| > \|g\|_\infty\}$ is locally μ -null, $\mu(A \cap \{g > \|g\|_\infty\}) = \lim_{n \rightarrow \infty} \mu(A_n \cap \{g > \|g\|_\infty\}) = 0$. Hence, the second term vanishes, so

$$\int_X |fg| d\mu = \int_X |fg| \mathbb{1}_A \mathbb{1}_{\{|g| \leq \|g\|_\infty\}} d\mu \leq \|g\|_\infty \int_X |f| \mathbb{1}_A d\mu = \|f\|_1 \|g\|_\infty.$$

□

Theorem 7.12 (Minkowski's inequality). *Let $(X, \mathcal{F}, \mu)$ be a measure space, and $p \in [1, \infty]$. Then,*

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

Proof. For $p = 1$ this follows from the triangle inequality and monotonicity and linearity of the integral:

$$\int_X |f + g| d\mu \leq \int_X |f| + |g| d\mu = \int_X |f| d\mu + \int_X |g| d\mu.$$

For $p = \infty$ this follows from properties of suprema (with care taken for the μ -null sets).

For $p \in (1, \infty)$, by the triangle inequality (line 1) then Hölder's inequality (line 2) we have

$$\begin{aligned} \int_X |f + g|^p d\mu &\leq \int_X |f| |f + g|^{p-1} d\mu + \int_X |g| |f + g|^{p-1} d\mu \\ &\leq \left(\|f\|_p + \|g\|_p \right) \left(\int_X |f + g|^{(p-1)q} d\mu \right)^{1/q} \\ &= \left(\|f\|_p + \|g\|_p \right) \left(\|f + g\|_p^p \right)^{1/q} \end{aligned}$$

so dividing on both sides we get the desired inequality. Note that in the second line we use that $(p-1)q = p$. □

Theorem 7.13. *Let $(X, \mathcal{F}, \mu)$ be a measure space. For $p \in [1, \infty]$, we get that $\mathcal{L}^p(X, \mathcal{F}, \mu)$ is a vector space over $\mathbb{R}$, and the maps $\|-\|_p : \mathcal{L}^p(X, \mathcal{F}, \mu) \rightarrow \mathbb{R}$ are seminorms.*

[TODO: proof]

Of course, given a vector space with a seminorm, we can take the quotient with respect to the subspace on which the seminorm is 0. Denote this subspace as

$$\mathcal{N}^p(X, \mathcal{F}, \mu) = \left\{ f \in \mathcal{L}^p(X, \mathcal{F}, \mu) : \|f\|_p = 0 \right\}.$$

Note that for $p \in [1, \infty)$, this is the set of functions which vanish a.e., and for $p = \infty$, this is the set of functions that vanish everywhere except on a locally μ -null set. (Of course, by our previous remark [TODO: number], these notions agree if μ is σ -finite.

Then, we can define L^p by the quotient of $\mathcal{L}^p$ by $\mathcal{N}^p$:

Definition 7.14. Let $(X, \mathcal{F}, \mu)$ be a measure space. For $p \in [1, \infty]$, define the normed vector space

$$L^p(X, \mathcal{F}, \mu) = \mathcal{L}^p(X, \mathcal{F}, \mu) / \mathcal{N}^p(X, \mathcal{F}, \mu).$$

Concretely, an element of L^p is an equivalence class $[f]$ of functions in $\mathcal{L}^p$ such that $f, g \in [f] \iff f - g \in \mathcal{N}^p$. Note that on L^p , the seminorm $\|-\|_p$ on $\mathcal{L}^p$ is now a genuine norm, since we have $\|f\|_p = 0$ iff $f = 0$.

Generally, we will simply say elements $f \in L^p$ are functions, even though to be precise we really mean to refer to an equivalence class of functions; in practice we get rid of this ambiguity by stating that all facts are true up to a set of measure 0, which is precisely the notion of equivalence that characterizes each equivalence class (at least for $p \in [1, \infty)$).

Theorem 7.15. *Let $(X, \mathcal{F}, \mu)$ be a measure space. Then, $L^p(X, \mathcal{F}, \mu)$ is complete, and hence a Banach space (complete normed vector space).*

We first need a technical lemma to get an easier criterion for proving that a normed vector space is complete:

Lemma 7.16. *A normed vector space $(V, \|-\|)$ is complete if and only if every absolutely convergent series converges: for all $w_n \in V$,*

$$\sum_{n=1}^{\infty} \|w_n\| < \infty \implies \left\| w - \sum_{n=1}^N w_n \right\| \rightarrow 0$$

for some $w \in V$ as $N \rightarrow \infty$.

Proof. In the forward direction, suppose V is complete. Pick an absolutely convergent sequence w_n . Then, let $v_N = \sum_{n=1}^N w_n$. By assumption, for any $M \leq N$,

$$\|v_N - v_M\| = \left\| \sum_{n=M+1}^N w_n \right\| \leq \sum_{n=M+1}^N \|w_n\|.$$

But since w_n converges absolutely, we can always find N, M making the LHS bounded above by whatever $\varepsilon > 0$ we choose. Thus v_N is Cauchy and hence converges by completeness. Then, it is straightforward to show that if $v_N \rightarrow v$, $v_n = \sum_{n=1}^{\infty} w_n$. [TODO: rigor]

In the backward direction, let v_n be a Cauchy sequence. Taking $\varepsilon = 2^{-k}$, we can find a sequence n_k such that

$$\sup_{m, m' \geq n_k} \|v_m - v_{m'}\| \leq \frac{1}{2^k}.$$

Let $w_k := v_{n_{k+1}} - v_{n_k}$; by construction we have that w_k is absolutely convergent:

$$\sum_{k=1}^{\infty} \|w_k\| = \sum_{k=1}^{\infty} \|v_{n_{k+1}} - v_{n_k}\| \leq \sum_{k=1}^{\infty} \frac{1}{2^k} = 1.$$

Note that by construction $v_{n_k} = \sum_{i=1}^k w_i$, so by assumption we have $v_{n_k} \rightarrow v_*$ converges.

Now, we show that $v_n \rightarrow v_*$: take any $\varepsilon > 0$. [TODO: later I don't feel bothered rn] □

Proof of Theorem 7.15. Let us first begin with $p \in [1, \infty)$. Let $f_n \in L^p$ be an absolutely convergent series:

$$\sum_{n=1}^{\infty} \|f_n\|_p < \infty.$$

We want to show that $\sum_{n=1}^N f_n$ converges in L^p .

Define $g: X \rightarrow \overline{\mathbb{R}}$ by

$$g(x) = \sum_{n=1}^{\infty} |f_n(x)|.$$

Note that this may be infinite. By monotone convergence and the triangle inequality (Minkowski's inequality),

$$\left(\int_X |g|^p d\mu \right)^{1/p} = \lim_{N \rightarrow \infty} \left\| \sum_{n=1}^N |f_n| \right\|_p \leq \lim_{N \rightarrow \infty} \sum_{n=1}^N \|f_n\|_p < \infty,$$

since f_n is an absolutely convergent series by assumption. Hence $\|g\|_p < \infty$, so $g(x) < \infty$ a.e.. Let $G = \{x \in X : g(x) < \infty\}$; note that $\mu(G^c) = 0$. Thus for $x \in G$, we have $s_N(x) := \sum_{n=1}^N |f_n(x)|$ as $N \rightarrow \infty$ converges, so $f_n(x)$ is an absolutely convergent series (in $\mathbb{R}$). Thus, we can define a function $s: X \rightarrow \mathbb{R}$ by

$$s(x) = \begin{cases} \lim_{N \rightarrow \infty} s_N(x) & x \in G \\ 0 & x \notin G. \end{cases}$$

Of course, since s_N is measurable and monotone in N , s is measurable.

We now know that $s_N \rightarrow s$ pointwise a.e., and we want to show that this is true in L^p . But, since $|s(x)| \leq g(x)$ and $|s_n(x)| \leq g(x)$ for all $x \in X$, and $\|g\|_p < \infty$, it follows that $\|s_n - s\|_p \leq \|2g\|_p < \infty$. Thus, by the dominated convergence theorem, it follows that $s_n \rightarrow s$ in L^p . □

When working with function spaces, it is often useful to know which subspaces are dense, since usually if some property holds or some map is defined on a dense subspace, one can argue that it holds everywhere, given some “closedness” of the desired property/map. (For example, a bounded/continuous linear map on a dense subspace extends to the full completion.)

Proposition 7.17. *Let $(X, \mathcal{F}, \mu)$ be a measure space, and let $\text{SF}(X, \mathcal{F}, \mu) \subseteq L^p(X, \mathcal{F}, \mu)$ be the set of (equivalence classes of) simple functions. Then, $\text{SF} \hookrightarrow L^p$ is dense.*

Proof. This follows simply by taking $f \in L^p$ as $f = f_+ - f_-$, then finding simple approximations $\varphi_{n+} \nearrow f_+$ and $\varphi_{n-} \nearrow f_-$ (pointwise), and defining $\varphi_n = \varphi_{n+} - \varphi_{n-}$. Then, by the monotone convergence theorem,

$$\begin{aligned} \lim_{n \rightarrow \infty} \|f - \varphi_n\|_p &\leq \lim_{n \rightarrow \infty} \|f_+ - \varphi_{n+}\|_p + \lim_{n \rightarrow \infty} \|f_- - \varphi_{n-}\|_p \\ &= \left(\int_X \lim_{n \rightarrow \infty} |f_+ - \varphi_{n+}|^p d\mu \right)^{1/p} + \left(\int_X \lim_{n \rightarrow \infty} |f_- - \varphi_{n-}|^p d\mu \right)^{1/p} \\ &= 0. \end{aligned}$$

□

Note: Midterm on Week 5 Tuesday, next class.

8. 5TH – L^p SPACES AND DUAL SPACES

8.1. Properties of L^p spaces.

Definition 8.1. A *step function* on $[a, b] \subseteq \mathbb{R}$ is a function of the form

$$f = \sum_{i=1}^n c_i \mathbb{1}_{[a_i, a_{i+1}i)}$$

where $c_i \in \mathbb{R}$ and $a = a_0 < a_1 < \dots < a_n = b$.

Note that step functions are simple, but not all simple functions are step functions.

Lemma 8.2. *Let $p \in [1, \infty)$. Then, on $[a, b] \in \mathbb{R}$,*

- (1) *Step functions are dense in $L^p([a, b], \mathcal{B}, \lambda)$,*
- (2) *Continuous functions are dense in $L^p([a, b], \mathcal{B}, \lambda)$.*

We prove this lemma by way of a more general construction, which we will see below.

Remark 8.3. Note that this does not hold for $p = \infty$: there is no continuous L^∞ -approximation to $\mathbb{1}_{[0, 1/2]}$ on $[0, 1]$, and there is no step function that is L^∞ -close to $f(x) = \sin(x^{-1})$ on $x \in (0, 1]$ and $f(0) = 0$.

Definition 8.4. Let X be a topological space equipped with its Borel σ -algebra $\mathcal{B}_X$. A measure μ on $(X, \mathcal{B}_X)$ is *outer regular* if for any $A \in \mathcal{B}$,

$$\mu(A) = \inf \{ \mu(U) : U \supseteq A, U \text{ open} \},$$

and μ is *inner regular* if for any $A \in \mathcal{B}$,

$$\mu(A) = \sup \{ \mu(K) : K \subseteq A, K \text{ compact} \}.$$

If μ is both inner and outer regular, we say it is *regular*.

Example 8.5. As we have shown before [TODO; reference], λ on $(\mathbb{R}^d, \mathcal{B}_{\mathbb{R}^d})$ is regular.

Proposition 8.6. *Let X be a normal (T_4) space with μ regular on $(X, \mathcal{B}_X)$.⁷ Then, for all $p \in [1, \infty)$ we have a dense embedding⁸*

$$C(X) \hookrightarrow L^p(X, \mathcal{B}_X, \mu).$$

Proof. It suffices to prove that for any $A \in \mathcal{B}_X$ with $\mu(A) < \infty$ and $\varepsilon > 0$, there exists a continuous $f \in C(X)$ such that

$$\|\mathbb{1}_A - f\|_p < \varepsilon.$$

By regularity, there exist $K \subseteq A \subseteq U$, with K compact and U open, so that

$$\mu(A) - \mu(K) \leq \frac{\delta}{2} \quad \text{and} \quad \mu(U) - \mu(A) \leq \frac{\delta}{2},$$

so $\mu(U) - \mu(K) \leq \delta$.

Using Urysohn's lemma,⁹ since K, U^c are closed, we can find a continuous function $f: X \rightarrow [0, 1]$ such that $f|_K = 1$ and $f|_{U^c} = 0$. Given such an f , note that $|f - \mathbb{1}_A| \neq 0$ only on $U \setminus K$, and $|f - \mathbb{1}_A| \leq 1$ everywhere, so

$$\|f - \mathbb{1}_A\|_p = \left(\int_X |f - \mathbb{1}_A|^p d\mu \right)^{1/p} \leq \delta^{1/p}.$$

Thus taking $\delta = \varepsilon^p$, we have our desired result. $\square$

Definition 8.7. A topological space X is *separable* if there exists a countable dense subset.

Proposition 8.8. *Let $(X, \mathcal{F}, \mu)$ be a measure space with μ σ -finite, $\mathcal{F}$ countably generated, and $p \in [1, \infty)$. Then, $L^p(X, \mathcal{F}, \mu)$ is separable.*

Note that the restriction of $p < \infty$ is needed here:

Example 8.9. Let $(X, \mathcal{F}, \mu) = ([0, 1], \mathcal{B}, \lambda)$. Then, $L^\infty([0, 1], \mathcal{B}, \lambda)$ is not separable, as

$$\mathcal{S} = \{\mathbb{1}_{[0, a)} : a \in [0, 1]\}$$

is an uncountable set which are pairwise L^∞ -distance 1.

Definition 8.10. Let X be a set and $A, B \subseteq X$. Then, the *symmetric difference*, denoted $A \triangle B$, is

$$A \triangle B = (A \setminus B) \cup (B \setminus A).$$

Lemma 8.11. *Let $(X, \mathcal{F}, \mu)$ be a finite measure space with $\mathcal{F} = \sigma(\mathcal{A})$, where $\mathcal{A}$ is an algebra. Then, for all $\varepsilon > 0$ and $A \in \mathcal{F}$, there exists an $A_0 \in \mathcal{A}$ such that*

$$\mu(A \triangle A_0) \leq \varepsilon.$$

⁷In particular, if X is a metric space or compact Hausdorff this holds.

⁸A linear map $f: X \rightarrow Y$ between vector spaces over $\mathbb{R}$ is a *dense embedding* if f is injective and $f(X) \subseteq Y$ is dense.

⁹If X is a metric space, we can use the function

$$f(x) = \max \left\{ 0, 1 - \frac{d(x, K)}{d(U^c, K)} \right\}.$$

In general, X is normal if and only if Urysohn's lemma applies: any pair of closed sets can be separated by a continuous function.

Proof. By the (uniqueness part of the) Carathéodory extension theorem,

$$\mu(A) = \inf \left\{ \sum_{i=1}^{\infty} \mu(A_i) : A \subseteq \bigcup_{i=1}^{\infty} A_i, A_i \in \mathcal{A} \right\}.$$

Thus, without loss of generality we can find a disjoint sequence A_i such that

$$\sum_{i=1}^{\infty} \mu(A_i) - \frac{\varepsilon}{2} \leq \mu(A).$$

Now, note that since $A \subseteq \bigcup_{i=1}^{\infty} A_i$, $A \setminus \bigcup_{i=1}^{\infty} A_i = \emptyset$, and hence we have the convergent sequence

$$\mu \left(A \setminus \bigcup_{i=1}^n A_i \right) \searrow 0$$

monotonically decreasing as $n \rightarrow \infty$, since the A_i were assumed to be disjoint. Pick $N > 0$ such that we have

$$\mu \left(A \setminus \bigcup_{i=1}^N A_i \right) \leq \frac{\varepsilon}{2},$$

so that defining $A_0 = \bigcup_{i=1}^N A_i$, we have

$$\mu(A \setminus A_0) \leq \frac{\varepsilon}{2}.$$

Now note that since $A_0 \subseteq \bigcup_{i=1}^{\infty} A_i$, we have $A_0 \setminus A \subseteq (\bigcup_{i=1}^{\infty} A_i) \setminus A$, so

$$\mu(A_0 \setminus A) \leq \mu \left(\left(\bigcup_{i=1}^{\infty} A_i \right) \setminus A \right) = \mu \left(\bigcup_{i=1}^{\infty} A_i \right) - \mu(A) \leq \left(\sum_{i=1}^{\infty} \mu(A_i) \right) - \mu(A) \leq \frac{\varepsilon}{2},$$

by our original assumption. Therefore, we have that

$$\mu(A \Delta A_0) \leq \mu(A \setminus A_0) + \mu(A_0 \setminus A) \leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

□

Proof of Proposition 8.8. Our first step is to make our life easy by making the generating set for $\mathcal{F}$ an algebra while keeping it countable. Let $\mathcal{C}$ be a countable set of generators: $\sigma(\mathcal{C}) = \mathcal{F}$. Let $B_i \uparrow X$, with $\mu(B_n) < \infty$. Without loss of generality, we can assume that $B_n \in \mathcal{C}$. Again without loss of generality, we can assume that $\mathcal{C}$ is closed under complements. Let $\mathcal{A}$ be the algebra generated by $\mathcal{C}$; note that $\mathcal{A}$ is still countable, since elements of $\mathcal{A}$ are finite unions of finite intersections of sets in $\mathcal{C}$.

Let $\mathcal{G}$ be the set of simple functions over $\mathcal{A}$ with rational coefficients:

$$\mathcal{G} = \left\{ \sum_{i=1}^n c_i \mathbb{1}_{A_i} : A_i \in \mathcal{A}, c_i \in \mathbb{Q}, n \geq 1 \right\}.$$

Note that $\mathcal{G}$ is still countable. It now suffices to prove that $\mathcal{G} \hookrightarrow \text{SF} \cap L^p$ is dense, since $\text{SF} \cap L^p \hookrightarrow L^p$ is dense.

Let $\varphi \in \text{SF}(X) \cap L^p(X)$. Then, $\varphi = \sum_{i=1}^m d_i \mathbb{1}_{E_i}$, for $d_i \in \mathbb{R}$ and $E_i \in \mathcal{F}$. Now, by density of $\mathbb{Q} \hookrightarrow \mathbb{R}$ and Lemma 8.11, we can find $\psi = \sum_{i=1}^m c_i \mathbb{1}_{A_i} \in \mathcal{G}$ with $|c_i - d_i| < \delta$ and

$\mu(A_i \triangle E_i) < \delta$.¹⁰ Then,

$$\begin{aligned}
\|\varphi - \psi\|_p &\leq \sum_{i=1}^m \|d_i \mathbb{1}_{E_i} - c_i \mathbb{1}_{A_i}\|_p \\
&\leq \sum_{i=1}^m \|d_i \mathbb{1}_{E_i} - d_i \mathbb{1}_{A_i}\|_p + \|d_i \mathbb{1}_{A_i} - c_i \mathbb{1}_{A_i}\|_p \\
&\leq \sum_{i=1}^m |d_i| \mu(A_i \triangle E_i)^{1/p} + |d_i - c_i| \mu(A_i)^{1/p} \\
&\leq \delta^{1/p}(|d_1| + \cdots + |d_m|) + \delta m \max \{\mu(A_i)^{1/p}\} \\
&\leq \delta^{1/p}(|d_1| + \cdots + |d_m|) + \delta m \max \{\delta + \mu(E_i)^{1/p}\} \\
&\leq C(\varphi) (\delta^{1/p} + \delta),
\end{aligned}$$

where $C(\varphi)$ is a constant only depending on φ . □

8.2. Dual spaces.

Definition 8.12. Let $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ be normed spaces. A linear map $T: X \rightarrow Y$ is *bounded* if

$$\sup_{\|x\| \leq 1} \|T(x)\|_Y = \sup_{\|x\|_X=1} \|T(x)\|_Y = \sup_{x \neq 0} \frac{\|T(x)\|_Y}{\|x\|_X} < \infty.$$

The space $L(X, Y)$ of bounded linear maps is a normed space with the norm

$$\|T\|_* = \sup_{\|x\|_X=1} \|T(x)\|_Y.$$

Lemma 8.13. Let $T: X \rightarrow Y$ be a linear map between normed spaces. Then, the following are equivalent:

- (1) T is continuous at 0,
- (2) T is continuous,
- (3) T is uniformly continuous,
- (4) T is Lipschitz continuous,
- (5) T is bounded.

Definition 8.14. Let $(X, \|\cdot\|)$ be a normed space. The *topological dual* of X , denoted X^* , is

$$X^* = L(X, \mathbb{R}) = \{T \in \text{Hom}_{\mathbb{R}}(X, \mathbb{R}) : \|T\|_* < \infty\}.$$

Note that X^* is distinct from the *algebraic dual* $\text{Hom}_{\mathbb{R}}(X, \mathbb{R})$, which may contain elements of infinite sup norm (and hence linear but not continuous functionals).

Example 8.15. Let $(X, \|\cdot\|) = (C([0, 1]), \|\cdot\|_{\infty})$ be a normed space with the resulting Borel σ -algebra, and let $\mu = \mu_+ - \mu_-$ be a finite signed measure (that is, μ_+ and μ_- are finite).

¹⁰There is a subtlety here since Lemma 8.11 was proved for finite spaces. But, we can cut off by the B_n then take a limit, which gives us the same result for σ -finite spaces.

Then, we have a linear functional $G_\mu: X \rightarrow \mathbb{R}$ by

$$G_\mu: f \mapsto \int_X f d\mu = \int_X f d\mu_+ - \int_X f d\mu_-,$$

which is bounded since

$$|G_\mu(f)| \leq \|f\|_\infty |\mu|([0, 1]) = \|f\|_\infty (\mu_+([0, 1]) + \mu_-([0, 1])).$$

[TODO: check the defns of signed measures etc] Hence $\|G_\mu\|_* \leq |\mu|([0, 1])$. Of course, in general the inequality is not strict: if $[0, 1]$ admits a “Hahn decomposition” $[0, 1] = P \sqcup N$ with

$$\mu_+(N^c) = 0 \quad \text{and} \quad \mu_-(P^c) = 0,$$

then taking $f = \mathbb{1}_P - \mathbb{1}_N$ gives us $|G_\mu(f)| = \|f\|_\infty |\mu|([0, 1])$, since of course $\|f\|_\infty = 1$.

Example 8.16. Let $(X, \mathcal{F}, \mu)$ be a measure space and $p, q \in (1, \infty)$ conjugate exponents. We have a linear pairing $L^p \times L^q \rightarrow \mathbb{R}$ by

$$(f, g) \mapsto \int_X fg d\mu.$$

Then, for each $g \in L^q$, the induced linear map $T_g: L^p \rightarrow \mathbb{R}$ by

$$T_g: f \mapsto \int_X fg d\mu$$

is bounded by Hölder’s inequality:

$$|T_g(f)| = \left| \int_X fg d\mu \right| \leq \int_X |fg| d\mu \leq \|f\|_p \|g\|_q.$$

This shows us that $\|T_g\|_* \leq \|g\|_q$; we now claim that we have equality. Let $f = (\text{sgn}(g))g^{q-1}$;¹¹ note that $f \in L^p$ because

$$\|f\|_p^p = \int_X |f|^p d\mu = \int_X |g|^{p(q-1)} d\mu = \int_X |g|^q d\mu = \|g\|_q^q < \infty.$$

Then,

$$\left| \int_X fg d\mu \right| = \left| \int_X |g|^q d\mu \right| = \|g\|_q^q = \|g\|_q \|g\|_q^{q-1} = \|g\|_q \|g\|_q^{q/p} = \|g\|_q \|f\|_p,$$

since $\|f\|_p^p = \|g\|_q^q$ as above.

Hence, we have showed that there is a linear isometric¹² embedding

$$L^q \hookrightarrow (L^p)^*.$$

In fact this is an isomorphism; we show this later.

9. 6Tu – SIGNED MEASURES

9.1. Signed measures.

¹¹The sign is defined by $\text{sgn}(x) = 1$ if $x > 0$, $\text{sgn}(x) = -1$ if $x < 0$, and $\text{sgn}(0) = 0$.

¹²A linear map $T: X \rightarrow Y$ between normed spaces is an *isometry* if the pullback of $\|\cdot\|_Y$ by T agrees with $\|\cdot\|_X$: for all $x \in X$,

$$\|T(x)\|_Y = \|x\|_X.$$

Definition 9.1. Let $(X, \mathcal{F})$ be a measure space, and let $\mu: \mathcal{F} \rightarrow \mathbb{R} \cup \{\infty\}$ or $\mu: \mathcal{F} \rightarrow \mathbb{R} \cup \{-\infty\}$. Then, μ is a *signed measure* if

- (1) $\mu(\emptyset) = 0$,
- (2) $\mu(\bigsqcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mu(A_i)$.

The triple $(X, \mathcal{F}, \mu)$ is called a *signed measure space*.¹³

Note that both terms in (2) are always well defined. Suppose $\mu: \mathcal{F} \rightarrow (-\infty, \infty]$ and let $J_- = \{i: \mu(A_i) < 0\}$. Then, of course since $\bigsqcup_{i \in J_-} A_i \in \mathcal{F}$, its measure is finite negative, so the overall sum cannot add up to $-\infty$.

Remark 9.2. We have several formulae that generalize the usual case:

- (1) If $A_i \uparrow A$, then $\mu(A_i) \rightarrow \mu(A)$,
- (2) If $A_i \downarrow A$ and $\mu(A_n) < \infty$ for some n , then $\mu(A_i) \rightarrow \mu(A)$.
- (3) To check that μ is countably additive, it suffices to show that it is finitely additive and that either (1) or (2) hold.

(1) and (2) both follow by taking consecutive differences and noticing they are disjoint, then using countable additivity.

Definition 9.3. Let $(X, \mathcal{F}, \mu)$ be a signed measure space. Then, $A \in \mathcal{F}$ is *positive for μ* if for all $B \subseteq A$, $B \in \mathcal{F}$, then $\mu(B) \geq 0$. Similarly, $A \in \mathcal{F}$ is *negative for μ* if A is positive for $-\mu$.

9.2. Decomposition theorems.

We now hope that we can decompose a signed measure into a difference of two positive measures. Our first technical tool is the following theorem:

Theorem 9.4 (Hahn decomposition theorem). *Let $(X, \mathcal{F}, \mu)$ be a signed measure space. Then, there exist $P, N \in \mathcal{F}$ such that*

$$X = P \sqcup N,$$

P is μ -positive, and N is μ -negative.

Example 9.5. Let $(X, \mathcal{F}, \mu_0)$ be a (unsigned) measure space, and $f: X \rightarrow \mathbb{R}$ with $\int_X f_- d\mu_0 < \infty$. Then

$$\mu(A) = \int_A f d\mu_0$$

is a signed measure.

Lemma 9.6. *Let $(X, \mathcal{F}, \mu)$ be a signed measure, and $A \in \mathcal{F}$ a set with finite negative mass: $-\infty < \mu(A) < 0$. Then, there exists a subset $B \in \mathcal{F}$ with $B \subseteq A$ such that B is μ -negative and $\mu(B) \leq \mu(A)$.*

That is, if we have a set of negative mass, we can isolate the negative mass to get a negative set, possibly passing to a subset by throwing away the portion with positive mass. But, to

¹³To be clear, we may refer to measures $\mu: \mathcal{F} \rightarrow [0, \infty]$ as *positive* measures.

be careful, we should throw away our positive mass bit by bit, then show that in the limit we get a set that is μ -negative.

Proof. Let $A \in \mathcal{F}$ with $-\infty < \mu(A) < 0$. If A is μ -negative, we are done, so assume that A is not μ -negative. Then,

$$\delta_1 = \sup \{ \mu(E) : E \in \mathcal{F}, E \subseteq A \} > 0.$$

Let $P_1 \subseteq A$ such that $\mu(P_1) \geq \min \{ \frac{\delta_1}{2}, 1 \}$. (The minimum is to take care of the case where $\delta_1 = \infty$.) Now let $B_1 = A \setminus P_1$. Then, we can inductively define

$$\delta_n = \sup \left\{ \mu(E) : E \subseteq B_{n-1} = A \setminus \bigcup_{i=1}^{n-1} P_i, E \in \mathcal{F} \right\},$$

then take $P_n \in \mathcal{F}$ with $\mu(P_n) \geq \min \{ \frac{\delta_n}{2}, 1 \}$, and $B_n = A \setminus \bigcup_{i=1}^n P_i$. This is of course a decreasing sequence, so we can take $P = \bigcup_{i=1}^{\infty} P_i$ and

$$B = \bigcap_{i=1}^{\infty} B_i = A \setminus P,$$

and we get that

$$0 > \mu(A) = \mu(B) + \mu(P) \geq \mu(B),$$

so $\mu(B) \leq \mu(A) < 0$, proving the second statement. Moreover, note that $\mu(P) < \infty$, since otherwise we would have $\mu(A) = \infty$, contradicting the assumption that $\mu(A) < 0$.

Now, note that by construction, P_i are disjoint, so

$$0 \leq \mu(P) = \sum_{i=1}^{\infty} \mu(P_i) < \infty,$$

so $\sum_{i=1}^{\infty} \delta_i < \infty$, and therefore $\delta_i \rightarrow 0$.

We now show that B is μ -negative. Assume for the sake of contradiction that there exists some $P_* \subseteq B$ with $\mu(P_*) > 0$. Since $B = \bigcap_{i=1}^{\infty} B_i$, we must have that $P_* \subseteq B_{n-1}$ for all n . Hence, by construction of δ_n ,

$$\delta_n \geq \mu(P_*) > 0,$$

which is a contradiction since $\delta_n \rightarrow 0$. □

Proof of Hahn decomposition theorem 9.4. Without loss of generality, let $\mu: \mathcal{F} \rightarrow (-\infty, \infty]$. Our proof strategy will go as follows:

- (1) Construct N to solve the optimization problem

$$\inf \{ \mu(A) : A \text{ } \mu\text{-negative} \}.$$

Take a sequence approaching the infimum, define N as the union of this sequence, then show that N is μ -negative, and that it achieves this infimum.

- (2) Define $P = N^c$, and show that P is μ -positive.

Step 1: N is μ -negative.

Let us define

$$L = \inf \{ \mu(A) : A \text{ } \mu\text{-negative} \}.$$

We claim that this infimum is achieved. Let N_k be a sequence of μ -negative sets such that $\mu(N_k) \rightarrow L$ as $k \rightarrow \infty$. Then, let $N = \bigcup_{k=1}^{\infty} N_k$. Of course, by countable subadditivity,

$$\mu(N) \leq \lim_{k \rightarrow \infty} \mu(N_k) = L.$$

We now show that N is μ -negative.

Let $B \subseteq N$; then, we can decompose B into $B_k = B \cap (N_k \setminus \bigcup_{i < k} N_i)$, so that $B = \bigsqcup_{k=1}^{\infty} B_k$ and $B_k \subseteq N_k$. Then since each N_k is μ -negative, $\mu(B_k) \leq 0$, and hence

$$\mu(B) = \sum_{k=1}^{\infty} \mu(B_k) \leq 0.$$

Thus, we have that $L \leq \mu(N)$, and so N achieves the infimum.

Step 2: $P = N^c$ is μ -positive.

Let $P = N^c$. Assume for the sake of contradiction that there exists an $A \subseteq P$ with $-\infty < \mu(A) < 0$. Then, by Lemma 9.6 there exists a $B \subseteq A$ that is μ -negative and $\mu(B) < 0$. But then we could take $N' = N \cup B$, and then noting that they are disjoint, see that

$$\mu(N') = \mu(N) + \mu(B) < L,$$

a contradiction. □

Remark 9.7. The Hahn decomposition is unique up to μ -null sets: sets $A \in \mathcal{F}$ such that for all $B \subseteq A$, $\mu(B) = 0$. Note that a set being μ -null is equivalent to being both μ -positive and μ -negative.

Indeed, if $X = P_1 \sqcup N_1 = P_2 \sqcup N_2$ are two Hahn decompositions, then the difference $D = (P_1 \cap N_2) \cup (P_2 \cap N_1)$ is both μ -positive and μ -negative. Hence (see below for the Jordan decomposition $\mu = \mu_+ - \mu_-$), we have $\mu_+(D) = \mu_-(D) = 0$.

Corollary 9.8 (Jordan decomposition theorem). *Let $(X, \mathcal{F}, \mu)$ be a signed measure space. Then, there exist positive measures μ_+, μ_- , at least one of which are finite, such that*

$$\mu = \mu_+ - \mu_-.$$

Proof. Let $X = P \sqcup N$ be a Hahn decomposition for X , and let

$$\mu_+(A) = \mu(A \cap P) \quad \text{and} \quad \mu_-(A) = -\mu(A \cap N).$$

This clearly implies that $\mu = \mu_+ - \mu_-$ by measurability. Then, since either $\pm\infty$ cannot occur as a value of μ , we must have either $\mu_+(P)$ or $\mu_-(N)$ are finite, which implies that at least one of the measures is finite.

Note that we have an equivalent definition:

$$\mu_+(A) = \sup \{ \mu(B) : B \in \mathcal{F}, B \subseteq A \} \quad \text{and} \quad \mu_-(A) = \sup \{ -\mu(B) : B \subseteq A \}.$$

We prove this for μ_+ . In the $\geq$ direction, for any $B \subseteq A$,

$$\mu(B) = \mu_+(B) - \mu_-(B) \leq \mu_+(B) \leq \mu_+(A).$$

In the other direction, taking $B = A \cap P$ achieves the supremum. □

Definition 9.9. Let $(X, \mathcal{F}, \mu)$ be a signed measure space. Then, the *variation* of μ is the (positive) measure

$$|\mu| = \mu_+ + \mu_-.$$

The *total variation* is

$$\|\mu\| = |\mu|(X).$$

We denote the space of measures with finite total variation by

$$M(X, \mathcal{F}) = \{\mu : (X, \mathcal{F}, \mu) \text{ signed measure space, } \|\mu\| < \infty\},$$

and note that $(M(X, \mathcal{F}), \|\cdot\|)$ is a normed space.

We can give this norm a characterization in terms of a sup norm:

Lemma 9.10. *Let $(X, \mathcal{F})$ be a measurable space, and $M(X, \mathcal{F})$ its space of signed measures with finite total variation. Then, we have the following equivalent characterizations of the norm:*

$$\|\mu\| = \sup_{\|\varphi\|_\infty \leq 1} \int_X \varphi d\mu = \sup \left\{ \sum_{i=1}^n |\mu(A_i)| : X = \bigsqcup_{i=1}^n A_i \right\} \leq 2 \sup \{ |\mu(A)| : A \in \mathcal{F} \}.$$

Proof. For the first formula, note that in one direction we have

$$\left| \int_X \varphi d\mu \right| = \left| \int_X \varphi d\mu_+ - \int_X \varphi d\mu_- \right| \leq \int_X |\varphi| d\mu_+ + \int_X |\varphi| d\mu_- \leq \mu_+(X) + \mu_-(X) = \|\mu\|,$$

and in the other direction take a Hahn decomposition $X = P \sqcup N$ and let $\varphi = \mathbb{1}_P - \mathbb{1}_N$.

[TODO: (b)] □

Lemma 9.11. *The space $(M(X, \mathcal{F}), \|\cdot\|)$ is a Banach space.*

Proof. The proof follows a straightforward strategy: pick a Cauchy sequence μ_n , define $\mu(A) = \lim_{n \rightarrow \infty} \mu_n(A)$, then check that μ is a signed measure with finite total variation.

Let μ_n be a Cauchy sequence, and fix $A \in \mathcal{F}$. Then, by definition,

$$|\mu_n(A) - \mu_m(A)| \leq \|\mu_n - \mu_m\|,$$

so $\mu_n(A)$ is Cauchy and hence converges in $\mathbb{R}$. Define $\mu : \mathcal{F} \rightarrow \mathbb{R}$ by

$$\mu(A) := \lim_{n \rightarrow \infty} \mu_n(A).$$

We show that in fact this convergence is uniform:

$$\lim_{n \rightarrow \infty} \sup_{A \in \mathcal{F}} |\mu_n(A) - \mu(A)| = 0.$$

Fix $\varepsilon > 0$, and take $N > 0$ such that $n, m \geq N$ implies that

$$\|\mu_n - \mu_m\| < \varepsilon.$$

Then for any $A \in \mathcal{F}$, we have

$$|\mu_n(A) - \mu_m(A)| < \varepsilon,$$

so taking $m \rightarrow \infty$ we can deduce that

$$|\mu_n(A) - \mu(A)| < \varepsilon.$$

Since this inequality does not depend on A , we get that

$$\sup_{A \in \mathcal{F}} |\mu_n(A) - \mu(A)| \leq \varepsilon,$$

then since $\varepsilon > 0$ was arbitrary, we get the desired result.

Hence, it follows that $\lim_{n \rightarrow \infty} \|\mu_n - \mu\| = 0$.

It now remains to show that μ is actually a signed measure. Obviously, $\mu(\emptyset) = 0$, and finite additivity follows from the fact that we can distribute limits over finite sums.

For next time: countable additivity!

□

10. 6TH – ABSOLUTE CONTINUITY AND RADON–NIKODYM THEOREM

Definition 10.1. Let μ, ν be measures on a measurable space $(X, \mathcal{F})$. We say that ν is *absolutely continuous with respect to μ* , denoted $\nu \ll \mu$, if for all $A \in \mathcal{F}$,

$$\mu(A) = 0 \implies \nu(A) = 0.$$

That is, ν is “smaller” than μ in the sense that μ -null sets are ν -null sets.

Let $(X, \mathcal{F}, \mu)$ be a measure space, and let $f: X \rightarrow [0, \infty)$ be a measurable function. Then, for $\nu = f\mu$, we have $\nu \ll \mu$, since for $\mu(A) = 0$,

$$\nu(A) = \int_A f d\mu = 0.$$

Indeed, the Radon–Nikodym theorem will establish the converse: if $\nu \ll \mu$, then there exists such a measurable f .

Lemma 10.2. Let $(X, \mathcal{F})$ be a measure space with two measures ν, μ . Then,

$$\nu \ll \mu \iff \forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } \mu(A) \leq \delta \rightarrow \nu(A) \leq \varepsilon,$$

with $\iff$ if ν is finite.

Proof. The $\Leftarrow$ direction is obvious. Suppose $\mu(A) = 0$. Then, [TODO]

In the $\implies$ direction, suppose by contradiction that there exists an $\varepsilon > 0$ such that for all $\delta > 0$, there exists a set $A \in \mathcal{F}$ such that $\mu(A) \leq \delta$ and $\nu(A) > \varepsilon$. Then, by absolute continuity for any $\varepsilon > 0$ we can find a $\delta > 0$ such that $\mu(B) \leq \delta \rightarrow \nu(B) \leq \varepsilon$. But,

Hence, we can create a sequence of sets $A_k \in \mathcal{F}$ with $\mu(A_k) \leq 2^{-k}$ but $\nu(A_k) > \varepsilon$. Let us take $B_n = \bigcup_{k \geq n+1} A_k$, and note that

$$\mu(B_n) \leq \sum_{k \geq n+1} 2^{-k} = 2^{-n}.$$

On the other hand, note that since $A_{n+1} \subseteq B_n$,

$$\nu(B_n) \geq \nu(A_{n+1}) \geq \varepsilon.$$

Now let $B_\infty = \bigcap_n B_n$, and note that since $\mu(B_n) \leq 2^{-n} < \infty$, we have that

$$\mu(B_\infty) = \lim_{n \rightarrow \infty} \mu(B_n) = 0,$$

but

$$\nu(B_\infty) = \lim_{n \rightarrow \infty} \nu(B_n) \geq \varepsilon,$$

where we use ν finite to justify the downwards limit.

□

Definition 10.3. Let ν, μ be measures on a measurable space $(X, \mathcal{F})$. Then, if there exists a function $f: X \rightarrow [0, \infty)$ such that $\nu = f\mu$, then f is called the *Radon–Nikodym derivative* of ν with respect to μ , often denoted

$$f = \frac{d\nu}{d\mu}.$$

Note that the Radon–Nikodym derivative is not unique, as we can change it up to a set of measure zero. But, this is the only obstruction to uniqueness, as shown by the following existence and uniqueness theorem.

Theorem 10.4 (Radon–Nikodym). *Let ν, μ be σ -finite measures on a measurable space $(X, \mathcal{F})$. Then, if $\nu \ll \mu$, then there exists a measurable $f: X \rightarrow [0, \infty)$ such that $\nu = f\mu$. Furthermore, f is unique up to a set of μ -measure zero.*

Example 10.5. If $F: \mathbb{R} \rightarrow \mathbb{R}$ is non-decreasing, right-continuous, and $\lim_{x \rightarrow -\infty} F(x) = 0$, then the formula

$$F(y) - F(x) = \mu((x, y])$$

defines a measure μ on intervals and hence can be extended to a full measure. Moreover, if $\mu \ll \lambda$, then we have

$$\mu((x, y]) = \int_x^y \frac{d\mu}{d\lambda} d\lambda = F(y) - F(x).$$

Proof of Radon–Nikodym Theorem 10.4. Let us first consider the case where ν and μ are both finite. We aim to solve the optimization problem of achieving the sup

$$\sup \left\{ \int_X \varphi d\mu : \varphi \in \text{Hom}_{\text{Meas}}(X, [0, \infty)), \varphi\mu \leq \nu \right\},$$

in the sense that $\int_A \varphi d\mu \leq \nu(A)$ for all $A \in \mathcal{F}$.

Let us denote this value by

$$M = \sup \left\{ \int_X \varphi d\mu : \varphi \in \text{Hom}_{\text{Meas}}(X, [0, \infty)), \varphi\mu \leq \nu \right\}.$$

We show that there exists a $\varphi \in \text{Hom}_{\text{Meas}}(X, [0, \infty))$ achieving this supremum. Let

$$\mathcal{S} = \{ \varphi \in \text{Hom}_{\text{Meas}}(X, [0, \infty)), \varphi\mu \leq \nu \}.$$

We first show that $\mathcal{S}$ is closed under maxima, then show that the supremum is achieved.

For the first claim, suppose $\varphi_1, \varphi_2 \in \mathcal{S}$. Let $\varphi = \max\{\varphi_1, \varphi_2\}$, and let $B_1 = \{\varphi_1 \geq \varphi_2\}$ and $B_2 = \{\varphi_2 > \varphi_1\}$. Then, $\varphi = \varphi_1 \mathbb{1}_{B_1} + \varphi_2 \mathbb{1}_{B_2}$, so that

$$\int_A \varphi d\mu = \int_{A \cap B_1} \varphi_1 d\mu + \int_{A \cap B_2} \varphi_2 d\mu \leq \nu(A \cap B_1) + \nu(A \cap B_2) = \nu(A).$$

So, $\varphi \in \mathcal{S}$.

For the second claim, let $\varphi_k \in \mathcal{S}$ be a sequence such that

$$\int_X \varphi_k d\mu \rightarrow M.$$

Without loss of generality, replacing φ_k with $\max\{\varphi_1, \dots, \varphi_k\}$, we can assume that φ_k is nondecreasing. Let $f = \lim_{k \rightarrow \infty} \varphi_k$.

Then, by the monotone convergence theorem,

$$M = \lim_{k \rightarrow \infty} \int_X \varphi_k d\mu = \int_X f d\mu,$$

so f achieves the supremum.

Finally, by Fatou's lemma, for any $A \in \mathcal{F}$,

$$\int_A f d\mu = \int_X \lim_{k \rightarrow \infty} \varphi_k \mathbb{1}_A d\mu \leq \liminf_{k \rightarrow \infty} \int_A \varphi_k d\mu \leq \nu(A),$$

so $f \in \mathcal{S}$ and hence the supremum is achieved by an element in $\mathcal{S}$.

We have thus proved the existence part of the theorem; now let us prove uniqueness. [TODO: this isn't quite true]

Define $\nu_0 = \nu - f\mu$, and note that since $f\mu \leq \nu$, ν_0 is a (positive) measure. So, it suffices to show that ν_0 has total mass 0: $\nu_0(X) = 0$. We do so by contradiction: if $\nu_0(X) > 0$ then we can find some set S such that $\nu \geq (f + \varepsilon \mathbb{1}_S)\mu$, contradicting the maximality of f .

Suppose for the sake of contradiction that $\nu_0(X) > 0$. Then since μ was assumed to be finite, there exists an $\varepsilon > 0$ such that $\nu_0(X) > \varepsilon\mu(X)$. Now, note that $\bar{\nu} = \nu_0 - \varepsilon\mu$ is a signed measure with $\bar{\nu}(X) \geq 0$. Let $X = P \sqcup N$ be a Hahn decomposition of X with respect to $\bar{\nu}$, and consider $f + \varepsilon \mathbb{1}_P$. We claim that $f + \varepsilon \mathbb{1}_P \in \mathcal{S}$ and $\square$, which would give a contradiction.

For the first claim notice that

$$\nu(A) - (f + \varepsilon \mathbb{1}_P)\mu(A) = \nu_0(A) - \varepsilon \mathbb{1}_P \mu(A) = \nu_0(A) - \varepsilon \mu(A \cap P)$$

[TODO]

Note that $\mu(P) > 0$: if we had $\mu(P) = 0$, then by absolute continuity we would have $\nu(P) = 0$, but then this would imply that $\nu_0(P) = 0$, which then implies that

$$\bar{\nu}(X) = \nu_0(X) - \varepsilon\mu(X) = \nu_0(N) - \varepsilon\mu(N) \leq 0,$$

a contradiction with our assumption that $\nu_0(X) > \varepsilon\mu(X)$. [TODO]

Now note that if $\nu = f\mu$ and ν is σ -finite, we must have $f = \infty$ on at most a set of μ -measure zero, and hence we can choose a Radon–Nikodym derivative f which is finite everywhere.

[TODO: sigma-finite]

Thus we have finished the finite case. [TODO]

For uniqueness, suppose f_1, f_2 are two Radon–Nikodym derivatives: $\nu = f_1\mu = f_2\mu$. Then, for all $A \in \mathcal{F}$, we have $\int_A f_1 - f_2 d\mu = 0$, so taking $A = \{f_1 \geq f_2\}$ and $\{f_1 \leq f_2\}$, we can gather that

$$(f_1 - f_2)_+ = (f_1 - f_2)_- = 0$$

μ -a.e., as desired. $\square$

10.1. Singularity.

Definition 10.6. Let $(X, \mathcal{F})$ be a measurable space, with μ a positive (resp. signed) measure. Then, μ is *concentrated* on $A \in \mathcal{F}$ if $\mu(A^c) = 0$ (resp. $|\mu|(A^c) = 0$).

Definition 10.7. Let μ, ν be measures on a measurable space $(X, \mathcal{F})$. Then, μ, ν are *mutually singular*, denoted $\mu \perp \nu$, if there exists an $A \in \mathcal{F}$ such that μ is concentrated on A and ν is concentrated on A^c .¹⁴

Definition 10.8. Let $(\mathbb{R}^d, \mathcal{B}_{\mathbb{R}^d}, \mu)$ be a measure space. Then, μ is said to be *singular* if μ is mutually singular with λ .

Example 10.9. Suppose $\{x_k\}_{k=1}^{\infty} \subseteq \mathbb{R}^d$ and $w_k \in \mathbb{R}$. Then,

$$\mu = \sum_{k=1}^{\infty} w_k \delta_{x_k}$$

is a singular (signed) measure.

Example 10.10. Let $\mu = \mu_+ - \mu_-$ be a Jordan decomposition of μ made according to a Hahn decomposition $X = P \sqcup N$. Then, μ_+ and μ_- are mutually singular.

Absolute continuity and mutual singularity are the two most polar opposite ways in which two measures could be related. In some sense, these two properties exhaust all of the possibilities, according to the following decomposition theorem.

Theorem 10.11 (Lebesgue decomposition theorem). *Let μ, ν be σ -finite positive measures on a measurable space $(X, \mathcal{F})$. Then, there is a unique decomposition*

$$\nu = \nu_a + \nu_s,$$

where $\nu_a \ll \mu$ and $\nu_s \perp \mu$.

11. 7TH – LEBESGUE DECOMPOSITION THEOREM AND FUNCTION SPACES

11.1. Lebesgue decomposition theorem.

Recall from last time the Lebesgue decomposition theorem (now written in slightly more generality):

Theorem 11.1 (Lebesgue decomposition theorem). *Let μ be a positive measure and ν a positive σ -finite or signed finite measure on a measurable space $(X, \mathcal{F})$. Then, there is a unique decomposition*

$$\nu = \nu_a + \nu_s,$$

where $\nu_a \ll \mu$ and $\nu_s \perp \mu$.

Proof. We carry out the proof for ν positive and finite; the other cases follow straightforwardly.

The idea is that we want to find E such that $\mu(E) = 0$ and $\nu_s(E^c) = 0$. Define

$$M = \sup \{ \nu(B) : B \in \mathcal{F}, \mu(B) = 0 \},$$

and note that since ν is finite, $M < \infty$. Then pick a sequence B_n such that $\mu(B_n) = 0$ and $\nu(B_n) \nearrow M$. Then, denoting

$$E = \bigcup_{n=1}^{\infty} B_n,$$

¹⁴If $(X, \mathcal{B}_X)$ is a topological space with its Borel σ -algebra, we can define the *support* of a measure μ to be the smallest closed set $C \subseteq X$ such that $\mu(C^c) = 0$, or alternatively the complement of the largest open set with μ -measure 0. Then, mutual singularity means that the support of the two measures are disjoint.

we get that

$$\mu(E) \leq \sum_{n=1}^{\infty} \mu(B_n) = 0 \quad \text{and} \quad \nu(E) \geq \limsup_{n \rightarrow \infty} \nu(B_n) = M.$$

Now define

$$\nu_s(A) = \nu(A \cap E) \quad \text{and} \quad \nu_a(A) = \nu(A \cap E^c).$$

We clearly have that $\nu_s \perp \mu$, since $\nu_s(E^c) = 0$ and $\mu(E) = 0$.

It remains to show that $\nu_a \ll \mu$. Let $A \in \mathcal{F}$ with $\mu(A) = 0$, and assume for the sake of contradiction that $\nu_a(A) = \nu(A \cap E^c) > 0$. Then, denote $\tilde{E} = E \cup A = E \sqcup (A \cap E^c)$, such that

$$\mu(\tilde{E}) = \mu(E) + \mu(A \cap E^c) = \mu(E),$$

since $\mu(A \cap E^c) \leq \mu(A) = 0$. But,

$$\nu(\tilde{E}) = \nu(E) + \nu(A \cap E^c) > M,$$

a contradiction.

Finally, uniqueness follows from the assertion that for any (positive/signed) measure λ , if $\lambda \perp \mu$ and $\lambda \ll \mu$, then $\lambda = 0$. $\square$

11.2. Function spaces.

Definition 11.2. Let $F: \mathbb{R} \rightarrow \mathbb{R}$ be a function, with $a, b \in \mathbb{R}$ with $a < b$. Then, the *variation* of F on $[a, b]$ is

$$V_F([a, b]) = \sup \left\{ \sum_{i=1}^n |F(t_i) - F(t_{i-1})| : \{t_i\} \in S_{a,b} \right\},$$

where $S_{a,b}$ is the set of partitions

$$S_{a,b} = \{ \{t_i\}_{i=0}^n : a = t_0 \leq \dots \leq t_n = b, n \in \mathbb{N} \}.$$

We can extend the above definition to half-open intervals $(-\infty, b)$ by letting t_0 be any real number, or to $\mathbb{R}$ by looking at the above supremum over all (finite) partitions of $\mathbb{R}$.

Definition 11.3. A function $F: \mathbb{R} \rightarrow \mathbb{R}$ is of *finite variation* if $V_F(\mathbb{R}) < \infty$.

Example 11.4. If $F: \mathbb{R} \rightarrow \mathbb{R}$ is nondecreasing and bounded, then

$$V_F(\mathbb{R}) = \sup F - \inf F.$$

Example 11.5. Let μ be a finite signed measure on $(\mathbb{R}, \mathcal{B})$. Then, its cdf

$$F_\mu(x) = \mu((-\infty, x])$$

is of finite total variation with

$$V_{F_\mu}(\mathbb{R}) \leq |\mu|(\mathbb{R}) = \|\mu\|.$$

Note that not all finite total variation measures are of this form, since F_μ is right continuous and its limit as $x \rightarrow -\infty$ is 0, but by the theorem below, these are essentially the only obstructions to every finite total variation measure arising as a cdf.

Theorem 11.6. Let X be the normed space of finite signed measures and Y be the normed space of functions with finite variation that are right continuous and vanish at $-\infty$, with the

variation norm. Then, $\Phi: X \rightarrow Y$ by

$$\Phi: \mu \mapsto F_\mu$$

is an isometric bijection.

Definition 11.7. A function $F: \mathbb{R} \rightarrow \mathbb{R}$ is *absolutely continuous* if for all $\varepsilon > 0$ there exists a $\delta > 0$ such that for all finite collections of intervals $\{(s_i, t_i)\}_{i=1}^n$ with $s_1 < t_1 < \cdots < s_n < t_n$, we have that

$$\sum_{i=1}^n |t_i - s_i| < \delta \implies \sum_{i=1}^n |F(t_i) - F(s_i)| < \varepsilon.$$

Theorem 11.8. Let $F: \mathbb{R} \rightarrow \mathbb{R}$ be absolutely continuous and $\lim_{x \rightarrow -\infty} F(x) = 0$. Then, there exists a function $f \in L^1(\mathbb{R}, \mathcal{B}, \lambda)$ such that

$$F(x) = \int_{-\infty}^x f(t) dt.$$

Intuitively, if we take $F = F_\mu$, then absolute continuity of F implies that $\mu \ll \lambda$, from which we can use the Radon–Nikodym theorem to get $\mu = f\lambda$, as desired.

11.3. Duality between L^p and L^q .

Recall that for conjugate exponents p, q , for each $g \in L^q(X, \mathcal{F}, \mu)$, we defined the map $T_g \in (L^p(X, \mathcal{F}, \mu))^*$ by

$$T_g: f \mapsto \int_X fg d\mu.$$

In [TODO; ref] we proved that this was a linear isometric embedding. We now prove that it is surjective, completing the proof of the below theorem.

Theorem 11.9. Let $(X, \mathcal{F}, \mu)$ be a (positive) measure space, and assume that either

- (1) $p \in [1, \infty)$ and μ is σ -finite, or
- (2) $p \in (1, \infty)$ and μ is arbitrary.

Then, $\Phi: L^q \rightarrow (L^p)^*$ defined by $\Phi: g \mapsto T_g$ is an isometric isomorphism.

Proof. As mentioned above, it remains to show surjectivity. Let $F \in (L^p)^*$. The strategy is as follows:

- (1) Find a measure ν such that $F(f) = \int_X f d\nu$.
- (2) Prove that $\nu \ll \mu$, so that by the Radon–Nikodym theorem we get a function g with $F(f) = \int_X fg d\mu$.
- (3) Prove that $F(f) = \int_X fg d\mu$.

For the sake of simplicity, we prove this result for μ finite.

(1): Define $\nu(A) = F(\mathbb{1}_A)$. We show that this is a measure.

Clearly, $\nu(\emptyset) = F(\mathbb{1}_\emptyset) = F(0) = 0$, since F is linear.

Now let $A_1, \dots, A_n$ be a sequence of disjoint sets. Then,

$$\nu \left(\bigsqcup_{i=1}^n A_i \right) = F \left(\sum_{i=1}^n \mathbb{1}_{A_i} \right) = \sum_{i=1}^n F(\mathbb{1}_{A_i}) = \sum_{i=1}^n \nu(A_i).$$

Now suppose A_i is a sequence of disjoint sets. Then, for $A = \bigsqcup_{i=1}^{\infty} A_i$, by the dominated convergence theorem we have that

$$\lim_{n \rightarrow \infty} \left\| \mathbb{1}_A - \sum_{i=1}^n \mathbb{1}_{A_i} \right\|_p = 0,$$

so since F is continuous,

$$\nu(A) = F(\mathbb{1}_A) = \lim_{n \rightarrow \infty} F \left(\sum_{i=1}^n \mathbb{1}_{A_i} \right) = \sum_{i=1}^{\infty} \nu(A_i).$$

(2): Take A such that $\mu(A) = 0$. Then, note that since $\mathbb{1}_A = 0$ a.e. and F is a functional on *equivalence classes* of functions in $\mathcal{L}^p$, we have

$$\nu(A) = F(\mathbb{1}_A) = F(0) = 0.$$

Hence by the Radon–Nikodym theorem, we have a function g such that $\nu = g\mu$.

(3): We now wish to show that

$$F(f) = \int_X f \, d\nu = \int_X fg \, d\mu.$$

This is true for f indicator functions, but to extend to all of L^p we wish to use the dominated convergence theorem. But, since we have no controls over g , we first take truncations of g to use the DCT, then pass to the limit.

Define $F_n(f) = F(f\mathbb{1}_{E_n})$, where

$$E_n = \{x \in X : |g(x)| \leq n\}.$$

We first wish to show that

$$F_n(f) = \int_X fg\mathbb{1}_{E_n} \, d\mu.$$

If $f = \mathbb{1}_A$ is an indicator function, then

$$F_n(f) = F(\mathbb{1}_{A \cap E_n}) = \nu(A \cap E_n) = \int_X \mathbb{1}_A \mathbb{1}_{E_n} g \, d\mu.$$

If $f = \sum_{i=1}^n c_i \mathbb{1}_{A_i}$ is simple, then by linearity,

$$F_n(f) = F_n \left(\sum_{i=1}^n c_i \mathbb{1}_{A_i} \right) = \sum_{i=1}^n c_i F_n(\mathbb{1}_{A_i}) = \sum_{i=1}^n c_i \int_X \mathbb{1}_{A_i} g \mathbb{1}_{E_n} \, d\mu = \int_X fg\mathbb{1}_{E_n} \, d\mu.$$

If $f \in L^p$, then taking a simple approximation $\varphi_k \rightarrow f$ in L^p , it follows that we have an L^p approximation $\varphi_k \mathbb{1}_{E_n} \rightarrow f \mathbb{1}_{E_n}$, so by continuity of F ,

$$F_n(f) = F(f\mathbb{1}_{E_n}) = \lim_{k \rightarrow \infty} F(\varphi_k \mathbb{1}_{E_n}) = \lim_{k \rightarrow \infty} \int_X \varphi_k g \mathbb{1}_{E_n} \, d\mu = \int_X fg\mathbb{1}_{E_n} \, d\mu,$$

where the last equality uses the DCT with the L^p estimate $\|\varphi_k - f\|_p \rightarrow 0$:

$$\left| \int_X f g \mathbb{1}_{E_n} - \varphi_k g \mathbb{1}_{E_n} d\mu \right| \leq n \int_X |f - \varphi_k| d\mu \leq n \|f - \varphi_k\|_p \mu(X)^{1/q}.$$

by Hölder's inequality with $|f - \varphi_k| \cdot 1$.

In conclusion, we have shown that

$$F_n(f) = F(f \mathbb{1}_{E_n}) = \int_X f g \mathbb{1}_{E_n} d\mu,$$

so that $F_n = T_{g \mathbb{1}_{E_n}}$. Of course, since we know that $\Phi: g \mapsto T_g$ is an isometric embedding,

$$\|g \mathbb{1}_{E_n}\|_q = \|T_{g \mathbb{1}_{E_n}}\|_{L^{p*}} = \|F_n\|_{L^{p*}} \leq \|F\|_{L^{p*}} < \infty.$$

Hence by monotone convergence of $g \mathbb{1}_{E_n} \nearrow g$ (for $q < \infty$), we have

$$\|g\|_q = \lim_{n \rightarrow \infty} \|g \mathbb{1}_{E_n}\|_q \leq \|F\|_{L^{p*}},$$

so indeed $g \in L^q$.

Finally, note that since $\|f \mathbb{1}_{E_n} - f\|_p = \|f \mathbb{1}_{E_n^c}\|_p \rightarrow 0$ by dominated convergence with f , it follows that

$$\lim_{n \rightarrow \infty} F_n(f) = F(f).$$

Then by Hölder's inequality,

$$\left| \int_X f g \mathbb{1}_{E_n} - f g d\mu \right| \leq \|F\|_{L^{p*}} \|f \mathbb{1}_{E_n^c}\|_p \rightarrow 0,$$

so in conclusion we have that

$$F(f) = \lim_{n \rightarrow \infty} F_n(f) = \lim_{n \rightarrow \infty} \int_X f g \mathbb{1}_{E_n} d\mu = \int_X f g d\mu.$$

□

12. STU – PRODUCT MEASURES AND FUBINI'S THEOREM

12.1. Product measures.

Definition 12.1. Let $(X_1, \mathcal{F}_1)$ and $(X_2, \mathcal{F}_2)$ be two measurable spaces. Their *product* $(X_1 \times X_2, \mathcal{F}_1 \times \mathcal{F}_2)$, is the measurable space where $X_1 \times X_2$ is the usual (set-theoretic) Cartesian product, and

$$\mathcal{F}_1 \times \mathcal{F}_2 = \sigma(\{A \times B: A \in \mathcal{F}_1, B \in \mathcal{F}_2\}).$$

Proposition 12.2. Let $(X_1, \mathcal{F}_1, \nu_1)$ and $(X_2, \mathcal{F}_2, \nu_2)$ be σ -finite measure spaces. Then there exists a unique product measure $\nu = \nu_1 \otimes \nu_2$ on $(X_1 \times X_2, \mathcal{F}_1 \times \mathcal{F}_2)$ such that for any rectangle $A \times B \in \mathcal{F}_1 \times \mathcal{F}_2$,

$$\nu(A \times B) = \nu_1(A) \nu_2(B).$$

Proof. Note that set

$$\mathcal{A} = \left\{ \bigsqcup_{i=1}^N A_i \times B_i: A_i \in \mathcal{F}_1, B_i \in \mathcal{F}_2 \right\}$$

is an algebra containing the generating set for $\mathcal{F}_1 \times \mathcal{F}_2$. On this algebra we may define

$$\nu \left(\bigsqcup_{i=1}^N A_i \times B_i \right) = \sum_{i=1}^N \nu_1(A_i) \nu_2(B_i).$$

Obviously $\nu(\emptyset) = 0$, so it suffices to check countable additivity. Suppose we have a collection $E_l = \bigsqcup_{i=1}^{N_l} A_i^l \times B_i^l$ such that $E = \bigsqcup_{l=1}^{\infty} E_l \in \mathcal{A}$, so that $E = \bigsqcup_{i=1}^M C_i \times D_i$. We thus aim to show that

$$\sum_{i=1}^M \nu_1(C_i) \nu_2(D_i) = \sum_{l=1}^{\infty} \sum_{i=1}^{N_l} \nu_1(A_i^l) \nu_2(B_i^l).$$

Reordering if necessary, we can assume that

$$\{(A_i^l, B_i^l) : 1 \leq i \leq N_l, l \geq 1\} = \{(F_k, G_k) : k \geq 1\},$$

such that for each $i \leq M$, $C_i \times D_i = \bigsqcup_{k \in S_i} F_k \times G_k$ for some indexing set S_i . Thus to prove the above formula it suffices to do so for each $C_i \times D_i$ individually. Thus it suffices to prove the following statement: if $C \in \mathcal{F}_1$ and $D \in \mathcal{F}_2$ and $F_k \in \mathcal{F}_1, G_k \in \mathcal{F}_2$ are such that

$$C \times D = \bigsqcup_{k=1}^{\infty} F_k \times G_k,$$

then the formula

$$\nu_1(C) \nu_2(D) = \sum_{k=1}^{\infty} \nu_1(F_k) \nu_2(G_k)$$

holds.

Note that $\mathbb{1}_{C \times D} = \sum_{k=1}^{\infty} \mathbb{1}_{F_k \times G_k}$. Fixing some $x, y \in X$, note that

$$\mathbb{1}_C(x) \mathbb{1}_D(y) = \mathbb{1}_{C \times D}(x, y) = \sum_{k=1}^{\infty} \mathbb{1}_{F_k \times G_k}(x, y) = \sum_{k=1}^{\infty} \mathbb{1}_{F_k}(x) \mathbb{1}_{G_k}(y).$$

Thus, for fixed x , we have that $\sum_{k=1}^N \mathbb{1}_{F_k}(x) \mathbb{1}_{G_k} \nearrow \mathbb{1}_C(x) \mathbb{1}_D$ as $N \rightarrow \infty$, so by the monotone convergence theorem we have

$$\nu_2(D) \mathbb{1}_C(x) = \int_X \mathbb{1}_C(x) \mathbb{1}_D(y) d\nu_2(y) = \sum_{k=1}^{\infty} \mathbb{1}_{F_k}(x) \int_X \mathbb{1}_{G_k}(y) d\nu_2(y) = \sum_{k=1}^{\infty} \mathbb{1}_{F_k}(x) \nu_2(G_k).$$

But then once again we have that $\sum_{k=1}^N \mathbb{1}_{F_k} \nu_2(G_k) \nearrow \mathbb{1}_C \nu_2(D)$ as $N \rightarrow \infty$, so again by the monotone convergence theorem,

$$\nu_1(C) \nu_2(D) = \sum_{k=1}^{\infty} \nu_2(G_k) \int_{X_1} \mathbb{1}_{F_k}(x) d\nu_1(x) = \sum_{k=1}^{\infty} \nu_1(F_k) \nu_2(G_k),$$

as desired. □

Theorem 12.3 (Fubini's theorem). *Let $(X_1, \mathcal{F}_1, \nu_1)$ and $(X_2, \mathcal{F}_2, \nu_2)$ be σ -finite measure spaces, and let $f: X_1 \times X_2 \rightarrow \mathbb{R}$ be a function that is either in $L^1(\nu_1 \times \nu_2)$ or $f \geq 0$.¹⁵ Then,*

$$\int_{X_1 \times X_2} f d(\nu_1 \otimes \nu_2) = \int_{X_1} \left(\int_{X_2} f(x, y) d\nu_2(y) \right) d\nu_1(x).$$

¹⁵In Cohn, the $f \geq 0$ case is called Tonelli's theorem.

Proof. Define

$$F(x) = \int_{X_2} f(x, y) d\nu_2(y).$$

The proof strategy is as follows:

- (1) $f(x, -): X_2 \rightarrow \mathbb{R}$ is measurable for ν_1 -a.e. $x \in X_1$, and is either $L^1(\nu_2)$ or ≥ 0 .
- (2) $F: X_1 \rightarrow \mathbb{R}$ is measurable, and is either $L^1(\nu_1)$ or ≥ 0 .
- (3) $\int_{X_1 \times X_2} f d(\nu_1 \otimes \nu_2) = \int_{X_1} F d\nu_1$.

In each of these steps, we use the monotone class theorem [TODO: ref]. As the proof in general is rather tedious, we limit our attention to proving the result for ν_1, ν_2 finite and f bounded.

Step 1: Let $\mathcal{H}$ be the set of measurable $f(-, -)$ such that $f(x, -)$ is measurable for all $x \in X_1$:

$$\mathcal{H} = \{f \in \text{Hom}_{\text{Meas}}(X_1 \times X_2, \mathbb{R}) : f(x, -) \in \text{Hom}_{\text{Meas}}(X_2, \mathbb{R}) \forall x \in X_1\}.$$

Clearly $\mathbb{1}_{X_1 \times X_2} \in \mathcal{H}$ (since ν_1, ν_2 are finite) and $\mathcal{H}$ is closed under linear combinations and monotone limits. It remains to check that $\mathbb{1}_A \in \mathcal{F}$ for all $A \in \mathcal{P}$, where $\sigma(\mathcal{P}) = \mathcal{F}_1 \times \mathcal{F}_2$. We pick the π -system

$$\mathcal{P} = \{A \times B : A \in \mathcal{F}_1, B \in \mathcal{F}_2\}.$$

But, note that

$$\mathbb{1}_{A \times B}(x, -) = \mathbb{1}_A(x) \mathbb{1}_B = \begin{cases} \mathbb{1}_B & x \in A \\ 0 & x \notin A, \end{cases}$$

and hence $\mathbb{1}_{A \times B} \in \mathcal{H}$.

Step 2: Now let $\mathcal{H}$ be the set of bounded measurable functions such that F is measurable:

$$\mathcal{H} = \left\{ f \in \text{Hom}_{\text{Meas}}(X_1 \times X_2, \mathbb{R}) : F_f = \int_{X_2} f(-, y) d\nu_2(y) \in \text{Hom}_{\text{Meas}}(X_1, \mathbb{R}) \right\}$$

Similarly to above, $\mathbb{1}_{X_1 \times X_2} \in \mathcal{H}$ follows from finiteness and linearity follows from linearity of the integral. Closedness under monotone limits follows from the monotone convergence theorem: if $h_n \nearrow h$, $h_n \in \mathcal{H}$, then by the monotone convergence theorem

$$F_{h_n}(x) = \int_{X_2} h_n(x, y) d\nu_2(y) \nearrow F_h(x),$$

so F_h is measurable (since it is a pointwise sup). Finally, using the same π -system as above,

$$F_{\mathbb{1}_{A \times B}}(x) = \int_{X_2} \mathbb{1}_{A \times B}(x, y) d\nu_2(y) = \mathbb{1}_A(x) \nu_2(X_2)$$

is measurable and hence $\mathbb{1}_{A \times B} \in \mathcal{H}$.

We leave step 3 and extending to σ -finite spaces to the reader. □

Remark 12.4. The above construction (product measures and Fubini's theorem) applies to any finite product of spaces.

Remark 12.5. If $(X_i, \mathcal{F}_i, \nu_i)$ is a countable sequence of finite measure spaces (e.g., probability spaces), then the product $(X, \mathcal{F}, \nu)$ is given by the Cartesian product

$$X = \prod_{i=1}^{\infty} X_i,$$

the *product σ -algebra*

$\mathcal{F} = \sigma(C_{A_1, \dots, A_n} : A_i \in \mathcal{F}_i, n \geq 1)$ where $C_{A_1, \dots, A_n} = \{x = (x_i)_{i=1}^{\infty} : x_i \in A_i \forall i \leq n\}$, and the *product measure* $\nu = \bigotimes_{i=1}^{\infty} \nu_i$ defined by

$$\nu(C_{A_1, \dots, A_n}) = \prod_{i=1}^n \nu_i(A_i).$$

12.2. Convolutions.

Proposition 12.6. *Let $f, g \in L^1(\mathbb{R}, \mathcal{B}, \lambda)$. Then,*

- (1) *For λ -a.e. $x \in \mathbb{R}$, the function $F_x(t) = f(x-t)g(t) \in L^1$,*
- (2) *For λ -a.e. $x \in \mathbb{R}$ the integral*

$$(f * g)(x) = \int_{\mathbb{R}} f(x-t)g(t) dt$$

is finite, and

- (3) *$f * g \in L^1$ with an estimate*

$$\|f * g\|_1 \leq \|f\|_1 \|g\|_1.$$

Proof. Note that the kernel

$$H(x, t) = f(x-t)g(t)$$

as a map $H: \mathbb{R}^2 \rightarrow \mathbb{R}$ is measurable, since each step

$$(x, t) \mapsto (x-t, t) \mapsto (f(x-t), g(t)) \mapsto f(x-t)g(t)$$

is measurable. Since $|H| \geq 0$, by Fubini's theorem,

$$\|f * g\|_1 = \int_{\mathbb{R}} \left| \int_{\mathbb{R}} f(x-t)g(t) dt \right| dx \leq \int_{\mathbb{R}} \int_{\mathbb{R}} |f(x-t)g(t)| dt dx = \int_{\mathbb{R}^2} |H(x, t)| d(\lambda \otimes \lambda)(x, t).$$

But on the other hand,

$$\begin{aligned} \int_{\mathbb{R}^2} |H(x, t)| d(\lambda \otimes \lambda)(x, t) &= \int_{\mathbb{R}} \left(\int_{\mathbb{R}} |f(x-t)| |g(t)| dx \right) dt \\ &= \int_{\mathbb{R}} |g(t)| \left(\int_{\mathbb{R}} |f(x-t)| dx \right) dt \\ &= \|f\|_1 \int_{\mathbb{R}} |g(t)| dt \\ &= \|f\|_1 \|g\|_1. \end{aligned}$$

□

12.3. Change of variables formula.

Definition 12.7. Let $F: (X, \mathcal{F}) \rightarrow (Y, \mathcal{G})$ be a measurable function, and let $(X, \mathcal{F}, \mu)$ be a measure space. The *pushforward measure* is defined by

$$F_{\#}\mu(A) = \mu(F^{-1}(A)).$$

Remark 12.8. If $(X, \mathcal{F}, \mu)$ is a probability space with $\mu(A) = \mathbb{P}(x \in A)$ and $F: (X, \mathcal{F}) \rightarrow (Y, \mathcal{G})$ is measurable, then the pushforward measure is characterized by

$$F_{\#}(A) = \mathbb{P}(x \in F^{-1}(A)) = \mathbb{P}(F(x) \in A).$$

Theorem 12.9 (Area formula). *Let $F: U \rightarrow V$ be a C^1 diffeomorphism onto its image, where $U, V \in \mathbb{R}^d$ are open. Then, for all $x \in V$,*

$$F_{\#}\lambda = |J_{F^{-1}}(x)| \lambda,$$

where the Jacobian is defined as

$$J_H(x) = \det(DH(x)),$$

where $DH(x): \mathbb{R}^d \rightarrow \mathbb{R}^d$ is the total derivative at x .

Corollary 12.10. *If $f \in L^1(U \rightarrow \mathbb{R})$ for $U \subseteq \mathbb{R}$ open, then for a C^1 diffeomorphism onto its image $F: U \rightarrow \mathbb{R}$,*

$$\int_U f(x) d\lambda(x) = \int_V f(F^{-1}(y)) |J_{F^{-1}}(y)| d\lambda(y).$$

13. 8TH – DIFFERENTIATION OF MEASURES

13.1. Density.

Let $\mathcal{O}, \mathcal{C} \subseteq 2^{\mathbb{R}^n}$ be the collections of *axis-aligned open/closed cubes*:

$$\mathcal{O} = \{(a_1, b_1) \times \cdots \times (a_n, b_n) \subseteq \mathbb{R}^n: a_i < b_i, |b_i - a_i| = |b_j - a_j|\}$$

$$\mathcal{C} = \{[a_1, b_1] \times \cdots \times [a_n, b_n] \subseteq \mathbb{R}^n: a_i < b_i, |b_i - a_i| = |b_j - a_j|\}$$

and for $C \in \mathcal{C}$ or $U \in \mathcal{O}$, define

$$\ell(C) = |b_i - a_i| = \ell(U)$$

to be the side length of a cube.

Definition 13.1. Let μ be a σ -finite Borel regular measure on $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n})$. Then, define the *upper and lower density* at $x \in \mathbb{R}^n$ by

$$\overline{D}\mu(x) = \limsup_{\varepsilon \rightarrow 0} \left\{ \frac{\mu(C)}{\lambda(C)} : x \in C, \ell(C) \leq \varepsilon \right\}$$

$$\underline{D}\mu(x) = \liminf_{\varepsilon \rightarrow 0} \left\{ \frac{\mu(C)}{\lambda(C)} : x \in C, \ell(C) \leq \varepsilon \right\}$$

We say that μ is *differentiable at x* if these agree, and in this case we define the *density*

$$D\mu(x) = \overline{D}\mu(x) = \underline{D}\mu(x).$$

Remark 13.2. Since $\mathbb{R}^n$ is G_δ (every closed set is a countable intersection of open sets), the upper/lower density can be equivalently defined in terms of $\mathcal{O}$ instead of $\mathcal{C}$; indeed, any basis for the topology suffices since $\mathbb{R}^n$ is second countable.

Theorem 13.3 (Density differentiation theorem). *Let μ be a σ -finite measure on $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n})$. Then, it is differentiable λ -a.e., with*

$$D\mu(x) = \frac{d\mu_a}{d\lambda}(x)$$

λ -a.e., where $\mu = \mu_a + \mu_s$.

Lemma 13.4. *Let μ be a σ -finite measure on $(\mathbb{R}^n, \mathcal{B}_{\mathbb{R}^n})$. Then $\overline{D}\mu$ and $\underline{D}\mu$ are Borel.*

Proof. For some $x \in \mathbb{R}^n$, suppose $\overline{D}\mu(x) < \infty$. Note that for

$$f_\varepsilon(x) = \sup_{U \in \mathcal{O}} \left\{ \frac{\mu(U)}{\lambda(U)} : x \in U, \ell(U) \leq \varepsilon \right\},$$

we have $f_\varepsilon \searrow \overline{D}\mu$ pointwise, so it suffices to show that f_ε is Borel. Note that

$$f_\varepsilon^{-1}((a, \infty)) = \bigcup \left\{ U \in \mathcal{O} : \ell(U) \leq \varepsilon, \frac{\mu(U)}{\lambda(U)} > a \right\},$$

which is open, so we are done. □

13.2. Vitali covering theorem.

A key technical tool in proving the above differentiation theorem [TODO: ref] is the following notion of a Vitali covering.

Definition 13.5. Let $A \subseteq \mathbb{R}^n$. A collection $\mathcal{V} \subseteq \mathcal{C}$ is a *Vitali covering* of A if $\bigcup_{C \in \mathcal{V}} C \supseteq A$ and for all $x \in A$ and $\delta > 0$, there exists $C \in \mathcal{V}$ such that $\ell(C) < \delta$ and $x \in C$.

Using the collection $\mathcal{C}$ is purely a matter of convention; in general we may take any collection of sets which is reasonably nice, in the sense that

$$\sup_{C \in \mathcal{C}} \frac{\text{diam}(C)^n}{\lambda(C)} < \infty.$$

Theorem 13.6 (Vitali covering theorem). *Let $\mathcal{V} \subseteq \mathcal{C}$ be a Vitali covering of $A \in \mathcal{B}_{\mathbb{R}^n}$. Then there exists a countable¹⁶ disjoint subcollection $\{C_i\}_{i=1}^\infty$ such that*

$$\lambda(A \setminus \bigcup_{i=1}^\infty C_i) = 0.$$

We now take the covering theorem on faith and prove the differentiation theorem [TODO; ref]. First, we prove the following two “comparison lemmas,” which will make concrete the intuition that $\overline{D}\mu$ and $\underline{D}\mu$ are densities.

Lemma 13.7 (Upper comparison lemma). *Let μ be a finite measure on $\mathbb{R}^n$, and let $A \in \mathcal{B}_{\mathbb{R}^n}$. Then, if $\overline{D}\mu(x) \geq b$ for all $x \in A$, then*

$$\frac{\mu(A)}{\lambda(A)} \geq b.$$

¹⁶Our convention is that *countable* includes both finite and countably infinite.

Proof. Fix $U \supseteq A$ open, and fix $\varepsilon > 0$. Let

$$\mathcal{V} = \left\{ C \in \mathcal{C} : C \subseteq U, \frac{\mu(C)}{\lambda(C)} > b - \varepsilon \right\}.$$

Note that this is a Vitali covering [TODO]. Hence by the Vitali covering theorem, there exists a countable disjoint collection $\{C_i\}_{i=1}^\infty$ such that $\lambda(A \setminus \bigcup_{i=1}^\infty C_i) = 0$ and $\bigcup_{i=1}^\infty C_i \subseteq U$. Then, we have

$$\begin{aligned} \mu(U) &\geq \mu\left(\bigcup_{i=1}^\infty C_i\right) \\ &= \sum_{i=1}^\infty \mu(C_i) \\ &\geq (b - \varepsilon) \sum_{i=1}^\infty \lambda(C_i) \\ &= (b - \varepsilon) \lambda\left(\bigcup_{i=1}^\infty C_i\right) \\ &\geq (b - \varepsilon) \lambda\left(A \cap \bigcup_{i=1}^\infty C_i\right) \\ &= (b - \varepsilon) \lambda(A) - (b - \varepsilon) \lambda\left(A \setminus \bigcup_{i=1}^\infty C_i\right) \\ &= (b - \varepsilon) \lambda(A). \end{aligned}$$

Hence taking $\varepsilon \rightarrow 0$, by Borel regularity we get that

$$\mu(A) = \inf_{U \supseteq A} \mu(U) \geq b \lambda(A),$$

as desired. $\square$

Lemma 13.8 (Lower comparison lemma). *Let μ be a finite measure on $\mathbb{R}^n$ such that $\mu \ll \lambda$, and let $A \in \mathcal{B}_{\mathbb{R}^n}$. Then, if $\underline{D}\mu(x) \leq b$ for all $x \in A$, then*

$$\frac{\mu(A)}{\lambda(A)} \leq b.$$

Proof. The argument is exactly the same as above, except this time we use absolute continuity to ensure that $\lambda(A \setminus \bigcup_{i=1}^\infty C_i) = 0$ implies that $\mu(A \setminus \bigcup_{i=1}^\infty C_i) = 0$. $\square$

Proof of density differentiation theorem TODO ??. First we consider the case where $\mu \perp \lambda$ and μ is finite, and we aim to show that $\overline{D}\mu = 0$, which would imply that $0 \leq \underline{D}\mu \leq \overline{D}\mu = 0$, so $D\mu = 0$.

Let $S \subseteq \mathbb{R}^n$ with $\lambda(S) = 0$ and $\mu(S^c) = 0$. Then, define

$$B_k = \left\{ x \in S^c : \overline{D}\mu(x) \geq \frac{1}{k} \right\}.$$

Then,

$$0 = \mu(B_k) \geq \frac{1}{k} \lambda(B_k),$$

so $\lambda(B_k) = 0$. But then note that

$$\bigcup_{k=1}^{\infty} B_k = \{x \in S^c : \overline{D}\mu(x) > 0\}$$

therefore has λ -measure 0, so $\overline{D}\mu = 0$ λ -a.e..

Second, we consider the case where $\mu \ll \lambda$ and μ is finite. Here we aim to show that x is differentiable a.e.. For each $p, q \in \mathbb{Q}$, $0 \leq p < q$, define

$$A(p, q) = \{x \in \mathbb{R}^n : \underline{D}\mu(x) \leq p < q \leq \overline{D}\mu(x)\}.$$

If we can show that $\lambda(A(p, q)) = 0$ for all $p < q$, then we can take the union over all rationals and get that $\underline{D}\mu < \overline{D}\mu$ on at most a set of λ -measure zero, as desired.

By the comparison lemmas, we have that

$$q\lambda(A(p, q)) \leq \mu(A(p, q)) \leq p\lambda(A(p, q)),$$

which implies that $\lambda(A(p, q)) = 0$ since μ is finite and $p < q$. Thus $D\mu$ exists λ -a.e..

By the Radon–Nikodym theorem, we have a measurable function $f = \frac{d\mu}{d\lambda}$. Pick $A \in \mathcal{B}_{\mathbb{R}^n}$ such that $D\mu(x) \leq a$ for all $x \in A$. Then, by the lower comparison lemma,

$$\int_A f d\lambda = \mu(A) \leq a\lambda(A) = \int_A a d\lambda,$$

so $f(x) \leq a$ for a.e. $x \in A$, else we could have modified A to get a contradiction. Hence $f(x) \leq D\mu(x)$. By using the upper comparison lemma we get the other bound, so $f(x) = D\mu(x)$ a.e..

Third, we combine these two cases to get the full (finite) case. Let $\mu = \mu_a + \mu_s$ be a finite measure on $\mathbb{R}^n$, and note that then $D\mu_a$ and $D\mu_s$ exist a.e. and

$$D\mu_a = \frac{d\mu_a}{d\lambda} \quad \text{and} \quad D\mu_s = 0,$$

so

$$D\mu = D\mu_a + D\mu_s = \frac{d\mu_a}{d\lambda}.$$

(Linearity here can be checked by combing through the definitions carefully.) □

Theorem 13.9. *Let $F: \mathbb{R} \rightarrow \mathbb{R}$ be a nondecreasing function. Then F is differentiable λ -a.e., and there exists $f \in L^1(\lambda)$ such that*

$$F(x) = \int_{-\infty}^x f(t) dt.$$

Intuitively, this result has the interpretation that each cdf has a pdf.

14. 9Tu – RIESZ–MARKOV–KAKUTANI THEOREM, DAY 1

Today and Thursday's lecture will be on the following representation theorem.

Theorem 14.1 (Riesz–Markov–Kakutani). *Let X be a LCH space and $L \in (C_c(X))^*$ be a positive bounded linear functional. Then, there exists a unique Borel regular measure μ representing L :*

$$L(f) = \int_X f \, d\mu.$$

Today's lecture will focus on the technical preliminaries and definitions, then next lecture we will prove this theorem.

14.1. Locally compact Hausdorff spaces.

Definition 14.2. Let X be a topological space. We say that X is *locally compact* if each point has a compact neighborhood: each $x \in X$ has an open neighborhood $U \ni x$ such that $\overline{U}$ is compact.

We say that X is *Hausdorff* if distinct points can be separated: for each $x, y \in X$ distinct, there exist $U, V \subseteq X$ open such that $U \cap V = \emptyset$, $x \in U$, and $y \in V$.

If X is locally compact and Hausdorff, we say X is *LCH*.

Example 14.3. Of course, $\mathbb{R}^d$ with its usual topology is LCH, since closed balls are compact.

As usual, when X is a topological space we endow it with its Borel σ -algebra; the situation of LCH spaces is no different.

Definition 14.4. Let $f: X \rightarrow \mathbb{R}$ be a continuous function. Its *support* is the set

$$\text{supp } f = \overline{\{x \in X : f(x) \neq 0\}}.$$

The space of *compactly supported continuous functions* is denoted $C_c(X)$.

Note that $C_c(X)$ with the uniform norm is a Banach space. Hence, its dual also has a Banach space structure with the sup norm, so that $L \in (C_c(X))^*$ is called *bounded* if

$$\sup_{\|f\|_\infty \leq 1} |L(f)| < \infty,$$

and L is *positive* if

$$f \geq 0 \rightarrow L(f) \geq 0.$$

When X is an LCH space, we recover many of the useful tools from smooth functions in Euclidean spaces, though in the continuous category. Namely, we get the existence of continuous bump/cutoff functions (Urysohn's lemma) and the existence of a partition of unity:

Theorem 14.5 (Urysohn's lemma). *Let X be LCH, $K \subseteq U \subseteq X$ with K compact and U open. Then, there exists $f \in C_c(U)$ such that $\mathbb{1}_K \leq f \leq \mathbb{1}_U$.*

In other words, $f = 1$ on K , $0 \leq f \leq 1$ on $U \setminus K$ and $f = 0$ outside of U .

Theorem 14.6 (Partition of unity). *Let $U_1, \dots, U_N \subseteq X$ be open, $f \in C_c(X)$, and $\text{supp } f \subseteq \bigcup_{i=1}^N U_i$. Then, there exist $f_i \in C_c(U_i)$ such that $f = f_1 + \dots + f_N$, and if $f \geq 0$ we can take $f_i \geq 0$.*

Another crucial fact we will need later is the following topology exercise:

Lemma 14.7. *Let X be Hausdorff and $K \subseteq X$ be compact. Then, K is closed.*

Proof. We wish to show that every $x \in X \setminus K$ has an open neighborhood disjoint from K ; this would show that $X \setminus K$ is open and hence K is closed.

For each $y \in K$, using the Hausdorff condition we get $U_y, V_y \subseteq X$ disjoint such that $x \in U_x$ and $y \in V_y$. Thus we get an open cover $K \subseteq \bigcup_{y \in K} V_y$, and hence we can extract a finite subcover $V_{y_1}, \dots, V_{y_N}$. Then, we have that $U = \bigcap_{i=1}^N V_{y_i}$ is a *finite* intersection of opens and hence open, and by construction U is disjoint from K , so we are done. $\square$

14.2. Regularity of measures.

Often when dealing with measures on $(X, \mathcal{B}_X)$ a topological space with its Borel σ -algebra, we want a measure to behave nicely with respect to the topological structure on X . This condition is encapsulated by regularity:

Definition 14.8. A measure μ on $(X, \mathcal{B}_X)$ is (*Borel*) *regular* if all of the following conditions hold:

- (1) (finite on compact sets): $\mu(K) < \infty$ for all $K \subseteq X$ compact,
- (2) (outer regularity of Borel by open): $\mu(A) = \inf \{\mu(U) : U \supseteq A \text{ open}\}$ and $A \in \mathcal{B}_X$,
- (3) (inner regularity of open by compact): $\mu(U) = \sup \{\mu(K) : K \subseteq U \text{ compact}\}$ and $U \subseteq X$ open.

Example 14.9. For example, $\mathbb{R}^d$ with the standard topology and Lebesgue measure is regular.

Recall the following definition:

Definition 14.10. A space X is *second countable* if it admits a countable basis.

We wish to use second countability of the underlying space (which is in some sense an upper bound on its size) to show that measures defined on the space cannot be too badly behaved. Indeed, this is (in some sense) true for LCH spaces:

Proposition 14.11. *Let X be LCH and second countable. Then, if μ is finite on compact sets, it is outer and inner regular.*

In particular, for LCH second countable spaces, being finite on compact sets is sufficient to show regularity.

Definition 14.12. Let $(X, \mathcal{B}_X)$ be a topological space equipped with its Borel σ -algebra. A set $A \in \mathcal{B}_X$ is called F_σ if it is the countable union of closed sets, and G_δ if it is the countable intersection of open sets.

Definition 14.13. A space $(X, \mathcal{B}_X)$ is called G_δ if each closed set is a G_δ .

Equivalently, we could impose the condition that each open set is a F_σ . That being said, there is a slightly stronger condition we can impose, asking for every open set to be the countable union of *compact* sets. This turns out to be true of LCH second countable spaces, and is the key technical result on the road to proving regularity.

Lemma 14.14. *Let X be LCH second countable. Then, every open set is a countable union of compact sets.*

Corollary 14.15. *Let X be LCH second countable. Then, every open set is F_σ and every closed set is G_δ , so X is G_δ .*

Proof. To prove this claim, let $\mathcal{O}$ be a countable basis for the topology on X . Then, let

$$\mathcal{O}_U = \{V \in \mathcal{O} : \bar{V} \subseteq U\}.$$

We claim that

$$U = \bigcup_{V \in \mathcal{O}_U} V;$$

this would prove our desired result, since $\mathcal{O}_U \subseteq \mathcal{O}$ is countable.

[TODO]

□

Lemma 14.16. *Let X be Hausdorff and G_δ . Then every finite measure μ is outer regular: for all $A \in \mathcal{B}_X$ Borel,*

$$\mu(A) = \inf \{\mu(U) : U \supseteq A \text{ open}\}.$$

Proof. Let $\mathcal{F} \subseteq \mathcal{B}_X$ be the collection of sets $A \in \mathcal{B}_X$ such that outer regularity by opens and inner regularity by closed sets holds:

$$\mu(A) = \inf \{\mu(U) : U \supseteq A \text{ open}\} \quad \text{and} \quad \mu(A) = \sup \{\mu(C) : C \subseteq A \text{ closed}\}.$$

We show that $\mathcal{F}$ is a σ -algebra containing all open sets, which gives the desired result. [TODO: do this lol]

It is useful to write

$$\mathcal{F} = \{A \in \mathcal{B}_X : \forall \varepsilon > 0 \exists C \text{ closed}, U \text{ open}, C \subseteq A \subseteq U, \mu(U \setminus C) \leq \varepsilon\}.$$

□

Proof of Proposition 14.11. Since X is LCH and second countable, it is G_δ . Hence, every open set U is a countable union of compact sets $U = \bigcup_{n=1}^\infty K_n$:

$$\mu(U) = \mu\left(\bigcup_{n=1}^\infty K_n\right) = \lim_{N \rightarrow \infty} \mu\left(\bigcup_{n=1}^N K_n\right) \geq \sup \{\mu(K) : K \subseteq U \text{ compact}\},$$

since $\bigcup_{n=1}^N K_n \subseteq U$ is compact (finite unions of compacts are compact).

To show inner regularity, we first show that μ must be σ -finite with respect to an exhaustion $U_n \uparrow X$ by open sets with compact closure.

First note that we can pick a countable basis where each set has compact closure, by taking an arbitrary countable basis $\{Q_i\}_{i=1}^\infty$ and taking finite intersections where at least one set has compact closure. This remains a countable basis, and now has the property that every basic open has compact closure. Hence taking $U_n = \bigcup_{i=1}^n Q_i$, we get an open exhaustion $U_n \uparrow X$ with each $\bar{U}_n$ compact.

Hence by assumption $\mu(U_n) < \infty$, so we can define the *finite* measure $\mu_n(A) = \mu(A \cap U_n)$. By Lemma 14.16, μ_n is outer regular. Now fix $A \in \mathcal{B}_X$, and by regularity we can find V_n

open such that

$$\mu_n(V_n) \leq \mu_n(A) + \frac{\varepsilon}{2^n},$$

so that $\mu((V_n \cap U_n) \setminus A) \leq \frac{\varepsilon}{2^n}$. Hence taking $V = \bigcup_{n=1}^{\infty} (V_n \cap U_n)$, we have

$$\mu(V \setminus A) \leq \sum_{n=1}^{\infty} \mu((V_n \cap U_n) \setminus A) \leq \sum_{n=1}^{\infty} \frac{\varepsilon}{2^n} = \varepsilon,$$

giving regularity. $\square$

15. 9TH – RIESZ–MARKOV–KAKUTANI THEOREM, DAY 2

Today's lecture will be focused on the proof of the Riesz–Markov–Kakutani representation theorem.

15.1. Uniqueness.

We first prove a preparatory lemma.

Lemma 15.1. *Let X be a LCH space and μ a regular Borel measure. Then for each $U \subseteq X$ open,*

$$\mu(U) = \sup \left\{ \int_U f d\mu : f \in C_c(U), f \leq \mathbb{1}_U \right\}.$$

Proof. The $\geq$ direction is obvious by monotonicity of the integral.

For the $\leq$ direction, let $\alpha \in \mathbb{R}$ be such that $0 < \alpha < \mu(U)$. Then, by regularity, there exists a compact $K \subseteq U$ such that $\mu(K) > \alpha$. By Urysohn's lemma, we have a cutoff function $f \in C_c(X)$ with $\text{supp } f \subseteq U$ and $f|_K = 1$. Hence,

$$\int_U f d\mu \geq \int_U \mathbb{1}_K d\mu = \mu(K) > \alpha.$$

Since α can be taken to be arbitrarily close to $\mu(U)$, the sup is attained. $\square$

Corollary 15.2 (Uniqueness of representation). *Let X be LCH and μ, ν Borel regular measures on $(X, \mathcal{B}_X)$ such that $\mu = \nu$ in $(C_c(X))^*$; that is,*

$$\int_X f d\mu = \int_X f d\nu$$

for all $f \in C_c(X)$. Then $\mu = \nu$ as measures.

Proof. By Lemma 15.1, we get that $\mu(U) = \nu(U)$ for each open set U , and hence $\mu = \nu$. $\square$

15.2. Existence.

We first prove three preparatory lemmas. The first lemma constructs the desired (outer) measure; the second lemma proves that it has the expected monotonicity properties with respect to L ; and the third lemma shows that the constructed outer measure indeed restricts to a Borel regular measure that agrees with L .

Lemma 15.3. *Let X be a LCH space and $L \in (C_c(X))^*$ a positive bounded linear functional. Then, define*

$$\tilde{\mu}^*(U) = \sup \{L(f) : f \in C_c(X), f \leq \mathbb{1}_U\}$$

and

$$\mu^*(A) = \inf \{ \tilde{\mu}^*(U) : U \supseteq A \text{ open} \}.$$

Then, μ^* is an outer measure and every Borel set is μ^* -measurable.

Proof. Since the two RHS expressions agree for U open, we can simply denote $\mu^*(U) = \tilde{\mu}^*(U)$.

Obviously $\mu^*(\emptyset) = 0$ since L is linear. Monotonicity similarly follows from monotonicity of the integral.

It remains to check subadditivity; we start with the case where each set is open. Let V_i be a sequence of open sets, and let $V = \bigcup_{i=1}^{\infty} V_i$. Let $f \in C_c(X)$ with $f \leq \mathbb{1}_V$. Then, since $\text{supp } f \subseteq V$ is compact, we can take a finite subcover, so that there exists an $N > 0$ such that $\text{supp } f \subseteq \bigcup_{i=1}^N V_i$. By the existence of a partition of unity, we can find $f_1, \dots, f_N$ such that $\text{supp } f_i \subseteq V_i$ and $f = \sum_{i=1}^N f_i$. Without loss of generality let us take $f \geq 0$, so that each f_i can be taken such that $f_i \geq 0$.

Then, by linearity,

$$L(f) = \sum_{i=1}^N L(f_i) \leq \sum_{i=1}^N \mu^*(V_i) \leq \sum_{i=1}^{\infty} \mu^*(V_i).$$

Hence by taking the supremum over all f , we get that μ^* is subadditive with respect to sequences of open sets.

For a general sequence A_i with $A = \bigcup_{i=1}^{\infty} A_i$, we can pick $V_i \supseteq A_i$ open such that $\mu^*(V_i) \leq \mu^*(A_i) + \frac{\varepsilon}{2^i}$. Define $V = \bigcup_{i=1}^{\infty} V_i$, and note that $A \subseteq V$, so that

$$\mu^*(A) \leq \mu^*(V) \leq \sum_{i=1}^{\infty} \mu^*(V_i) \leq \varepsilon + \sum_{i=1}^{\infty} \mu^*(A_i).$$

Taking $\varepsilon \rightarrow 0$, we get the desired result.

We now show that Borel sets are μ^* -measurable. Since open sets generate $\mathcal{B}_X$, it suffices to show that each open set is μ^* -measurable. More specifically, it suffices to check that for each $U \subseteq X$ open and $A \subseteq X$ with $\mu^*(A) < \infty$,

$$\mu^*(A) \geq \mu^*(A \cap U) + \mu^*(A \cap U^c).$$

We first show this for $A = V \subseteq X$ open.

Fix $\varepsilon > 0$, and pick $f_1 \in C_c(X)$ with $f_1 \leq \mathbb{1}_{V \cap U}$ such that $L(f_1) \geq \mu^*(V \cap U) - \varepsilon$. Then let $K = \text{supp } f_1 \subseteq U \cap V$. Then clearly $V \cap K^c \supseteq V \cap U^c$, so we can pick $f_2 \in C_c(X)$ with $f_2 \leq \mathbb{1}_{V \cap K^c}$ and $L(f_2) \geq \mu^*(V \cap K^c) - \varepsilon \geq \mu^*(V \cap U^c) - \varepsilon$.

Now let $f = f_1 + f_2$, and note that $\text{supp } f_1 \cap \text{supp } f_2 = \emptyset$ and $f \leq \mathbb{1}_V$. By linearity,

$$\mu^*(V) \geq L(f) \geq L(f_1) + L(f_2) \geq \mu^*(V \cap U) + \mu^*(V \cap U^c) - 2\varepsilon,$$

so taking $\varepsilon \rightarrow 0$ we get the desired result.

Now let $A \subseteq X$ be an arbitrary set. Then, fixing $\varepsilon > 0$, we can find $V \supseteq A$ open such that $\mu^*(V) \leq \mu^*(A) + \varepsilon$. Then, by above we can compute that

$$\mu^*(A) \geq \mu^*(V) - \varepsilon \geq \mu^*(V \cap U) + \mu^*(V \cap U^c) - 3\varepsilon \geq \mu^*(A \cap U) + \mu^*(A \cap U^c) - 3\varepsilon,$$

from which the desired result follows by taking $\varepsilon \rightarrow 0$. $\square$

Lemma 15.4. *Let X be a LCH space and $L \in (C_c(X))^*$ a positive bounded linear functional, and μ^* as above. Then, for any $f \in C_c(X)$ and $A \subseteq X$,*

- (1) $f \geq \mathbb{1}_A \implies L(f) \geq \mu^*(A)$,
- (2) $0 \leq f \leq \mathbb{1}_A \implies L(f) \leq \mu^*(A)$.

Proof. In the first case, let $f \in C_c(X)$ with $f \leq \mathbb{1}_A$. Then, since f is continuous,

$$U_\varepsilon = f^{-1}((1 - \varepsilon, \infty)) = \{x \in X : f(x) > 1 - \varepsilon\}$$

is an open set containing A . Now let $g \in C_c(X)$ be such that $g \leq \mathbb{1}_{U_\varepsilon}$ and $L(g) \geq \mu^*(U_\varepsilon)$. Note that (pointwise) $(1 - \varepsilon)g \leq f$, so by monotonicity of L (since L is positive),

$$\mu^*(A) \leq \mu^*(U_\varepsilon) \leq L(g) \leq \frac{1}{1 - \varepsilon} L(f).$$

Now, the result follows by taking $\varepsilon \rightarrow 0$.

In the second case, take $U \supseteq A$. Then, $f \leq \mathbb{1}_U$, so $L(f) \leq \mu^*(U)$. Since U was an arbitrary open set containing A , it follows that $L(f) \leq \mu^*(A)$.

□

Lemma 15.5. *Let X be a LCH space and $L \in (C_c(X))^*$ a positive bounded linear functional, and μ^* as above. Then $\mu := \mu^*|_{\mathcal{B}_X}$ is Borel regular and*

$$L(f) = \int_X f d\mu$$

for all $f \in C_c(X)$.

Proof. Outer regularity holds by definition, so it remains to check inner regularity and finiteness on compact sets.

In particular, for inner regularity it suffices to check that for all $U \subseteq X$ open,

$$\mu^*(U) \leq \sup \{\mu^*(K) : K \subseteq U \text{ compact}\}.$$

Pick $\varepsilon > 0$, and let $f \in C_c(X)$ such that $f \leq \mathbb{1}_U$ and $L(f) \geq \mu^*(U) - \varepsilon$. Let $K = \text{supp } f$, and then note that by Lemma 15.4, we have $L(f) \leq \mu^*(K)$, so that

$$\mu^*(K) \geq L(f) \geq \mu^*(U) - \varepsilon,$$

which gives the desired result by taking $\varepsilon \rightarrow 0$.

To show finiteness on compact sets, note that for any compact $K \subseteq X$, by Lemma 15.4 we can take any $f \in C_c(X)$ with $f \geq \mathbb{1}_K$, so that $\mu^*(K) \leq L(f) < \infty$.

Now we prove that $L(-) = \int_X - d\mu$ on $C_c(X)$. By linearity, it suffices to check this for $f \geq 0$.

Let $f \in C_c(X)$ such that $f \geq 0$. Pick $\varepsilon > 0$, then define the “layer-cake decomposition”

$$f_n(x) = \begin{cases} 0 & f(x) < n\varepsilon \\ \min\{f(x) - n\varepsilon, \varepsilon\} & f(x) \geq n\varepsilon. \end{cases}$$

Clearly $f_n \in C_c(X)$, and since $\|f\|_\infty < \infty$, $N = \left\lfloor \frac{\|f\|_\infty}{\varepsilon} \right\rfloor < \infty$, so we can take $f(x) = \sum_{n=0}^N f_n(x)$. Now define the “cake-sheets”

$$K_n = \{x \in X : f(x) \geq n\varepsilon\} \supseteq \text{supp } f_n,$$

which are compact since they are closed subsets of $\text{supp } f$. Then, $\varepsilon \mathbb{1}_{K_{n+1}} \leq f_n \leq \varepsilon \mathbb{1}_{K_n}$

[TODO]

□

Proposition 15.6. *Let X be a LCH space and $L \in (C_c(X))^*$ a positive bounded linear functional. Then there exists a positive Borel regular measure μ on $(X, \mathcal{B}_X)$ such that*

$$L(f) = \int_X f d\mu$$

for all $f \in C_c(X)$.

[TODO]

Proof. For each $U \subseteq X$ open, define

$$\tilde{\mu}^*(U) = \sup \{L(f) : f \in C_c(U), f \leq \mathbb{1}_U\}$$

and for each $A \subseteq X$ arbitrary, define

$$\mu^*(A) = \inf \{\tilde{\mu}^*(U) : U \supseteq A \text{ open}\}.$$

We first prove that $\tilde{\mu}^* = \mu^*$, then we show that μ^* is an outer measure.

□

16. 10TU – REVIEW

16.1. Topological facts.

Definition 16.1. Let X be a topological space, and let $A, B \subseteq X$ be disjoint subsets. Then, a function $f : X \rightarrow [0, 1]$ *separates A and B* if $f|_A = 0$ and $f|_B = 1$. In this case we say f *separates A and B* , or that A and B are *separated by a continuous function*.

Definition 16.2. Let X be a topological space. We say that X is T_4 , or *normal*, if for every pair of disjoint closed sets $C_1, C_2 \subseteq X$, there exist disjoint open $U_1, U_2 \subseteq X$ such that $C_1 \subseteq U_1$ and $C_2 \subseteq U_2$.

Theorem 16.3 (Urysohn lemma). *Let X be a topological space. Then, the following are equivalent:*

- (1) X is T_4 ,
- (2) For all $C \subseteq X$ closed and $U \subseteq X$ open with $C \subseteq U$, there exist $C' \subseteq X$ closed and $U' \subseteq X$ open such that $C \subseteq U' \subseteq C' \subseteq U$.
- (3) Every pair $C_0, C_1 \subseteq X$ of closed disjoint sets is separated by a continuous function.
- (4) For every $C \subseteq U \subseteq X$, there exists $f \in C(X \rightarrow [0, 1])$ with $f|_C = 1$ and $\text{supp } f \subseteq U$.

Proof. Obviously, (1) $\iff$ (2) and (3) $\iff$ (4) by taking complements.

(3) $\implies$ (1) is straightforward: given a function $f: X \rightarrow [0, 1]$ separating $C_0, C_1 \subseteq X$, take

$$U_0 = \left\{ x \in X: f(x) < \frac{1}{4} \right\} \quad \text{and} \quad U_1 = \left\{ x \in X: f(x) > \frac{3}{4} \right\}.$$

The nontrivial direction is that taking (2) $\implies$ (4).

Let $C_1 = C$ and $U_0 = U$ be given, with $C_1 \subseteq U_0$. Take $U_1 = \emptyset$ and $C_1 = X$. By (2), we can find closed and open sets such that $U_1 \subseteq C_1 \subseteq U_{\frac{1}{2}} \subseteq C_{\frac{1}{2}} \subseteq U_0 \subseteq C_0$. Applying (2) again, we can get closed and open sets such that

$$U_1 \subseteq C_1 \subseteq U_{\frac{3}{4}} \subseteq C_{\frac{3}{4}} \subseteq U_{\frac{1}{2}} \subseteq C_{\frac{1}{2}} \subseteq U_{\frac{1}{4}} \subseteq C_{\frac{1}{4}} \subseteq U_0 \subseteq C_0,$$

and so on.

Recall that $\mathbb{Z}[\frac{1}{2}] \subseteq \mathbb{Q}$ are the *dyadic rationals*: $\mathbb{Z}[\frac{1}{2}] = \left\{ \frac{a}{2^k}: k \in \mathbb{Z}_{\geq 0}, a \in \mathbb{Z} \right\}$. Then, for all $q \in \mathbb{Z}[\frac{1}{2}] \cap [0, 1]$, we get

$$U_q \subseteq C_q \quad \text{and} \quad C_q \subseteq U_{q'} \text{ for } q > q'.$$

Then define $f: X \rightarrow [0, 1]$ by the “2-adic valuation”:

$$f(x) = \sup \left\{ q \in \mathbb{Z} \left[\frac{1}{2} \right] \cap [0, 1]: x \in U_q \right\} = \inf \left\{ q \in \mathbb{Z} \left[\frac{1}{2} \right] \cap [0, 1]: x \in X \setminus C_q \right\}.$$

Then, since

$$\{x \in X: f(x) > t\} = \bigcup_{q>t} U_q \quad \text{and} \quad \{x \in X: f(x) < t\} = \bigcup_{q<t} (X \setminus C_q)$$

are both open, f is continuous. □

Remark 16.4. If X is compact Hausdorff, then it is normal. To see this, note that closed sets are compact and hence we can find open sets separating each pair of points, and then taking a finite subcover we get the desired separation.

Corollary 16.5. *Let X be LCH. Then for all $K \subseteq U \subseteq X$ with K compact and U open, there exists a bump function $f \in C_c(U \rightarrow [0, 1])$ with $f|_K = 1$.*

Proof. At each point $x \in K$ we can find a neighborhood V_x disjoint from U^c . We can extract a finite subcover, which then gives us an open neighborhood $K \subseteq V \subseteq \bar{V} \subseteq U$ with $\bar{V}$ compact. Then, applying Urysohn’s lemma to K and V^c we get a function separating K and V^c , and hence a function equal to 1 on K and vanishing outside of V , so that it is compactly supported. □

Definition 16.6. Let X be a topological space. Then, a continuous function $f: X \rightarrow \mathbb{R}$ *vanishes at infinity* if for all $\varepsilon > 0$ there exists a $K \subseteq X$ compact such that

$$\sup_{x \in K} |f(x)| \leq \varepsilon.$$

The space of functions vanishing at infinity is denoted $C_0(X)$.

Proposition 16.7. *Let X be LCH and μ a Borel regular measure. Then,*

- (1) $C_c(X) \hookrightarrow L^p(X, \mu)$ is dense,
- (2) $C_c(X) \hookrightarrow C_0(X)$ is dense in the L^∞ norm.

Proof. (1): Let $A \in \mathcal{B}_X$ with $\mu(A) < \infty$, so that $\|\mathbb{1}_A\|_p < \infty$. By regularity, we can find $U \supseteq A$ such that $\mu(U \setminus A) \leq \varepsilon$. Again by regularity, we can find a $K \subseteq U$ such that $\mu(U \setminus K) \leq \varepsilon$. By Urysohn's lemma, we can find a cutoff function $f \in C_c(X)$ such that $f|_K = 1$ and $\text{supp } f \subseteq U$.

This gives us that

$$\|\mathbb{1}_A - \mathbb{1}_U\|_p \leq \varepsilon^{1/p} \quad \text{and} \quad \|\mathbb{1}_U - f\|_p \leq \|\mathbb{1}_U - \mathbb{1}_K\|_p \leq \varepsilon^{1/p},$$

and hence by the triangle inequality

$$\|\mathbb{1}_A - f\|_p \leq \|\mathbb{1}_A - \mathbb{1}_U\|_p + \|\mathbb{1}_U - f\|_p \leq 2\varepsilon^{1/p}.$$

(2): Obviously, if $f_n \in C_c(X)$ and $f_n \rightarrow f$ in the sup norm, then f is continuous and vanishes at infinity, since $\text{supp } f = \bigcup_{n=1}^{\infty} \text{supp } f_n$. Hence $\overline{C_c(X)} \subseteq C_0(X)$.

In the other direction, suppose $f \in C_0(X)$. Then, for $\varepsilon > 0$ pick $K \subseteq X$ compact such that $\sup_K |f| \leq \varepsilon$. Then, let $h \in C_c(X)$ be a bump function with $h|_K = 1$ and $\text{supp } h$ compact. Then, $\tilde{f} = hf$ is compactly supported and continuous, and

$$\|\tilde{f} - f\|_{\infty} \leq \varepsilon,$$

as desired. □

Corollary 16.8. *Let X be LCH. Then, $(C_0(X), \|\cdot\|_{\infty})$ is a Banach space.*

Definition 16.9. Let X be LCH and μ a signed measure on $(X, \mathcal{B}_X)$. Then, μ is (Borel) *regular* if any of the following equivalent definitions hold:

- (1) $|\mu|$ is regular,
- (2) μ_+, μ_- are regular,
- (3) $\mu = \sum_{i=1}^m a_i \mu_i$ for $a_i \in \mathbb{R}$ and μ_i positive regular.

We denote the space of regular measures

$$M_{\text{reg}}(X) = \{\mu \in M(X) : \mu \text{ regular}\}.$$

Note that if $\mu \in M_{\text{reg}}(X)$, we get a functional $\phi_{\mu} \in (C_0(X))^*$ by

$$\phi_{\mu}(f) = \int_X f d\mu.$$

Indeed, we have a converse, the Riesz representation theorem for signed measures:

Theorem 16.10 (Riesz representation theorem for signed measures). *The map $\Phi: M_{\text{reg}}(X) \rightarrow (C_0(X))^*$ defined by $\Phi(\mu) = \phi_{\mu}$ is an isometric isomorphism.*

Proof. We first show that Φ is an isometry. Note that

$$|\phi_{\mu}(f)| \leq \left| \int_X f d\mu_+ \right| + \left| \int_X f d\mu_- \right| \leq \|f\|_{\infty} \|\mu\|.$$

Hence $\|\phi_\mu\|_* \leq \|\mu\|$. In the other direction, we can pick a finite set $A_1, \dots, A_n$ of disjoint Borel sets such that

$$\sum_{i=1}^n |\mu(A_i)| \geq \|\mu\| - \varepsilon.$$

By regularity (of $|\mu|$) we may pick $K_i \subseteq A_i$ compact with $|\mu(A_i \setminus K_i)| \leq |\mu|(A_i \setminus K_i) \leq \frac{\varepsilon}{n}$, so

$$\sum_{i=1}^n |\mu(K_i)| \geq \|\mu\| - 2\varepsilon.$$

Since K_i are disjoint we may pick U_i disjoint opens with $K_i \subseteq U_i$. Pick h_i bump functions identically 1 on K_i and supported in U_i , and let $f_i = \text{sgn}(\mu(K_i))h_i$. Then let $f = \sum_{i=1}^n f_i$. Then,

$$\int_X f d\mu = \sum_{i=1}^n \int_{K_i} f_i d\mu + \int_{\bigcap_{i=1}^n K_i^c} f d\mu \geq \sum_{i=1}^n |\mu(K_i)| + \left| \mu \left(\bigcap_{i=1}^n K_i^c \right) \right|$$

[TODO; finish proof of bound]

Now we show that Φ is surjective. If $L \in (C_0(X))^*$ is nonnegative, by Riesz–Markov–Kakutani we have a Borel measure μ such that $L(f) = \int_X f d\mu$ for all $f \in C_c(X)$. Then, since $C_c(X) \hookrightarrow C_0(X)$ is dense and L is bounded, this extends to a functional on the full space $C_0(X)$.

In general we claim that if $L \in (C_0(X))^*$, then we can decompose $L = L_+ - L_-$ where L_+, L_- are nonnegative. Then, by above we can recover positive measures μ_+ and μ_- representing L_+ and L_- , and then the desired signed measure is $\mu = \mu_+ - \mu_-$.

To find this decomposition, we will find L_+ and then define $L_- = L_+ - L$. For $f \in C_0(X)$, $f \geq 0$,

$$L_+(f) = \sup \{L(g) : g \in C_0(X), 0 \leq g \leq f\}.$$

Intuitively, this supremum makes $g = 0$ on the parts where the measure is negative and $g = f$ on the parts where the measure is positive. Clearly L_+ is positive, since the sup is bounded below by $L(0) = 0$. [TODO: prove L_+ is linear]

□

17. 10TH – STONE–WEIERSTRASS AND VITALI COVERING LEMMA

17.1. Stone–Weierstrass theorem. We first prove the Weierstrass approximation theorem for polynomials, and then using this prove the general result: the Stone–Weierstrass theorem on a compact Hausdorff space.

Theorem 17.1 (Weierstrass). *Let $P([a, b])$ be the space of polynomials on $[a, b]$. Then, $P([a, b]) \hookrightarrow C([a, b])$ is dense.*

Let X be a compact Hausdorff space. We say that $\mathcal{A} \subseteq C(X)$ is an *algebra* if it is a vector subspace closed under max and min. Furthermore, we say that $\mathcal{A}$ *separates points* if for all points $x, y \in X$, there exist $h_x, h_y \in \mathcal{A}$ with $h_x(x) \neq h_y(y)$.

Theorem 17.2 (Stone–Weierstrass theorem). *Let X be a compact Hausdorff space, and $\mathcal{A} \subseteq C(X)$ an algebra. Then, $\mathcal{A} \hookrightarrow C(X)$ is dense; that is, $\overline{\mathcal{A}} = C(X)$ in the uniform topology.*

Note that polynomials on $[a, b]$ form an algebra, so this is a genuine generalization of the Weierstrass polynomial approximation theorem.

Proof of Theorem 17.1. [TODO: fill in details later]

Step 1: WLOG we can assume $\mathcal{A} = \overline{P([a, b])}$ is closed. First show that $|x|$ can be arbitrarily well approximated by a polynomial p : $\|p(x) - |x|\|_\infty \leq \varepsilon$. Then for any $f, g \in \mathcal{A}$, can show that $\|p(f(x) - g(x)) - |f(x) - g(x)|\|_\infty \leq \varepsilon$, hence $\min\{f, g\}, \max\{f, g\} \in \mathcal{A}$.

Step 2: Fix $f \in C(X)$. For all $x, y \in X$, using the separation property and taking linear combinations we can find h_x, h_y such that $f(x) = h_x(x)$ and $f(y) = h_y(y)$, then cook up functions h_{xy} that are ε -close to f on an open set $V_{xy} \supseteq \{x, y\}$. The sets V_{xy} form an open cover of X , so take finite subcovers in each index to cook up an aggregate function h by min-max which well-approximates f .

Probabilistic proof: WLOG take $[a, b] = [0, 1]$, $f \in C([0, 1])$. Take $X_n \sim \text{Bin}(n, p)^{\frac{1}{n}}$, so that each $X_n = \frac{1}{n} \sum_{i=1}^n B_{i_n}$, where $B_{i_j} \sim \text{Bern}(p)$ iid. Take $Q_f(p) = \mathbb{E}(f(X_n))$; note that $p = \mathbb{E}(B_{i_j})$, so by linearity $p = \mathbb{E}(X_n)$. We want to show that

$$\sup_{p \in [0, 1]} |f(p) - Q_f(p)| \leq \varepsilon;$$

note that this is equivalent to showing that

$$\lim_{n \rightarrow \infty} |f(\mathbb{E}(X_n)) - \mathbb{E}(f(X_n))| \leq \varepsilon.$$

But, this follows from the law of large numbers. □

17.2. Vitali covering lemma. Let (X, d) be a metric space and $B = \overline{B_r(x)} \subseteq X$ a closed ball centered at $x \in X$ of radius $r > 0$. For any $\lambda > 0$, denote $\lambda B = \overline{B_{\lambda r}(x)}$.

In the following we use general metric spaces, but our primary application will be the case of $\mathbb{R}^d$ with metric induced by the norm.

Theorem 17.3 (Vitali 3-times finite covering). *Let X be a metric space, and let $\{B_i\}_{i=1}^n$ be a finite collection of balls. Then, there exists a finite disjoint subcollection $\{B_{i_j}\}_{j=1}^m$ such that*

$$\bigcup_{i=1}^n B_i \subseteq \bigcup_{j=1}^m 3B_{i_j}.$$

Proof. [TODO: refine]

Step 1: Find the subcollection: greedy algorithm. First pick the biggest ball, then continue iterating by finding the next biggest ball that does not intersect the previously chosen balls. Since n is finite this process terminates in finitely many steps.

Step 2: Prove the 3-times covering. For any B_i in the original covering, find the biggest ball B_{i_j} in the subcollection intersecting B_i . By construction the radius of $B_i \leq$ the radius of B_{i_j} . Hence $B_i \subseteq 3B_{i_j}$, by the triangle inequality. □

Theorem 17.4 (Vitali 5-times infinite covering). *Let X be a metric space, and let $\mathcal{C}$ be a collection of closed balls in X with $\sup_{B \in \mathcal{C}} \text{rad}(B) < \infty$. Then, there exists a subcollection*

$\mathcal{C}' \subseteq \mathcal{C}$ of disjoint balls such that

$$\bigcup_{B \in \mathcal{C}} B \subseteq \bigcup_{B' \in \mathcal{C}'} 5B'.$$

If in addition X is separable, $\mathcal{C}'$ is countable.

The primary application of this lemma will be for separable metric spaces (or more generally Polish spaces, which are topological spaces homeomorphic to a separable metric space; think the difference between a metric space and metrizable space). The last point follows from the straightforward fact that a collection of disjoint balls in a separable space must be at most countable.

As a second remark, the constant 5 is not optimal: any $3 + \varepsilon$ for $\varepsilon > 0$ will do.

Proof. TODO

Proof is similar, modulo an additional fudge factor that forces a constant of $3 + \varepsilon$ when doing the greedy algorithm, since we have to split up the collection $\mathcal{C}$ into subcollections of increasing radius. $\square$

In applications, it is often useful to define the following notion of a covering by closed balls which is “fine” in the sense that any point is contained in arbitrarily small balls.

Definition 17.5. Let $A \subseteq \mathbb{R}^d$ be a subset. A *Vitali covering* of A is a collection $\mathcal{V}$ of closed balls such that for all $\varepsilon > 0$ and $x \in A$, there exists a ball $B \in \mathcal{V}$ such that $x \in B$ and $(B) < \varepsilon$.

If A is Borel, then such a cover can be refined to a countable disjoint cover of A up to a set of measure zero:

Theorem 17.6 (Vitali covering theorem). *Let $A \in \mathcal{B}_{\mathbb{R}^d}$ and $\mathcal{V}$ a Vitali covering of A . Then, there exists a countable disjoint subcollection $\mathcal{C} \subseteq \mathcal{V}$ such that*

$$\lambda \left(A \setminus \bigcup_{B \in \mathcal{C}} B \right) = 0.$$